

GE Healthcare

Signa EXCITE to HDx 3.0T System Upgrade

Technical Publication



5166006
Rev 7

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

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GE Medical Systems, a General Electric Company, going to market as GE Healthcare
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USA

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WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

REVISION HISTORY

REV	DATE	PRIMARY REASON FOR CHANGE
1 ...	Mar 2, 2006	Initial Release.
2 ...	Jul 11, 2006	Section 2–3 – Added CAUTION note to upgrade flowchart that refers to the IIS Key and SaveINFO operation. Revised s/w load instructions and added LCD monitor adjustments (MRIhc15805, MRIhc16251). Added blocks for restoreINFO, ICW, and update of eLicense (MRIhc15873, MRIhc15892, MRIhc16209) Section 4 – Added Note 14 for CAT6 cable. Section 7: Added note informing the installer not to connect additional monitors to host computer prior to operating system software load (MRIhc15870). Clarified the installation of jumpers when setting up the DVD module is the SCSI tower.
3 ..	Feb 22, 2007	Section 2: Reversed flowchart procedures <i>Software Option Install w/eLicense</i> and <i>Install and Calibration Wizard</i> . Section 4: Added note for instructions on storing Service Cable Kit (CQA 13094026). Section 6: Removed PAC2B part number. Section 7: Added information on cable disconnection from existing GOC. For installing Run 1323, added notes for routing with Run 1322 and that CAT6 cable must replace existing CAT5 cable. Added note that Humidity iButton can be installed in either direction. Section 8: Updated MGD chassis install information. Revised installation location of ICN brackets (CQA 13078452). Updated board installation instructions for upgrading to 16 channels. Added instructions for storing Service Cable Kit.
4 ..	May 24, 2007	Section 7: Added iButton installation and Environmental Sensors Block Diagram to Section 7–6, Temperature and Humidity Sensor Kit Installation. Section 8: Updated APS Board part number from 2294300–14 to 2294300–16 (ECO 2044030). Adding existing cables connected to Term Server (CQA13112624).
5 ..	Jun 17, 2008	Section 4: Updated cable map to show OW component and cable upgrades. Updated cable maps to show new Temperature and humidity Sensor installation and cables. Section 7: Revised entire section for replacing the PC computer in the GOC, and other components, if required. Also revised temperature/humidity sensor installation to reflect the new design.
6 ..	Sep 18, 2008	Section 1: Added HDxt (Release 15.x) information to Introduction. Section 8: Revised section to incorporate the new SUN ICN modules for upgrades.
7 ...	Jul 27, 2010	Section 2: Updated eLicense model type from Signa to MR HDx (per Direction 2397845). Section 4: Updated cable maps for removal of OW components and installation of Z400 GOC (ECO 2096124). Section 7: Updated instructions for removal of obsolete components and GOC. Added instructions for installing Z400 GOC (ECO 2096124). Section 8: Removed second set of ICN brackets in cabinet because new Sun 4170 ICN replaces two ICN's installed in previous upgrades (ECO 2101939).

LIST OF EFFECTIVE PAGES

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Damage in Trans.	–	2–3 to 2–4	1	4–3 to 4–10	7	OPERATOR			
Direction 2128126 ...	1*	2–5	7			WORKSPACE		TAB 9	
A	7			TAB 5		2164904–7	–	RF ACCESSORY	
i	1			PENETRATION PANEL		7–1 to 7–26	7	CABINET	
				2164904–5	–			2164904–9	–
TAB 1		TAB 3		5–1 to 5–2	1	TAB 8		9–1 to 9–2	1
GETTING STARTED		POWER OFF				SYSTEM CABINET		TAB 10	
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1–1 to 1–6	6	3–1	1	MAGNET ROOM		8–1 to 8–6	6	2164904–11	–
				2164904–6	–	8–7 to 8–9	7	10–1 to 10–2	1
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SECTION 1 – GETTING STARTED

1-1 UPGRADE INTRODUCTION

This publication documents the mechanical installation information for upgrading a Signa® 3.0T EXCITE (Release G3, EXCITE II) to Signa HDx 3.0T (Release 14.x) or HDxt (Release 15.x). Refer to illustration 1-1 for pictorial overview of upgraded system.



FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Tools and test equipment to install and calibrate the Signa 3.0T HDx system are listed in the *Signa HDx 3.0T Upgrades Pre-Installation* manual, Section 10, TOOLS AND TEST EQUIPMENT.

Service installation tools used in this manual that may not be included in above references lists include:

- 46-301450G1 Fiber Optic Cable Connector Repair kit

1-3 PRE-DELIVERY FDO CHECKS FOR SIGNA EXCITE

Before the shipment is made, Be sure that FDO review has been completed by consultation with the Signa HDx 3.0T Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1-4 UPGRADE SCHEDULE CONSIDERATIONS

FMI's, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. The IIS key will need to be re-generated. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1-5 PRODUCT DELIVERY INSTRUCTIONS AND PACKING

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short. Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Fixed Site Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1-6 DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1-800-548-3366 and use option 8.
- Contact your local service coordinator for more information on this process.

1-7 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-7-1 MATERIAL POLICY OVERVIEW

1-7-1-1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly either to GoldSeal or Renewable Resources. Do not remove or swap any parts from the material to be returned.

1-7-1-2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-7-2 MATERIAL TO BE RETURNED

Return all parts, cables, modules, and equipment deinstalled for the HDx (Release 14.x) upgrade to GoldSeal which includes:

- GOC and Operator Workspace components
- Driver Module
- SRI 2
- 8 Channel Switch
- LPCA
- Front and Rear Bridge
- Exciter / Receiver Boards

1-7-3 RETURN KIT

Packaging and/or transportation details can be obtained through the local Installation Specialist. If unavailable, call GoldSeal. The following material may have been provided to facilitate return of material.

- Shipping labels
- Waybill
- Address labels
- Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.
- Packing material

1-7-4 DEINSTALLATION PROCEDURES

Refer to instructions in this manual for deinstallation procedures of site.

1-7-5 REGION SPECIFIC RETURN INFORMATION**1-7-5-1 Sites in Americas**

A GoldSeal return pre-printed BAX waybill and mailing label(s) may be included with the upgrade kit. The upgrade returns kit (p/n 2352804) consists of:

- 2358604 – Waybill
- 2357240 – GoldSeal Return Labels
- 2357241– Upgrade Return Envelope (waybill and return labels)

Gold Seal Address and contact for coordination of transportation arrangements and packaging:

GE Medical Systems – GoldSeal
3114 North Grandview Blvd
Waukesha, WI 53188
Phone: 1-877-251-4468 or 262-513-4147

1-7-5-1 Sites in Americas (Continued)

For non-GoldSeal Returns, utilize GEMS Renewable Resources. For coordination of transportation arrangements and packaging:

Phone: (414)–747–6997 or Dial Comm 8*579–6997

Fax: (414)–747–6855

Field ASPEN: 1–800–525–1516, Box #26041

1-7-5-2 Sites in Asia

At this time, system components are not returned for systems in Asia, but whole systems must be returned from complete system (forklift) upgrades.

Return procedure for GEMS–Asia countries:

Please contact your Regional Service Headquarters to get instructions.

1-7-5-3 Sites in Europe

The GoldSeal Returns Coordinator contacts each site individually and reviews the material to be returned and the logistics for the return. A list of material and a printable shipping label is sent to the site's primary FE, service providers, and forwarders.

GoldSeal Europe return information:

GEMS–E – GOLD SEAL OPERATION & RECYCLING CENTER

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL

Quai 45 à 49 – 91090 LISSES – FRANC

Tel: (+33) 1–60–91–57–35 or (+33) 1 64 97 55 41

Fax: (+33) 1–60–86–12–31

For non-Gold Seal returns in Europe:

European Renewable Resources Center

GEODIS LOGISTICS – CD 26 – Pièce de la Remise – Bâtiment EVL2

Quai 45 à 49 – 91090 LISSES – FRANCE

TEL = (+33)–1–64–97–55–41 FAX = (+33)–1–60–86–12–31

1-7-6 PRODUCT LOCATOR INFORMATION

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1–8.

1-8 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-9 UPGRADE SEQUENCE

Follow the system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to install the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

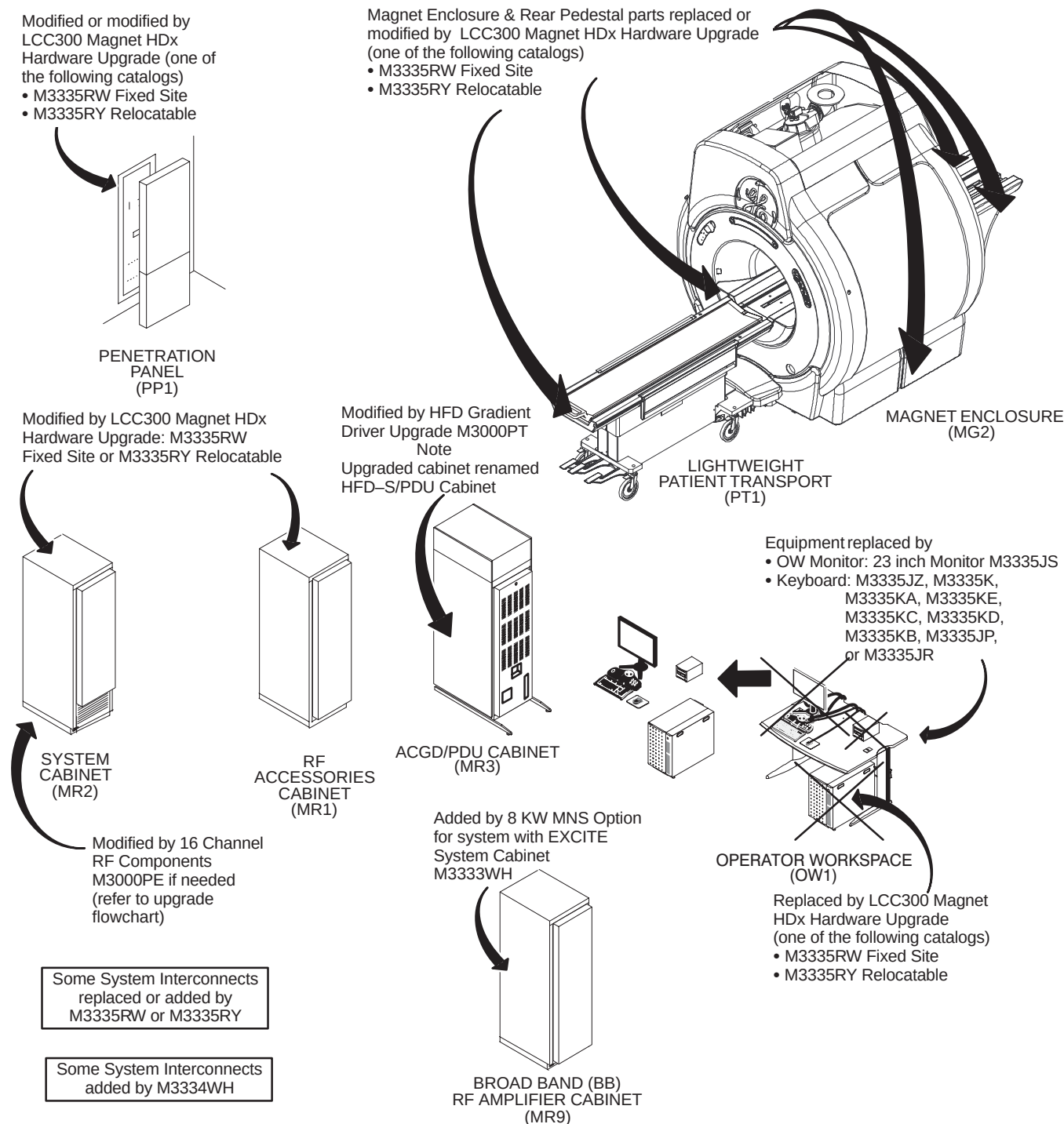
- Verify that all required upgrades have arrived complete and site is ready to shut down.
- Shut site down.
- Remove obsolete equipment and cables.

Aside: Removing and installing cables is more efficient and accurate when the equipment room is un-cluttered. It is strongly recommended that this be done after old equipment is removed and before new equipment is positioned. This house cleaning will make the installation tasks much easier.

- Route new cables
- Install new equipment and modify existing equipment.
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Load software.
- Perform Functional Checks, System calibrations, and System performance checks
- Replace covers, complete final tasks, and return site to customer.

1-10 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment replaced and modified by Signa 3.0T EXCITE (11.x) to HDx (14.x) or HDxt (15.x) Upgrade.



Release 11.x To Release 14.x SYSTEM WITH LCC300 MAGNET HDx Upgrade Modified Equipment

ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa HDx 3.0T Release 14.x.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

This upgrade flowchart has been developed assuming that all Signa HDx equipment has been delivered together. **Make sure that every part required is available before starting.**

2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

 Procedures to be Completed by Field Engineer

MARK OFF
PERFORMED
ITEMS

IMPORTANT : PER ISO
9001 REQUIREMENTS

CAUTION

Do not proceed without an IIS Key on the
SaveINFO media. Installation calibration
wizard will not permit the exit from
Installation mode in 14.x without IIS Key.

A

START

Signa 3.0T EXCITE II to EXCITE HDx
UGRADE CHECKLIST

☐ RE-GENERATE IIS KEY.
This must be done before upgrade starts to
make sure that sytem meets specifications.

☐ PERFORM "saveINFO" PROCEDURE

 Procedures to be Completed by FE and/or Mechanical Installers

Section 1
☐ Section 1-5
PRODUCT DELIVERY INSTRUCTIONS

Section 1
☐ Section 1-6
DAMAGE IN TRANSPORTATION

Section 1
☐ Section 1-8
PRODUCT LOCATOR

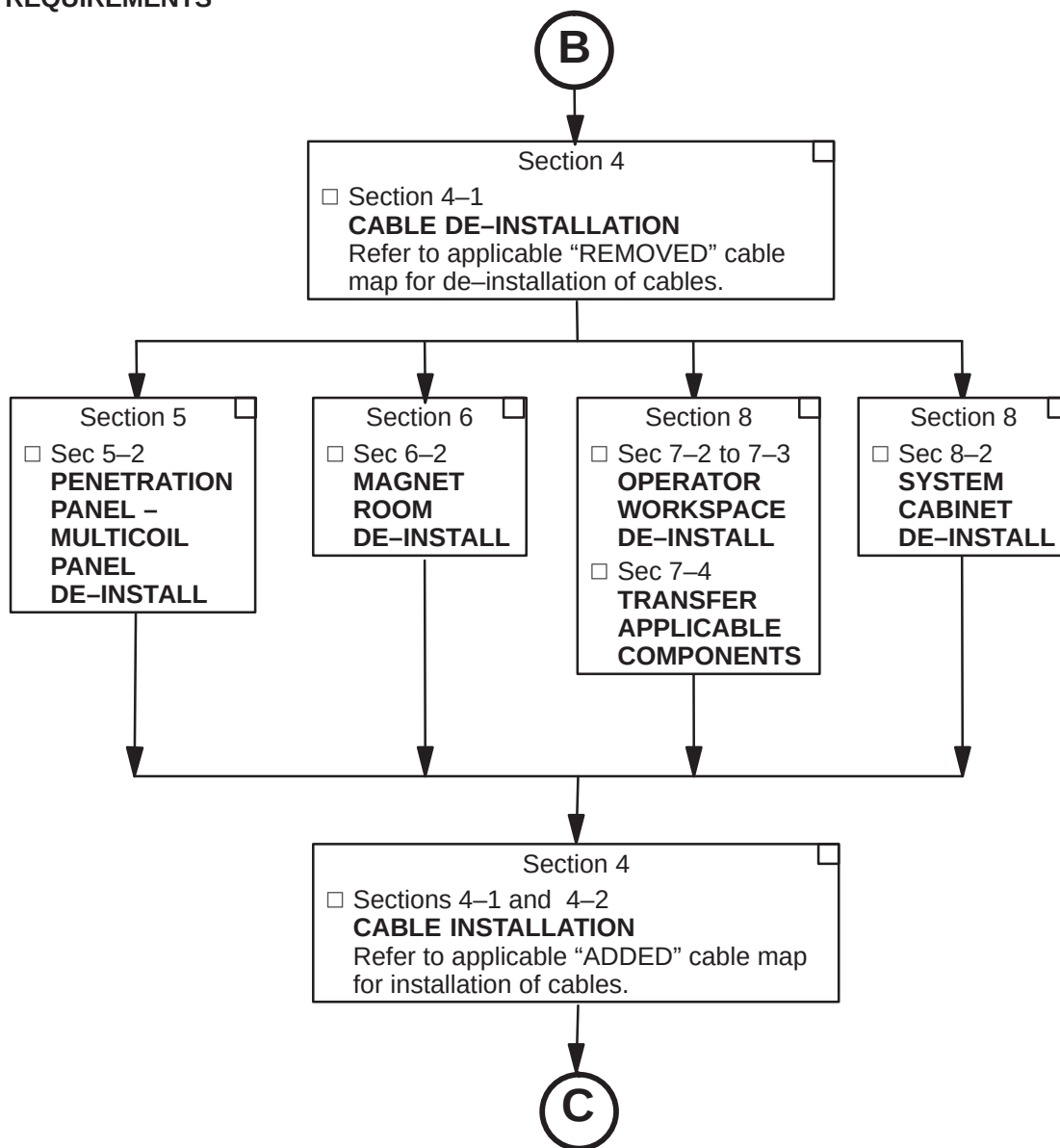
Section 3
☐ Section 3-1
SYSTEM POWER SHUTDOWN
☐ LOCKOUT & TAGOUT MAIN
DISCONNECT

B

De-Installation Procedures to be Completed by FE and/or Mechanical Installers

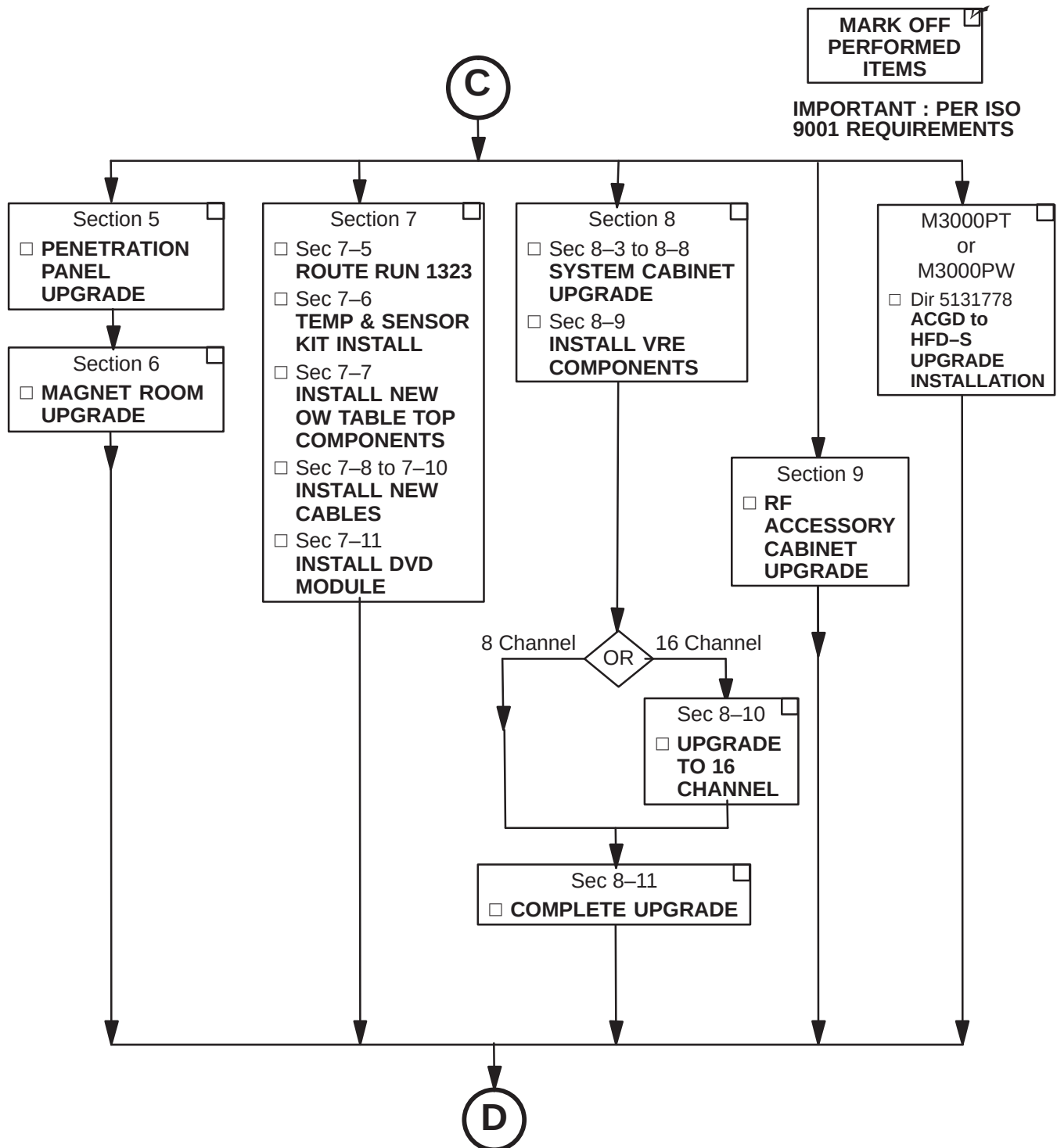


IMPORTANT : PER ISO
9001 REQUIREMENTS



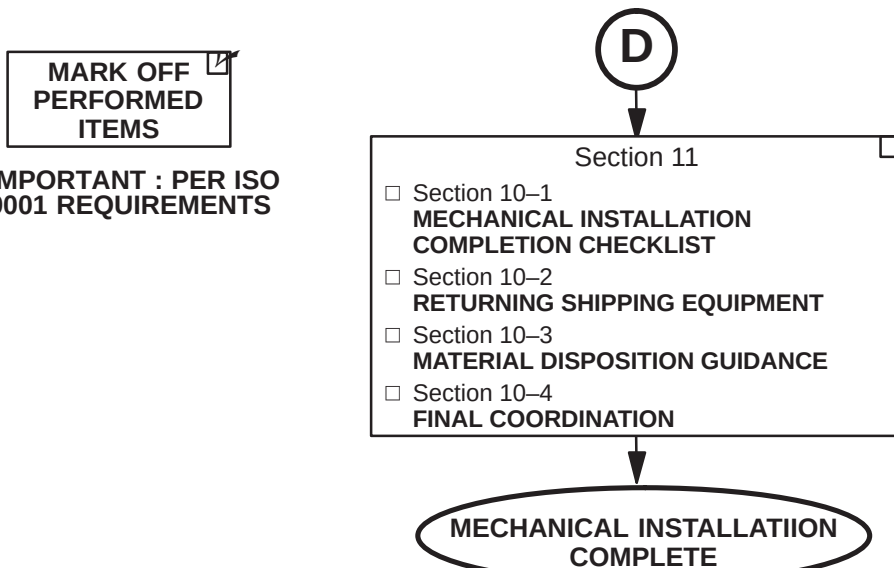
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers

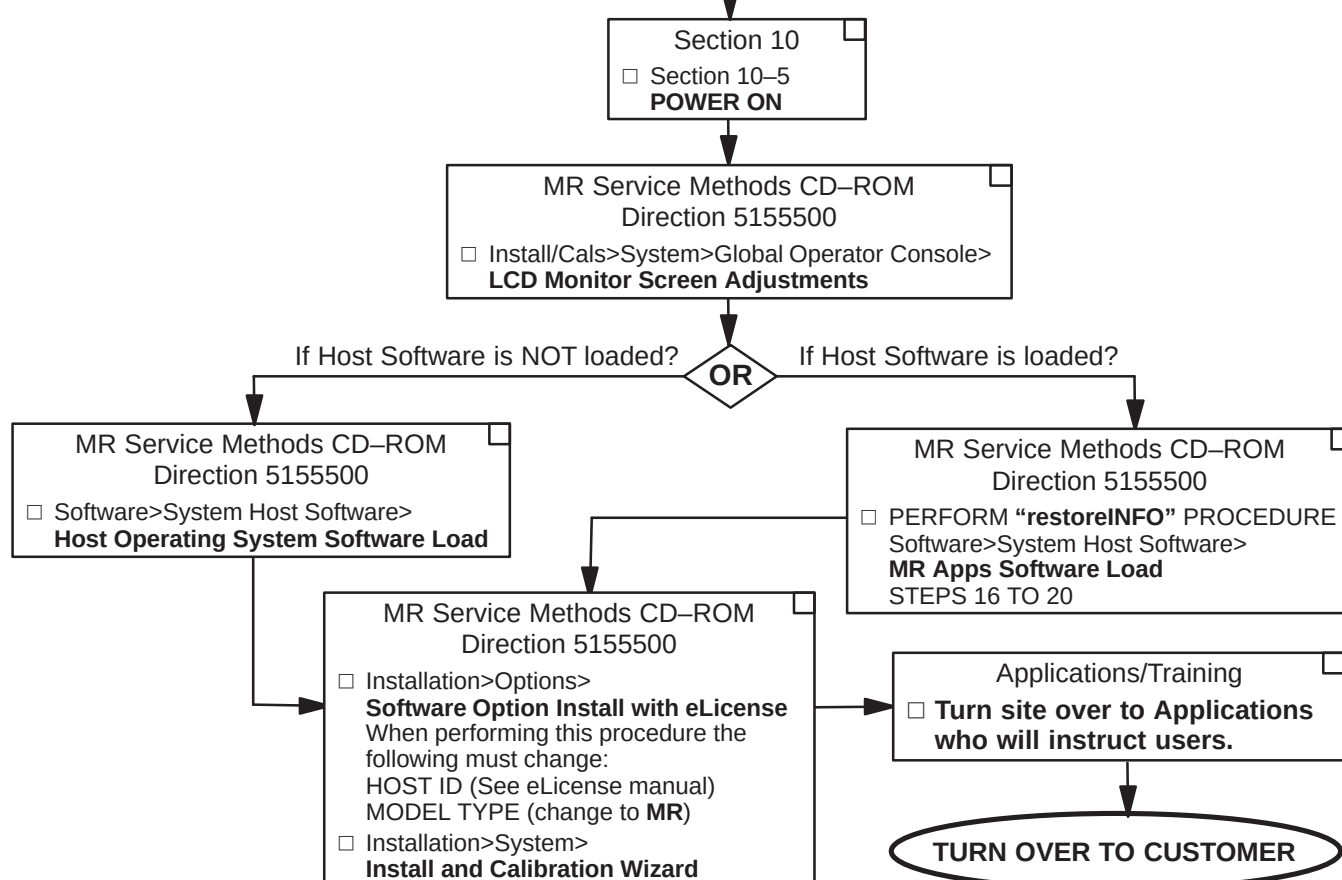


2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Make sure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – SYSTEM UPGRADE CABLES

4-1 INTRODUCTION

This section provides information on removing, relabeling and installing system cables for upgrading to Signa HDx 3.0T Release 15.x.

A double set of “Removed” and “Added” cables maps is provided. One set for the equipment room and the other for the magnet room. Tape maps in a convenient location to check off affected “Removed” and “Added” cables.

The Affected “Removed” Cable maps are located on the following pages.

- Page 4-3 Affected (removed) Cables for Upgrading to Signa HDx 3.0T Release 15.x (Equip. Room)
- Page 4-5 Affected (removed) Cables for Upgrading to Signa HDx 3.0T Release 15.x (Magnet Room)

The Affected “Added” Cable maps are located on the following pages.

- Page 4-7 Affected (added) Cables for Upgrading to Signa HDx 3.0T Release 15.x (Equip. Room)
- Page 4-9 Affected (added) Cables for Upgrading to Signa HDx 3.0T Release 15.x (Magnet Room)

Upon completion of connecting cables, return fold-out map to the binder for future reference.

4-2 SORTING AND ROUTING CABLES



Ty-wraps must be cut flush with no protruding sharp edges or points. Failure to do so can result in numerous laceration hazards when servicing the equipment.

System power, ground, and data cables are color coded at each end by destination per colors shown on Cable Map.

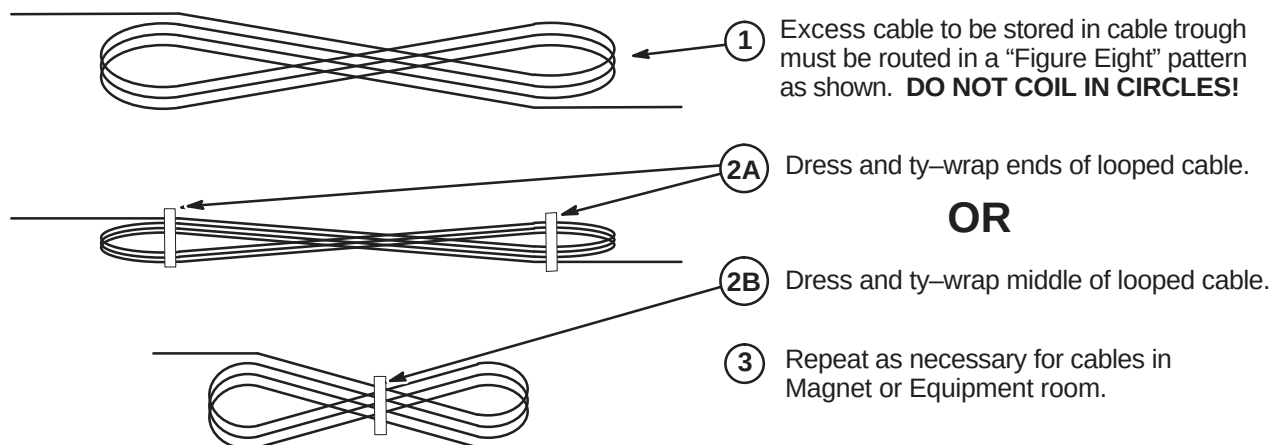
Note: The Service Cable Kit (5160683) is a collection of service cables delivered to site in a plastic bag. Refer to Section 8-10 for instructions on storage of this kit within System cabinet.

1. Sort cables by destination. Normally, it is best to sort cables originating from the farthest cabinet or component from the PDU first. This will make it easier to pull cables through the troughs.

Note: Carefully review architectural drawings to insure that all ducts are properly installed and that all signal and power cables are accounted for before starting to route cable runs.

2. Unroll coiled cables along the intended route to insure that they will lay freely without twists or kinks. Route cables in accordance with the applicable System Interconnect Diagram.
3. Plan for storage location of coiled cable excess length if cables are to remain at delivered length. See Illustration 4-1. Some cables are usually cut to length such as gradient cables, power cables, Head and Body RF, and Fiber-optic cables.

4-2 SORTING AND ROUTING CABLES (Continued)



PROPER STORAGE OF EXCESS CABLES

ILLUSTRATION 4-1

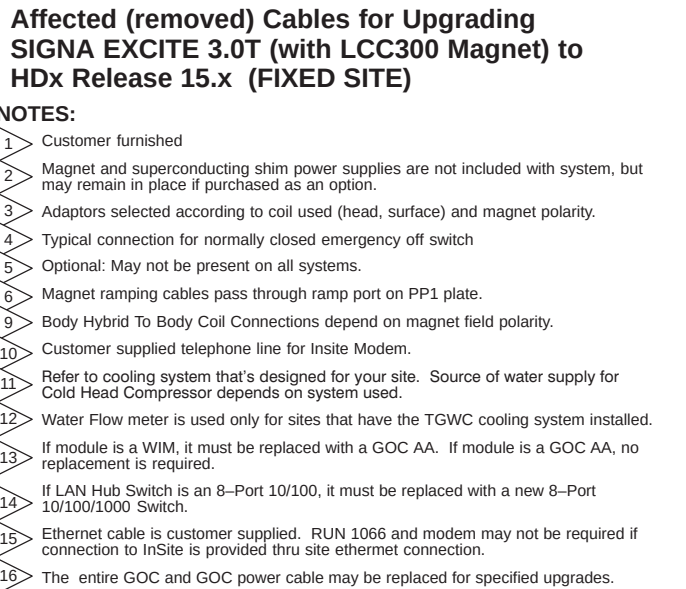
4. Place cables in provided troughs, conduit, or ducts. Leave sufficient slack for later connection when cabinets are in final position according to site architectural plans. Route cables inside exam room to Rear Pedestal area with sufficient length remaining to complete routing and connecting within Magnet Enclosure.
5. Rear Pedestal Cables – It is desired to keep the gradient cables separated from the signal and bias cables.

4-3 CABLE RE-LABELING

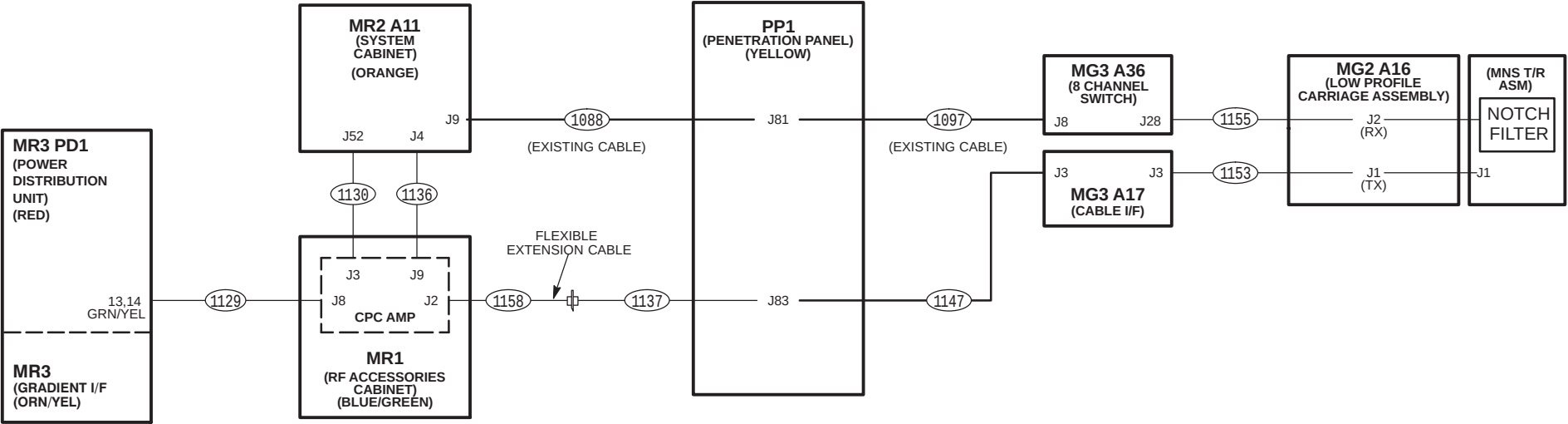
Relabel existing cables as described below:

- The new GOC upgrades the WIM to a GOCAA. This requires that Run 1084 be relabeled as Run 1255. Attach blank label to each end of cable and mark with information below.
- Relabel existing equipment room cable Run 1030 as instructed below.

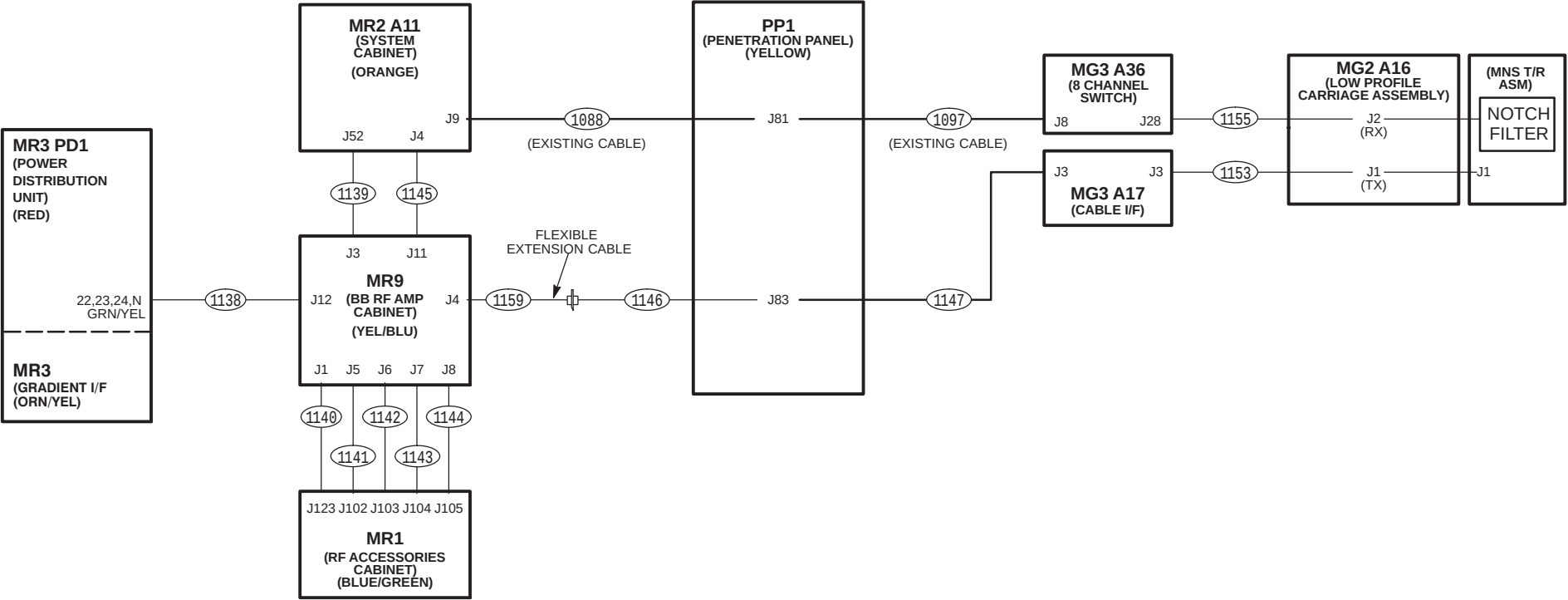
ONE END		OTHER END	
OLD LABEL	NEW LABEL	OLD LABEL	NEW LABEL
RUN #1084 MR1-A11-J26 2354601	RUN #1255 MR2-A11-J26 5120763-3	RUN #1084 OW1-J21-J6 VIA RUN #1062 2354601	RUN #1255 OW1-A21-J18 5120763-3
RUN #1030 PP1-J78 2329624	RUN #1329 PP1-J78 5160765	RUN #1030 MR2-A11-J61 2329624	RUN #1329 MR2-DVR-J10 5160765

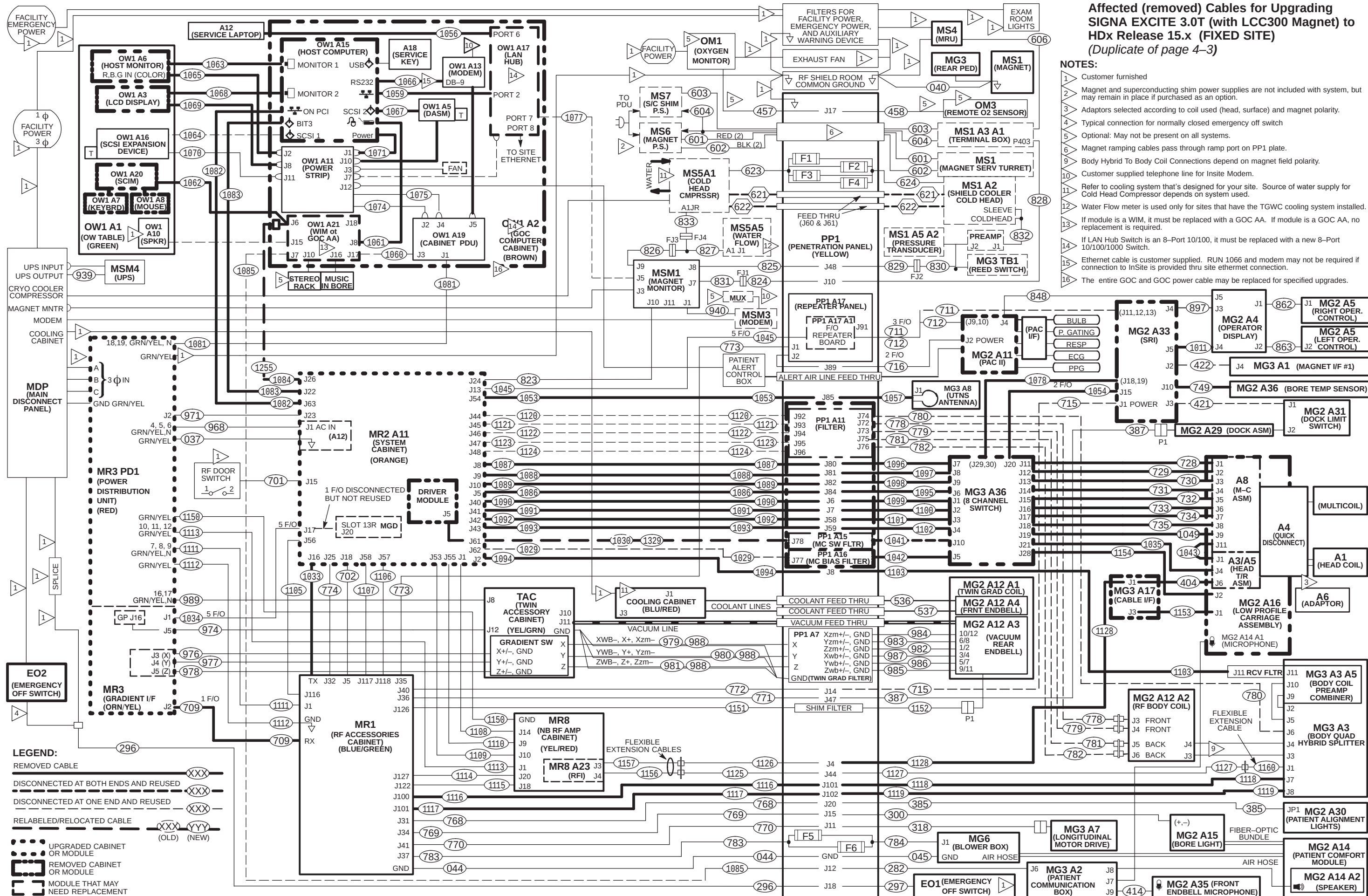


SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

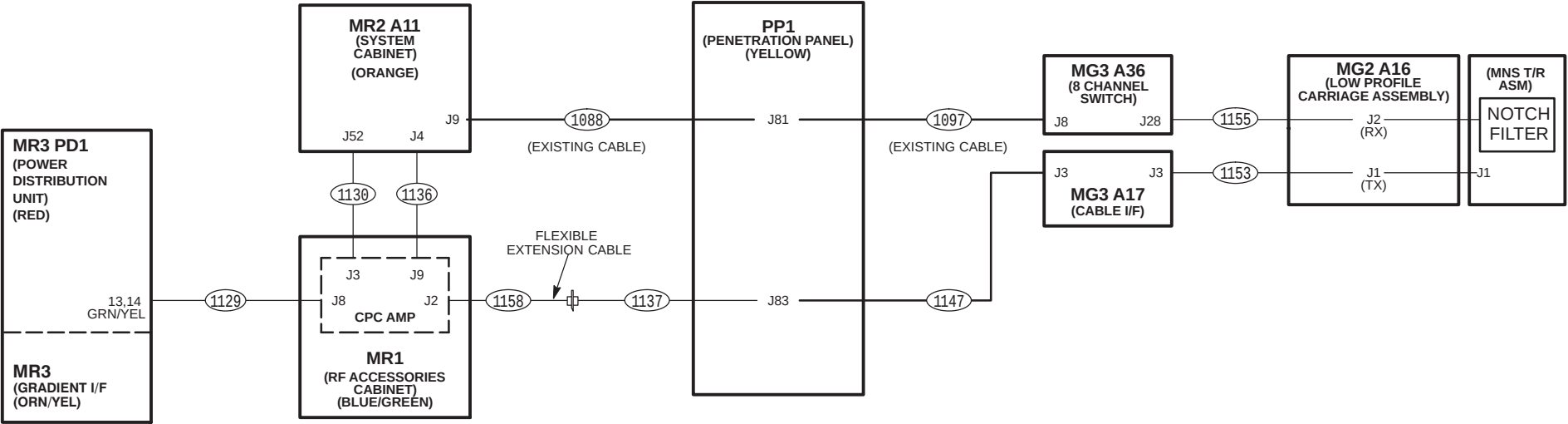


SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

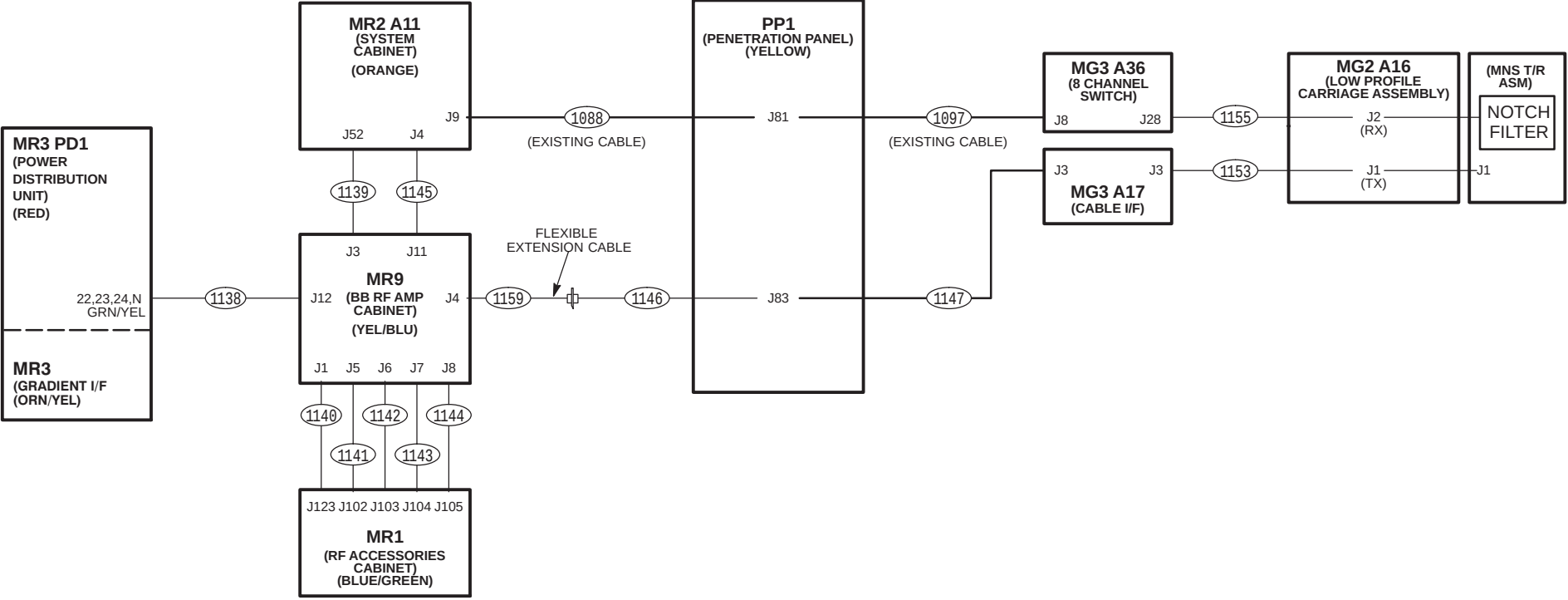


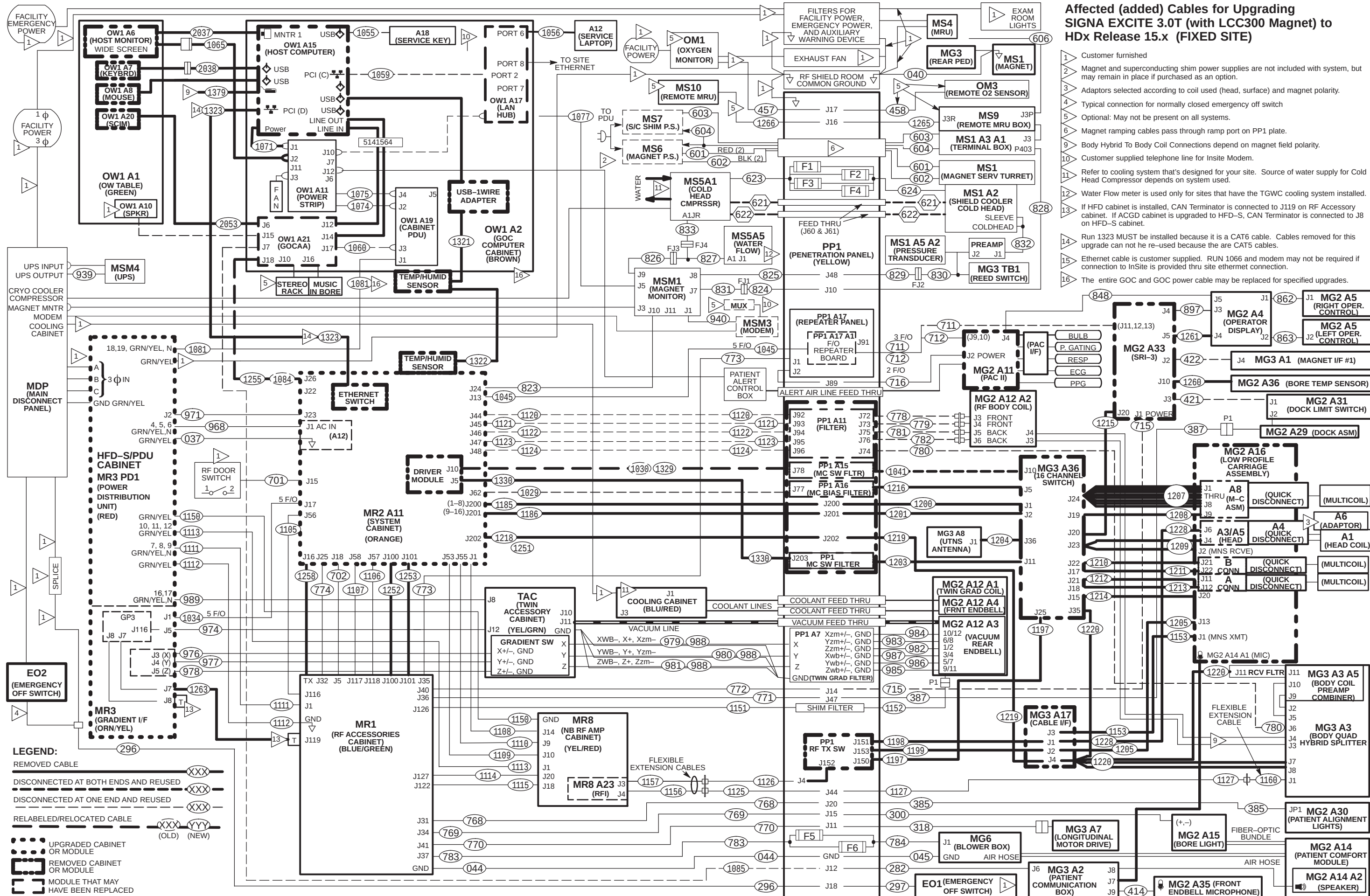


SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

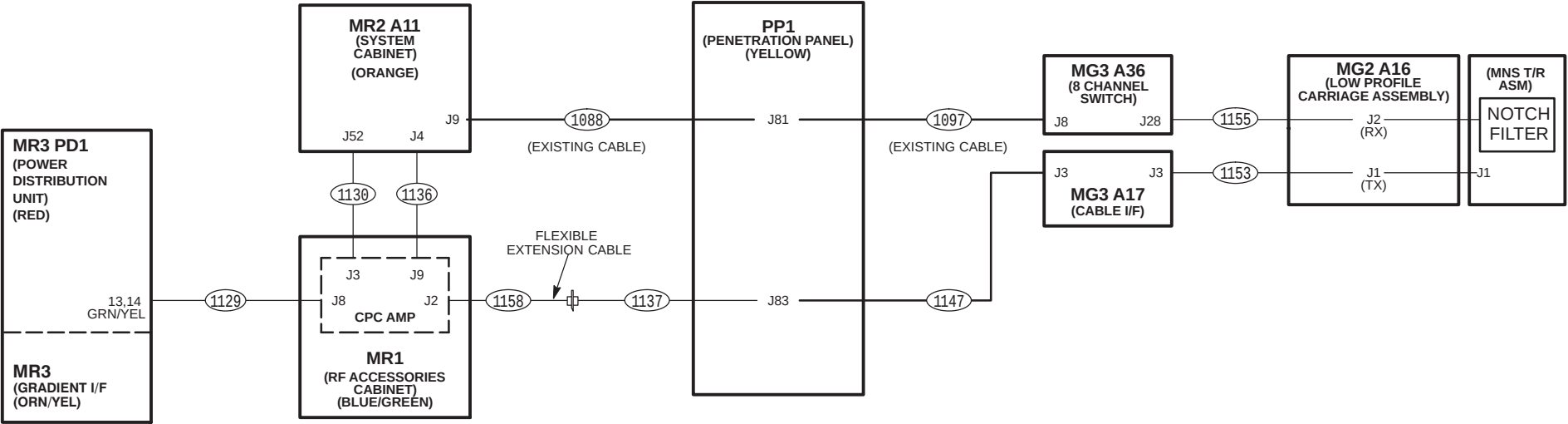


SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

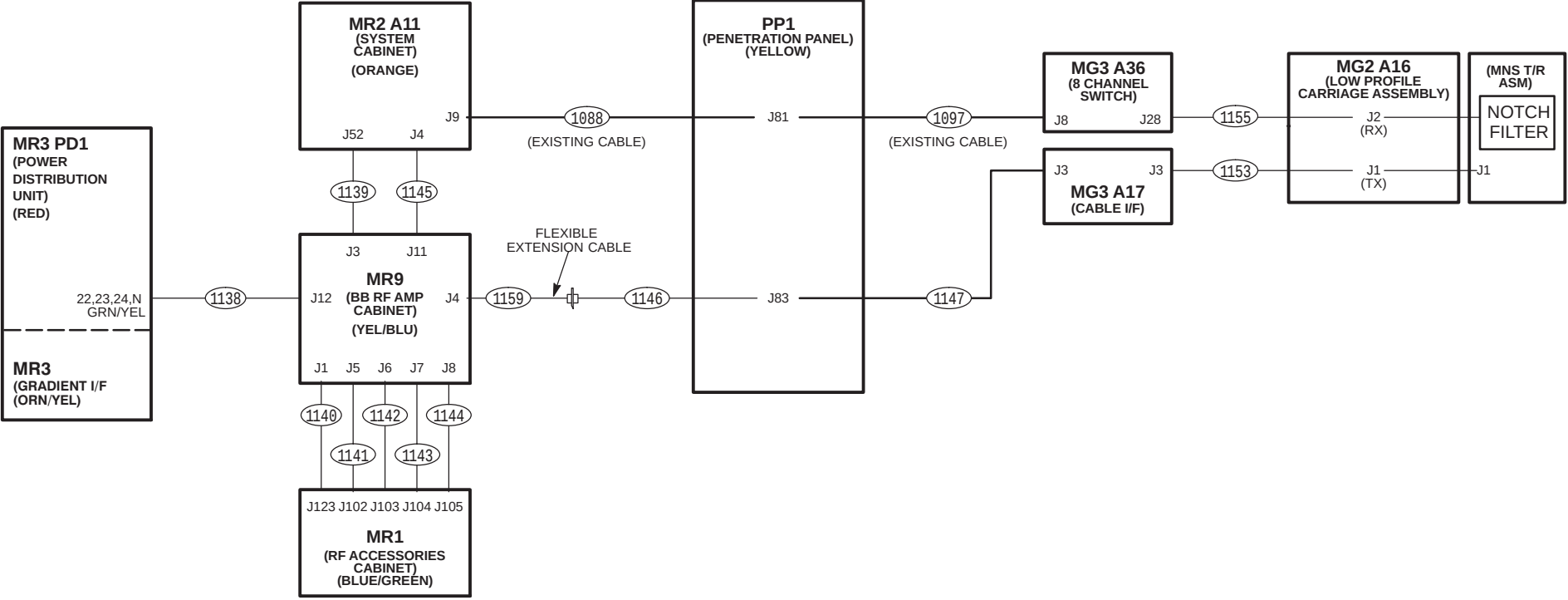


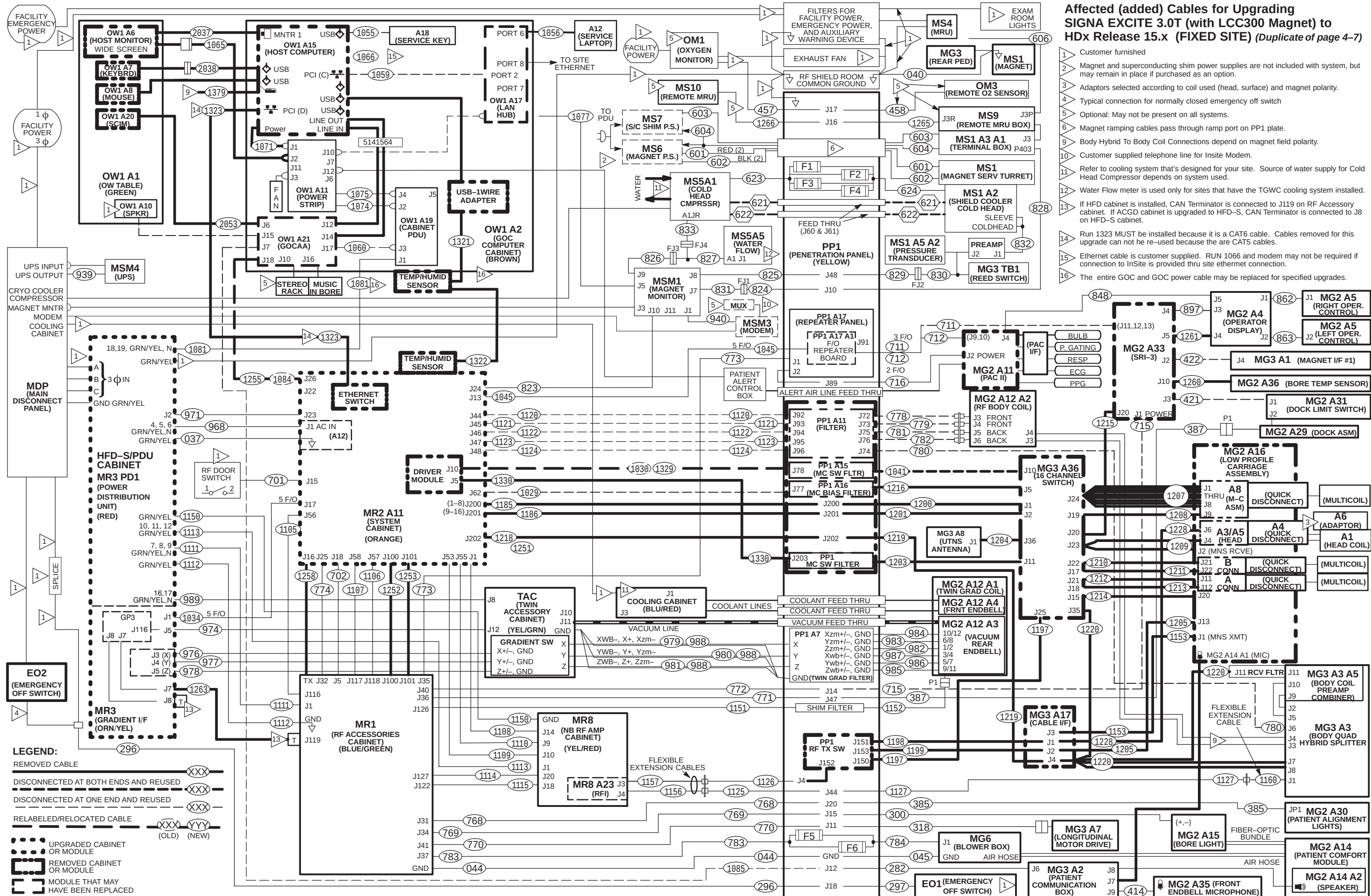


SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

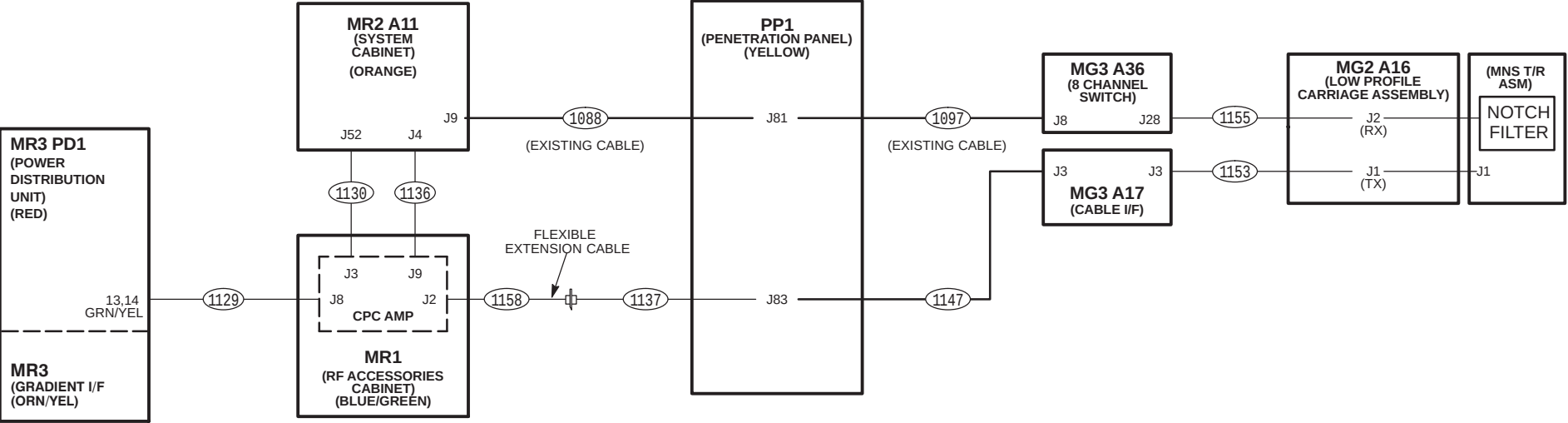


SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS

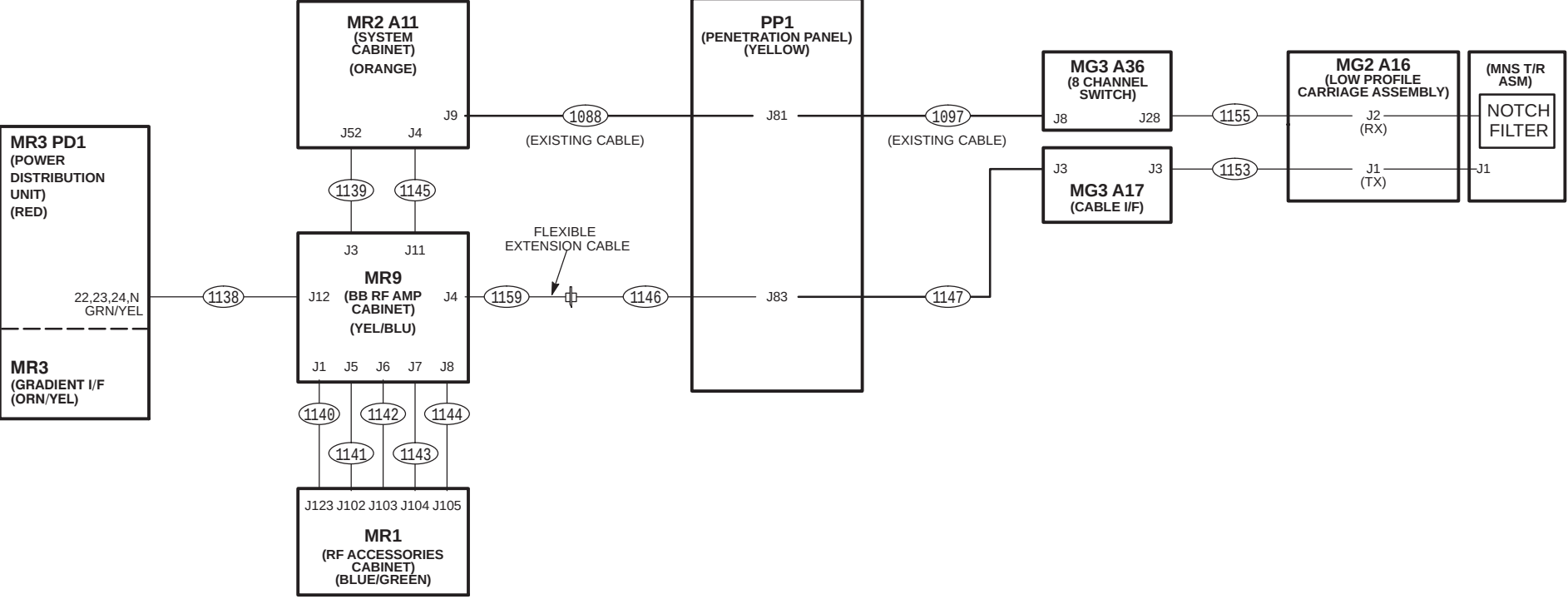




SIGNA EXCITE 3.0T: 4KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



SIGNA EXCITE 3.0T: 8KW MNS OPTION
ADDITIONAL SYSTEM INTERCONNECTS



SECTION 5 – PENETRATION PANEL

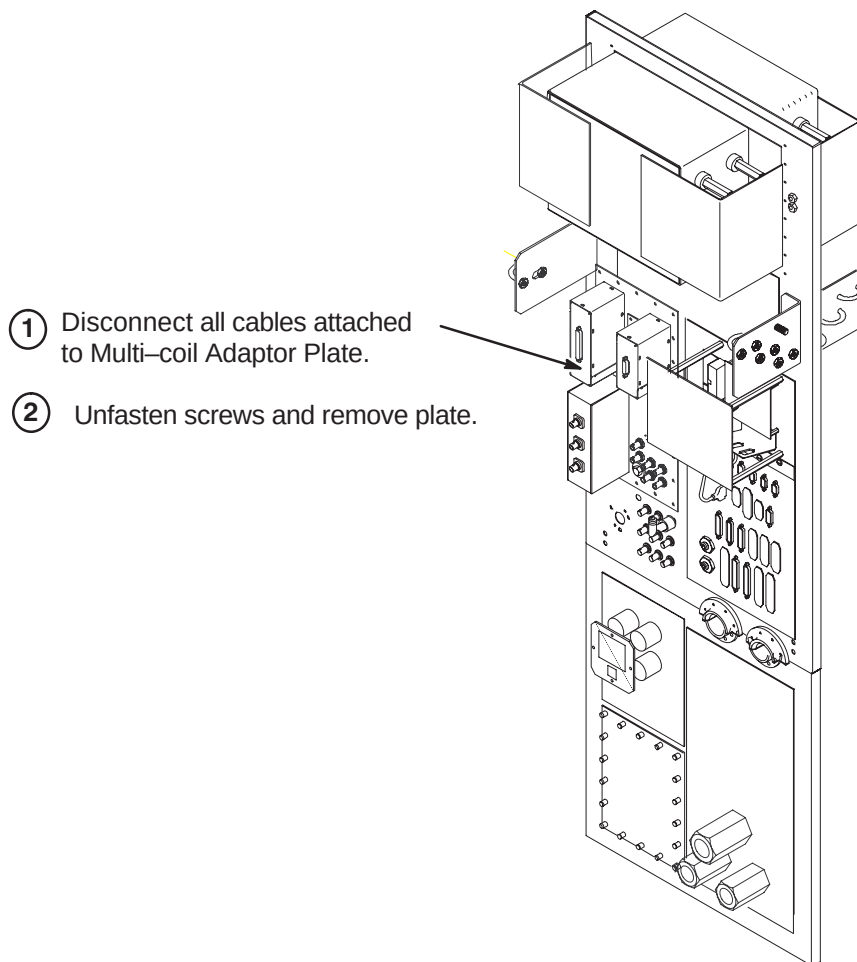
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

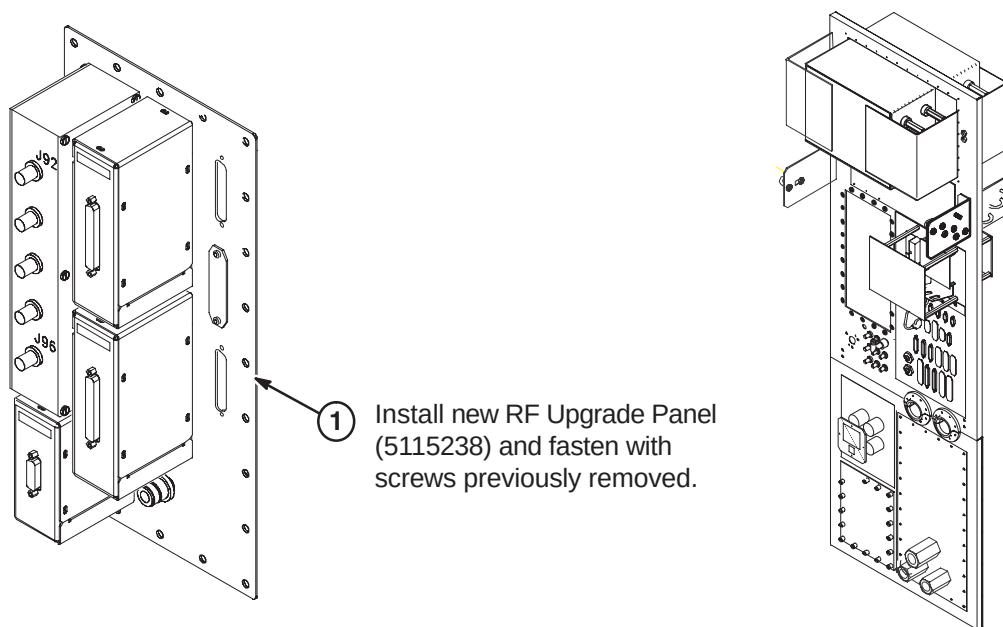
5-1 INTRODUCTION

The Multi-coil Adaptor Plate will be replaced by the RF Upgrade Panel (5115238).

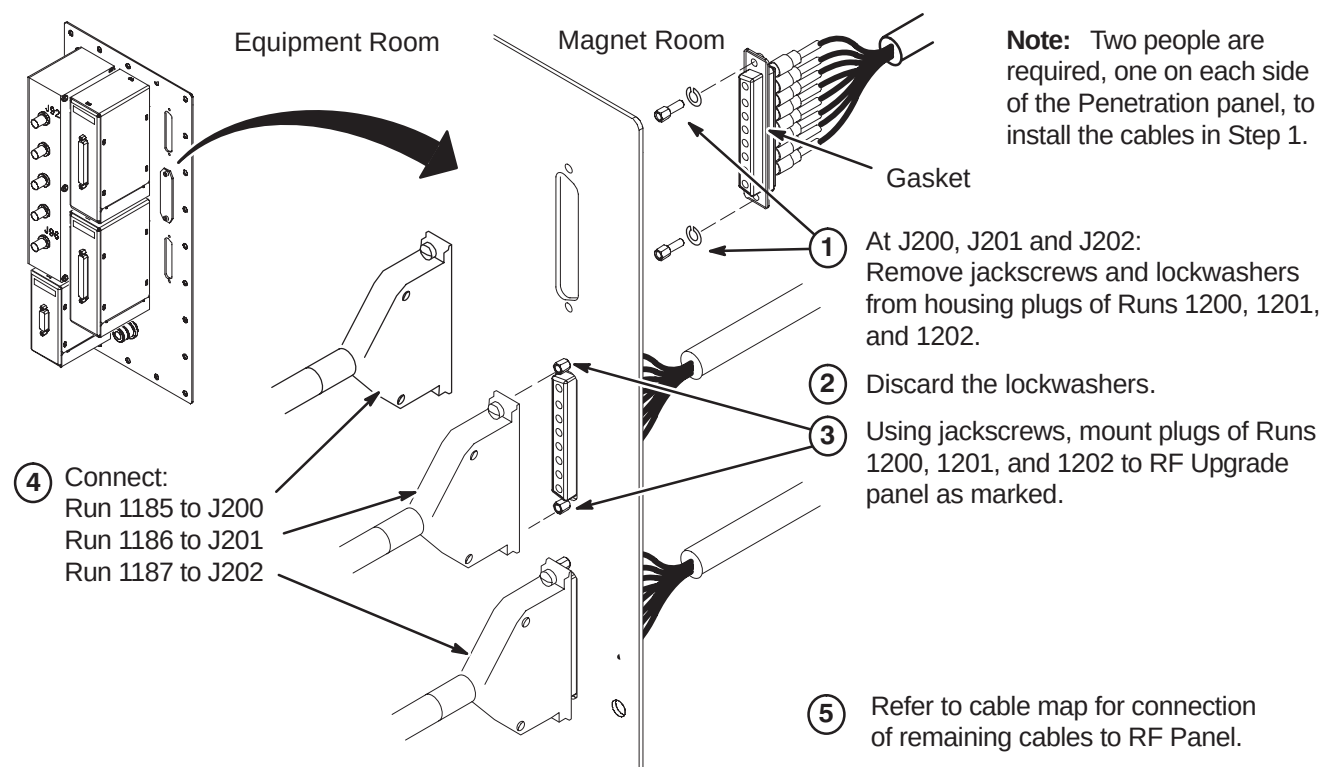
5-2 REMOVE MULTI-COIL ADAPTOR PLATE FROM PENETRATION PANEL



5-3 INSTALL NEW EXCITE RF PANEL



5-4 INSTALLATION OF CABLES AT J200, 201, AND J202



SECTION 6 – MAGNET ROOM

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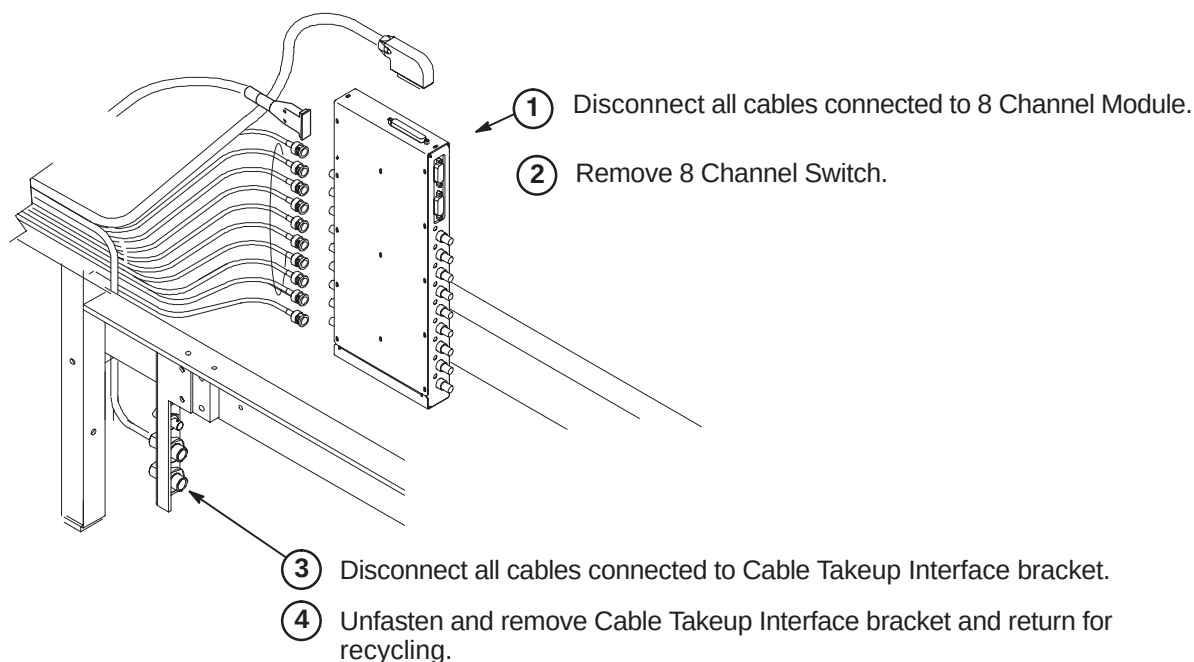
CAUTION

MAKE SURE POWER TO SYSTEM IS OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

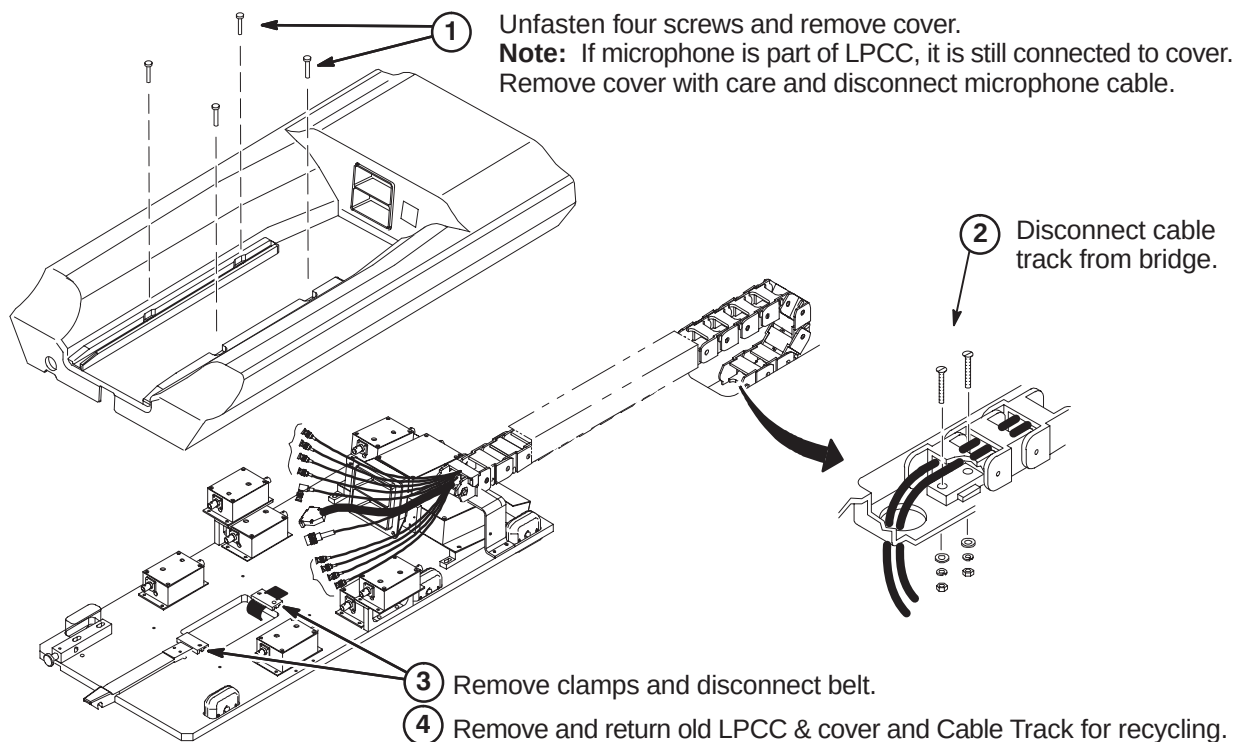
6-1 INTRODUCTION

Modules on the Rear Pedestal, The Front and Rear Split Bridge, LPCA and Cable Track, SRI, and PAC will be replaced.

Several cables will be removed and replaced with new cables, some cables will be relabeled and reused, and new cables will be routed and connected. Refer to Section 4 cable maps for information on removing and installing cables

6-2 DE-INSTALLATION PROCEDURES**6-2-1 DISCONNECT CABLE TRACK CABLES FROM REAR PEDESTAL MODULES**

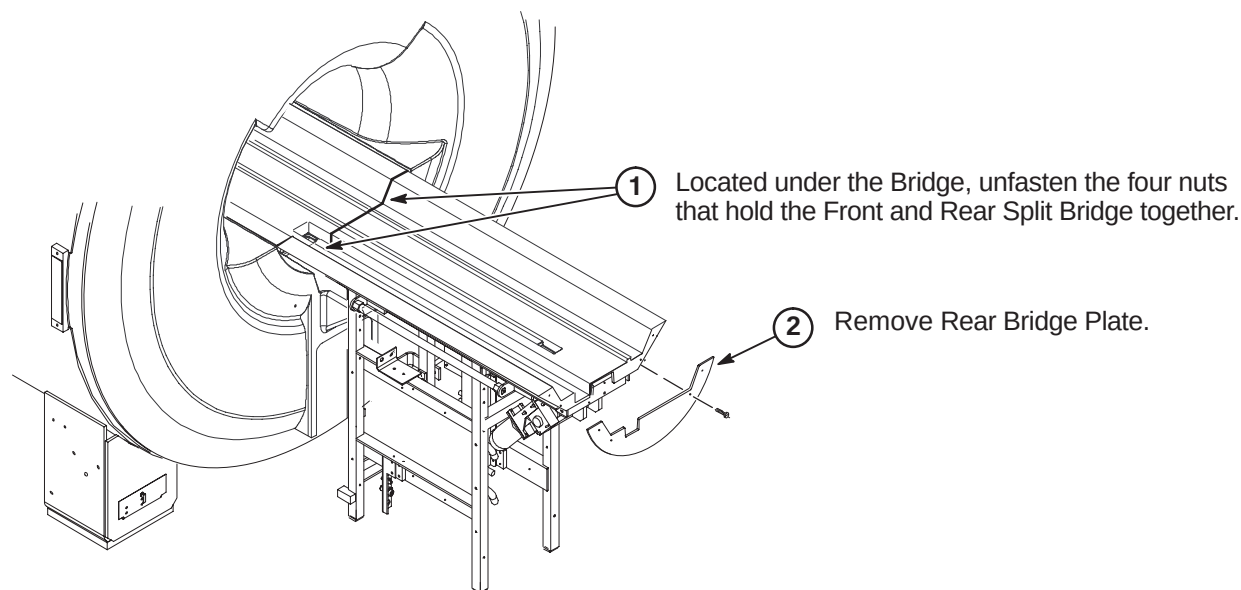
6-2-2 DE-INSTALL EXISTING LOW PROFILE CARRIAGE COVER & CABLE TRACK



6-2-3 DE-INSTALL BRIDGE

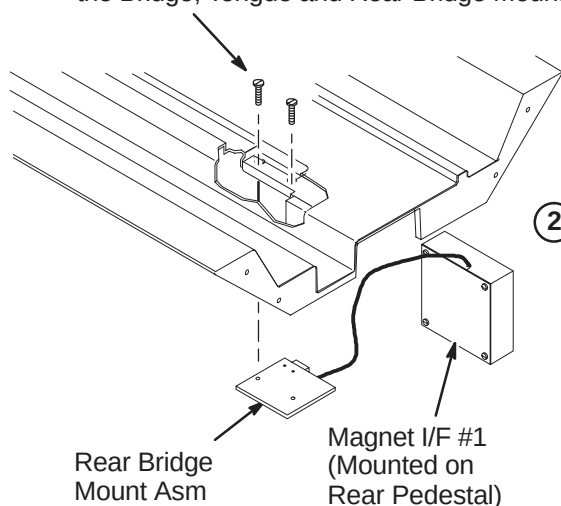
Upgrading to EXCITE TwinSpeed Release 12.x requires a new Front and Rear Split Bridge.

6-2-3-1 Unfasten Rear Split Bridge

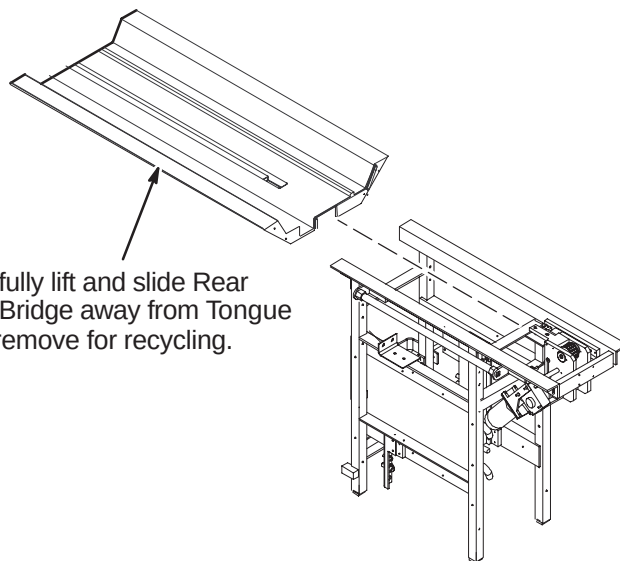
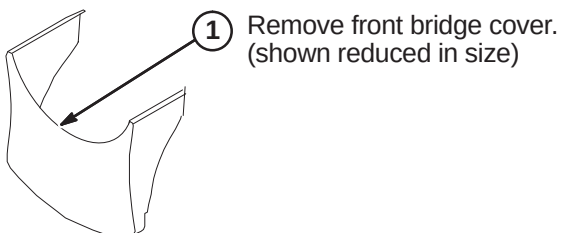


6-2-3-2 Remove Rear Split Bridge

- ① Unfasten and remove the two screws connecting the Bridge, Tongue and Rear Bridge Mount Asm.

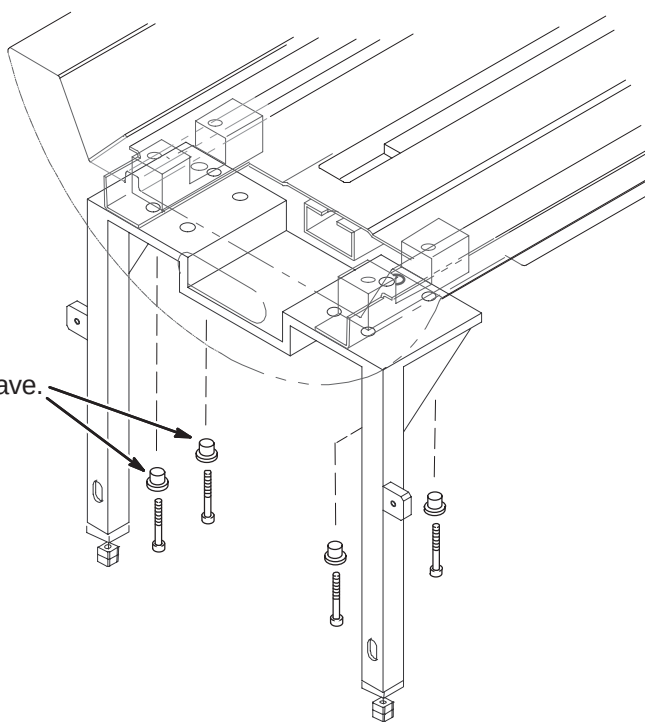


- ② Carefully lift and slide Rear Split Bridge away from Tongue and remove for recycling.

**6-2-3-3 Remove Front Split Bridge**

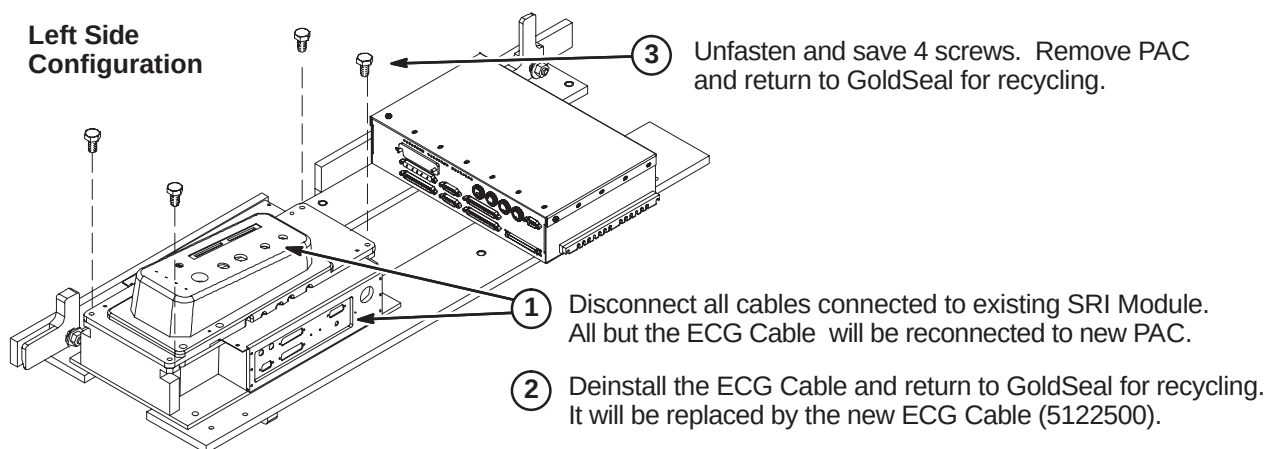
- ② Unfasten the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings and save. These will be reused for installing the new Bridge.

- ③ Carefully lift and remove Front Split Bridge and return for recycling.



6-2-4 DE-INSTALL PAC MODULE

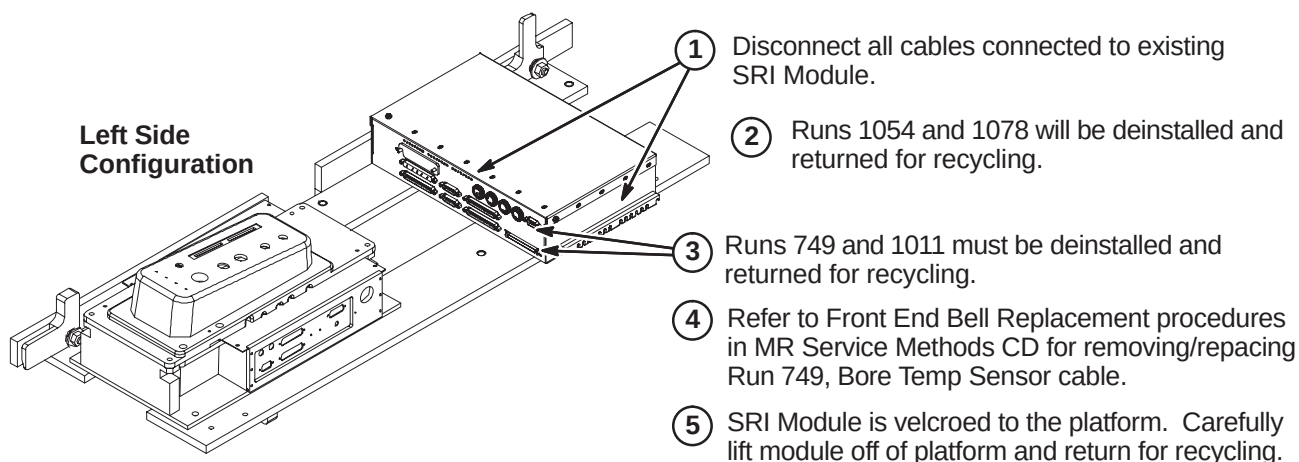
The existing PAC Module will be replaced by a new PAC Module with independent vector gating.



6-2-5 DE-INSTALL SRI MODULE

The existing SRI Module will be replaced by the SRI-3 Module.

NOTE: Existing Runs 749 and 1011 must be removed and replaced with new Runs 749 (to be relabeled in the future as Run 1260) and 1011 (to be relabeled in the future as Run 1261). There is a functionality difference.

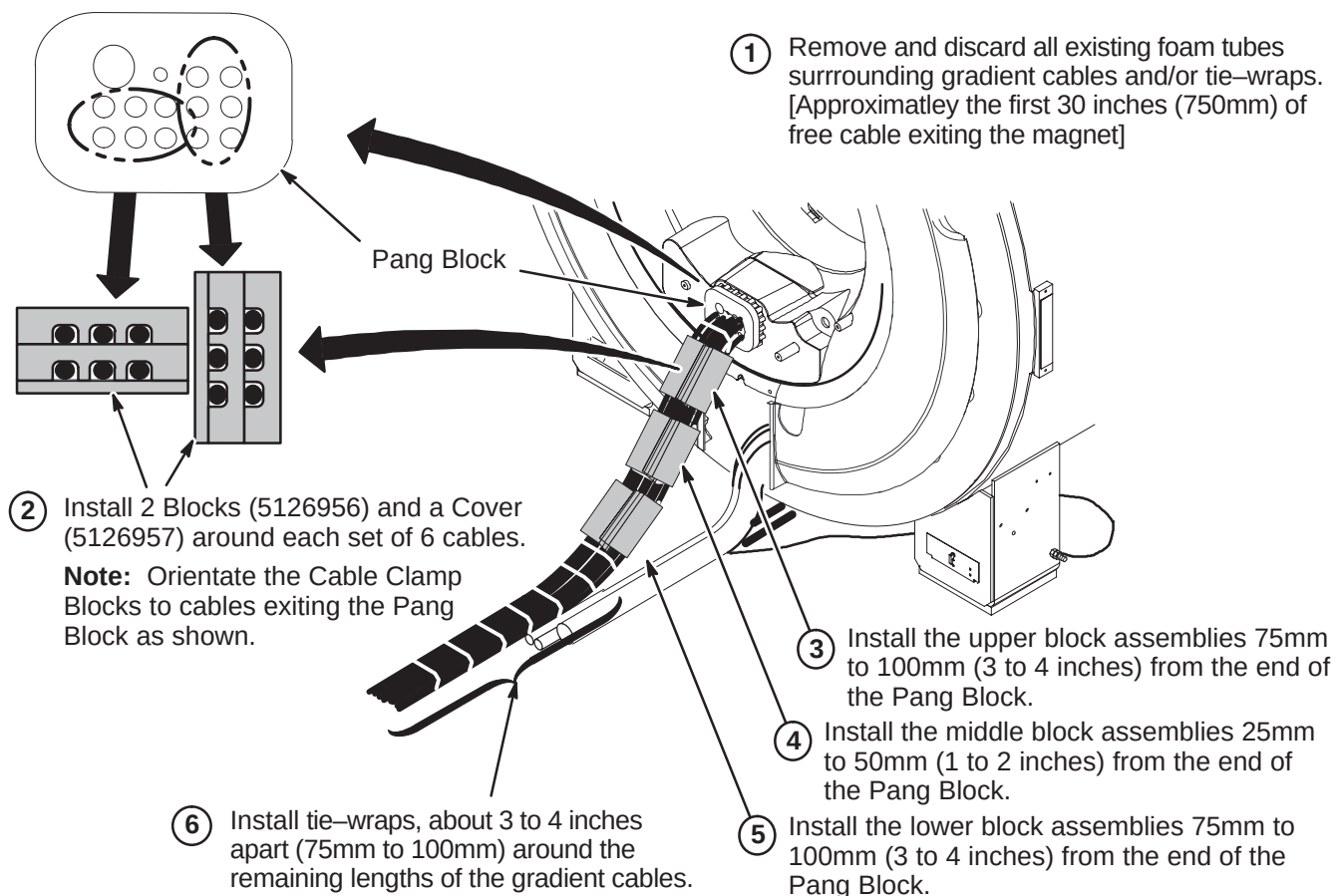


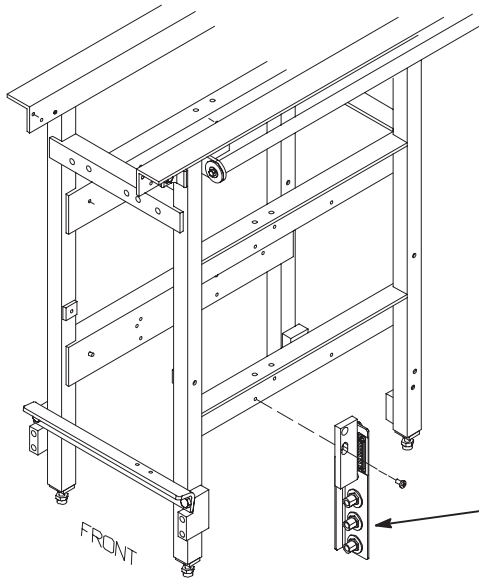
6-3 INSTALLATION OF NEW PARTS

6-3-1 REPLACE GRADIENT CABLE FOAM TUBES WITH CLAMP BLOCKS

The existing foam tubes that are wrapped around the gradient cables at the rear of the magnet are to be removed and replaced with cable clamp blocks.

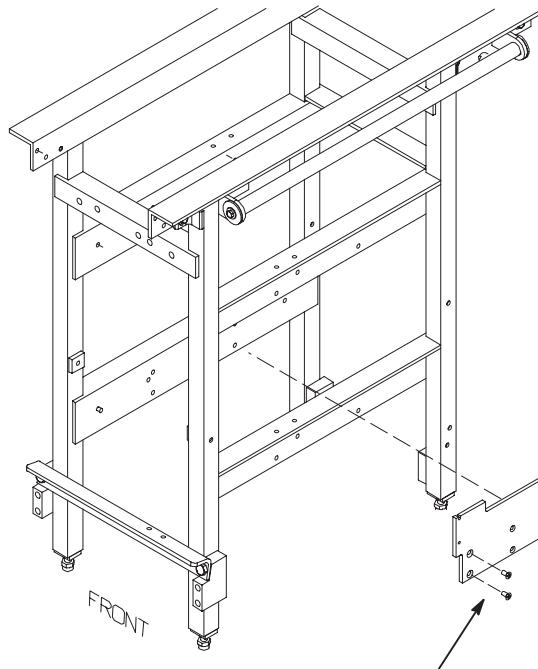
Note: Twelve blocks (5126956) and six covers (5126957) will be used, leaving six unused covers.



6-3-2 INSTALL NEW BULKHEAD INTERFACE

NOTE: Components and cables not shown on Rear Pedestal for clarity.

- ① Install new Bulkhead Interface (5113250) using Loctite 242 (46-170686P2) and two M6 x 12mm pane Head Screws (2109873-24).

6-3-3 INSTALL 16 CHANNEL SWITCH

NOTE: Components and cables not shown on Rear Pedestal for clarity.

Part of Rear Pedestal

- ③ Connect Run 1204 (5116105) as shown.

- ① Install 16 Channel Switch Mounting Plate (5110428) using Loctite 242 (46-170686P2) and depending on style of Rear Pedestal:

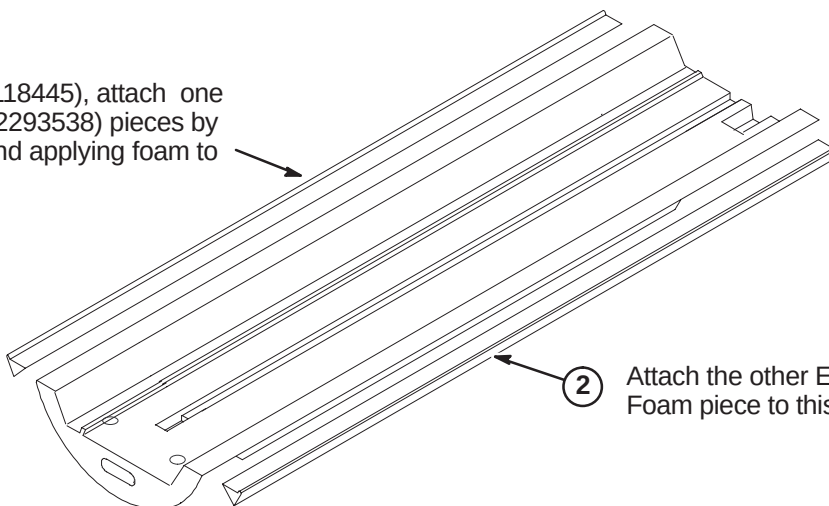
- Four M6 x 12mm Slotted Flat Head Screws (25120369-86) in the four larger outside holes.
- or
- Two screws installed in the two center holes.

- ② Attach 16 Channel Switch (5109645) to Mounting Plate using Loctite 242 (46-170686P2) and four M3 x 12mm Pan Head screws (46-312342P24).

6-3-4 INSTALLATION OF FRONT BRIDGE

6-3-4-1 Attach Edge Foam

- ① Starting at the front of Bridge (5118445), attach one of the Twin Bridge Edge Foam (2293538) pieces by removing the protective cover and applying foam to bottom side of Bridge surface.



- ② Attach the other Edge Foam piece to this side.

6-3-4-2 Install Bridge

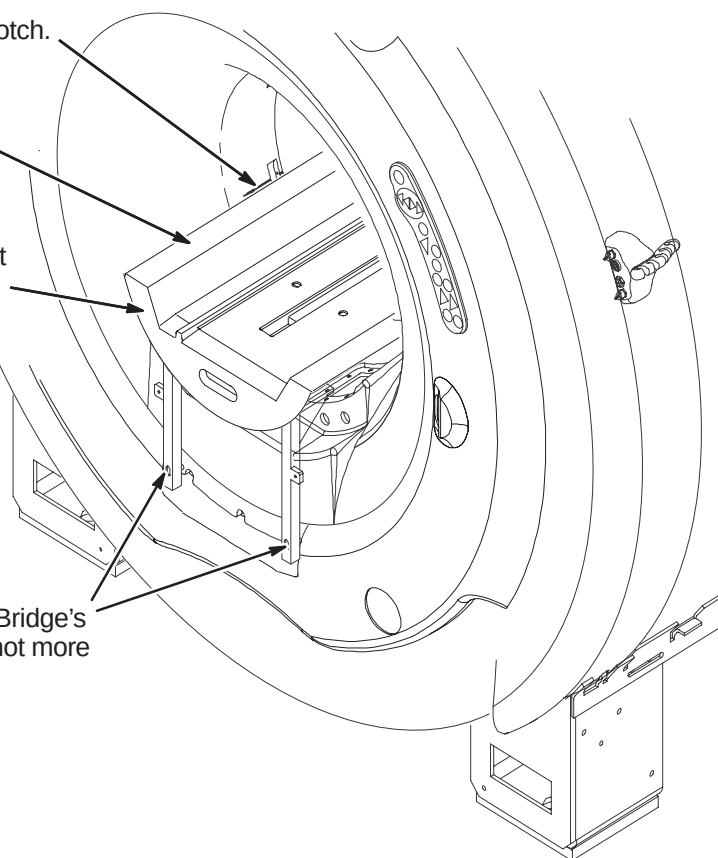
- ① Install new Front Bridge Asm into magnet and attach Bridge to front Mounting Bracket using the four (4) M6 x 35mm long Socket Head screws (2109867-17) and Support Bushings removed in Procedure 6-2-3-3.

End Bell notch.

Top of Bridge

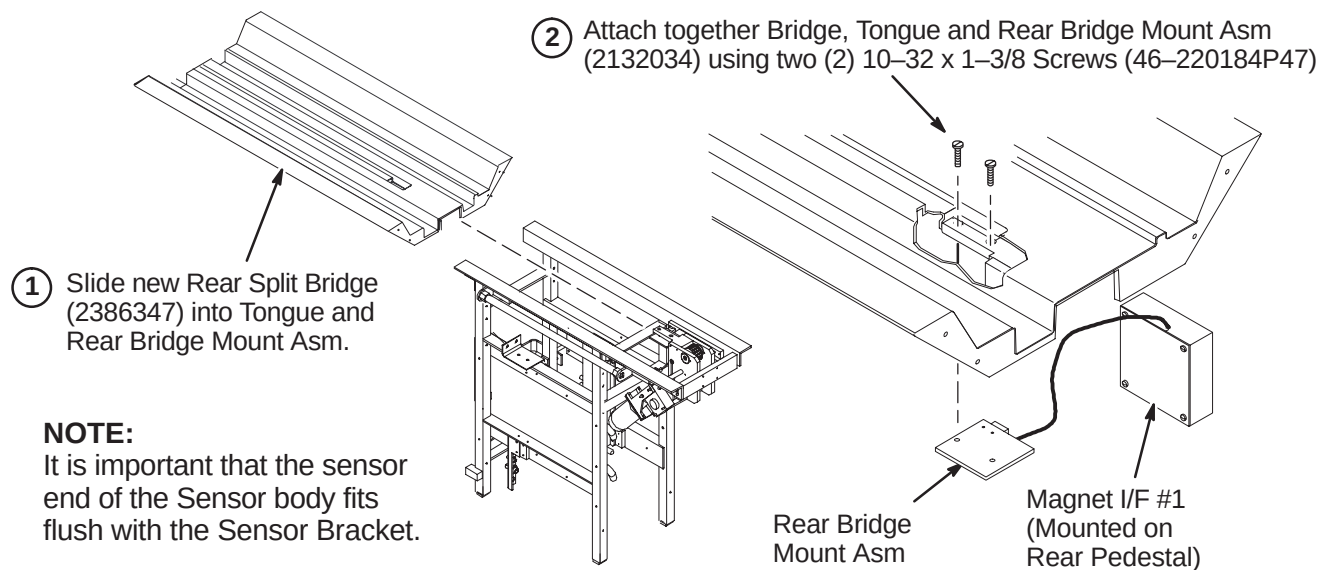
- ② Adjust the support hardware until the Bridge's top surface is not less than 2mm and not more than 4mm above the End Bell notch.

- ③ Check left-right levelness.

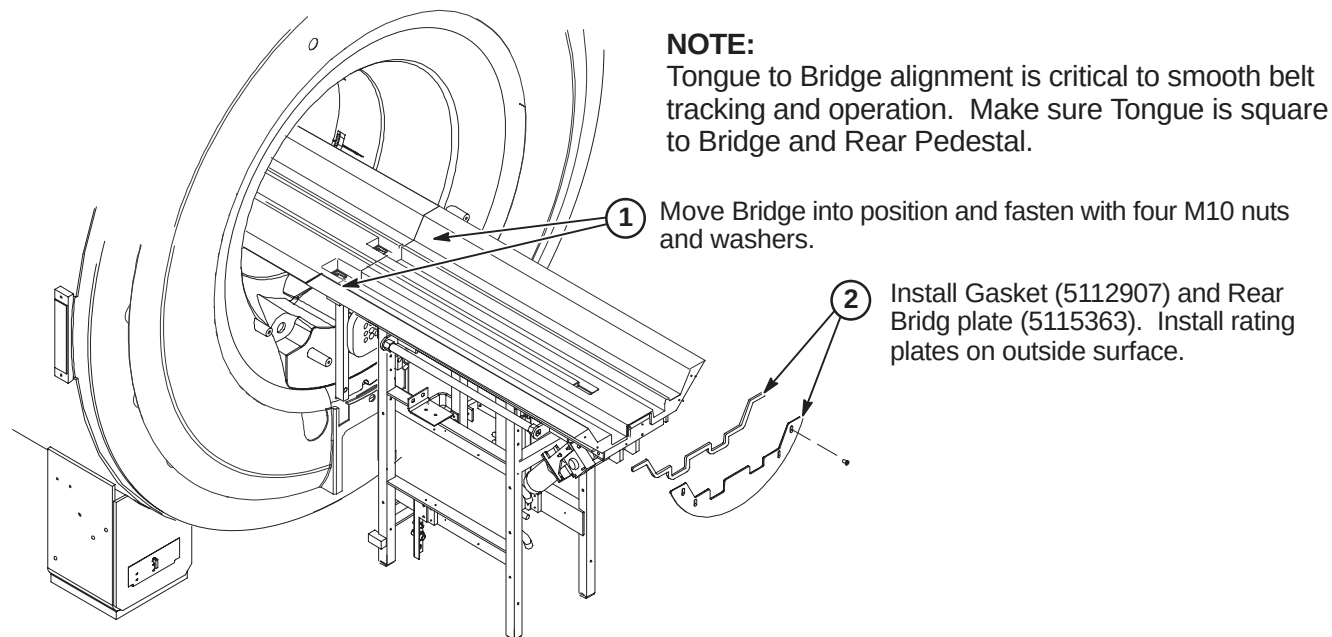


6-3-5 INSTALLATION OF REAR SPLIT BRIDGE

6-3-5-1 Install Rear Split Bridge on Rear Pedestal



6-3-5-2 Connect Rear Split Bridge to Front Split Bridge

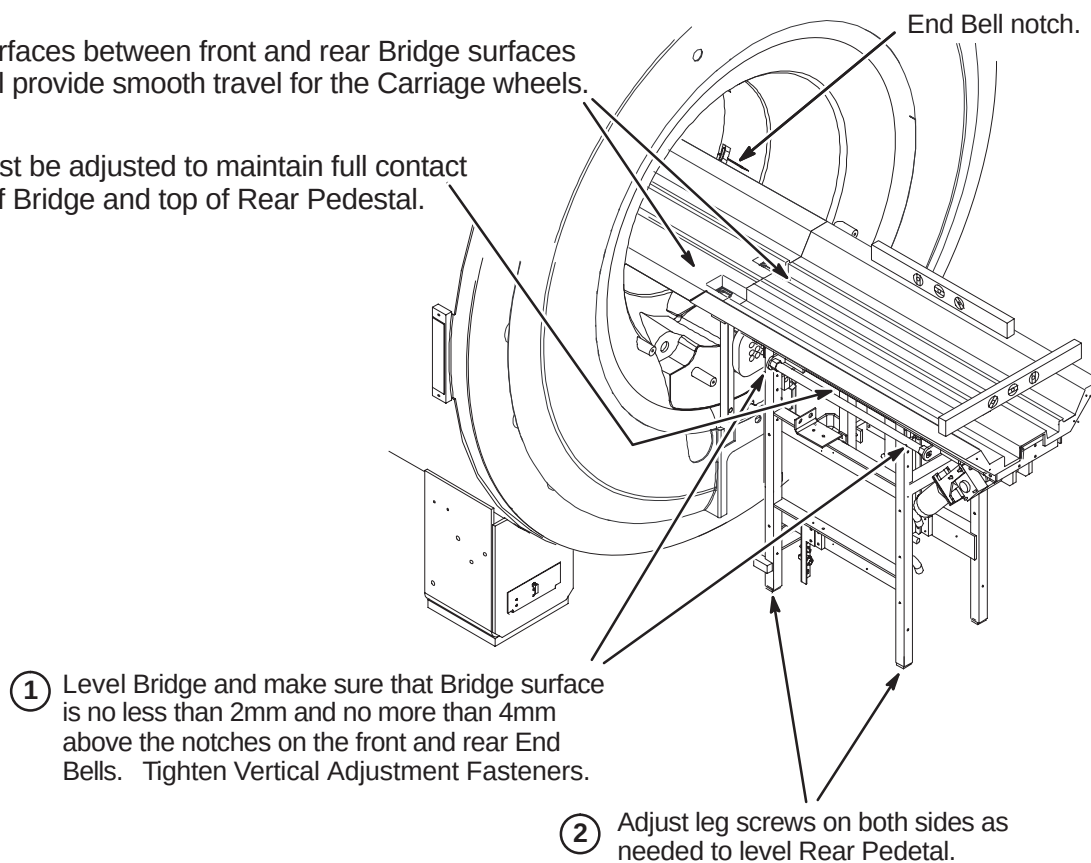


6-3-5-3 Level Bridge**NOTE:**

Make sure that surfaces between front and rear Bridge surfaces are even. This will provide smooth travel for the Carriage wheels.

NOTE:

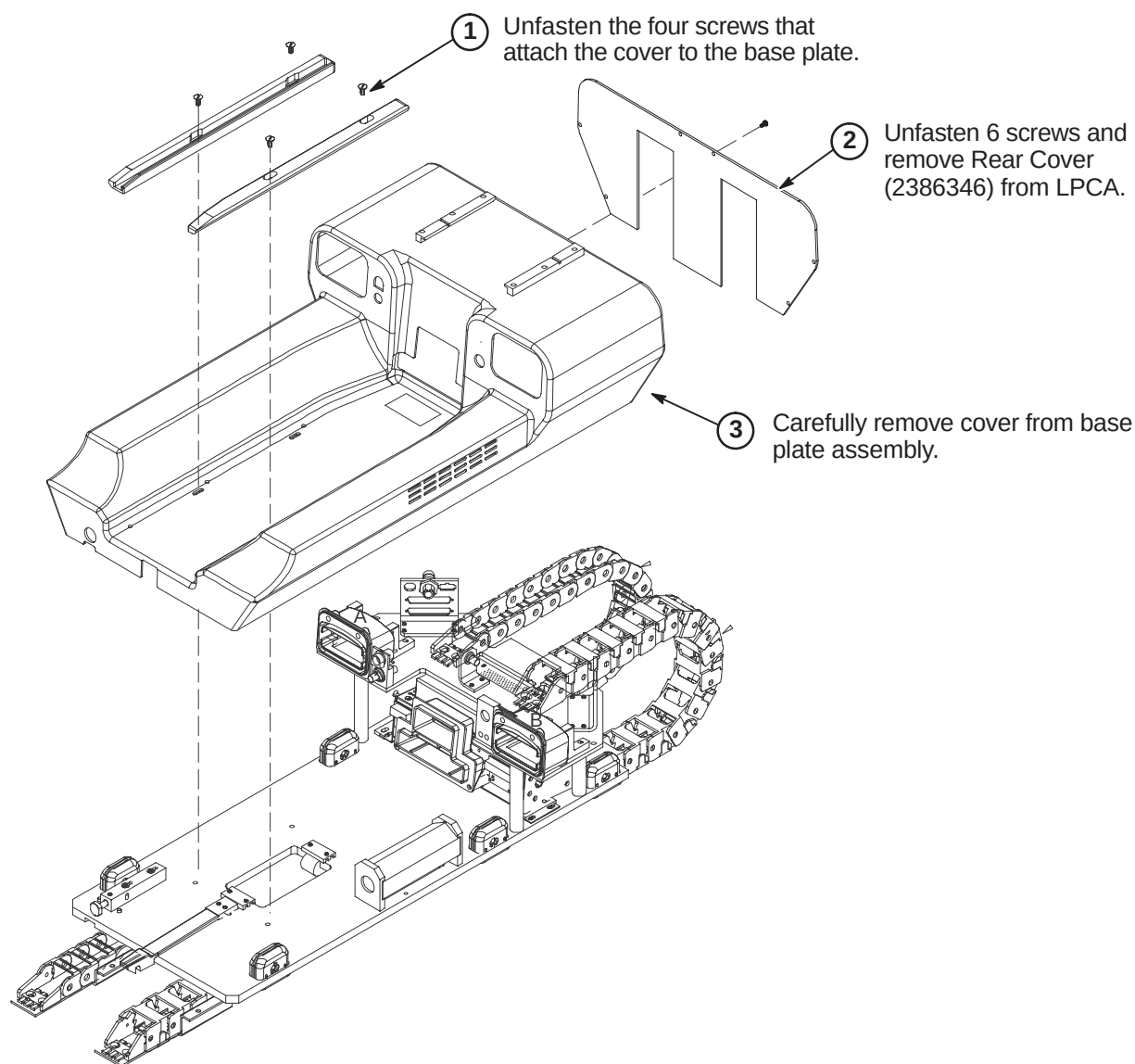
Rear Pedestal must be adjusted to maintain full contact between bottom of Bridge and top of Rear Pedestal.



6-3-6 INSTALL LPCA AND CABLE TRACK(S)

The WideOpen style enclosure will be upgraded to an LPCA with a dual cable track.

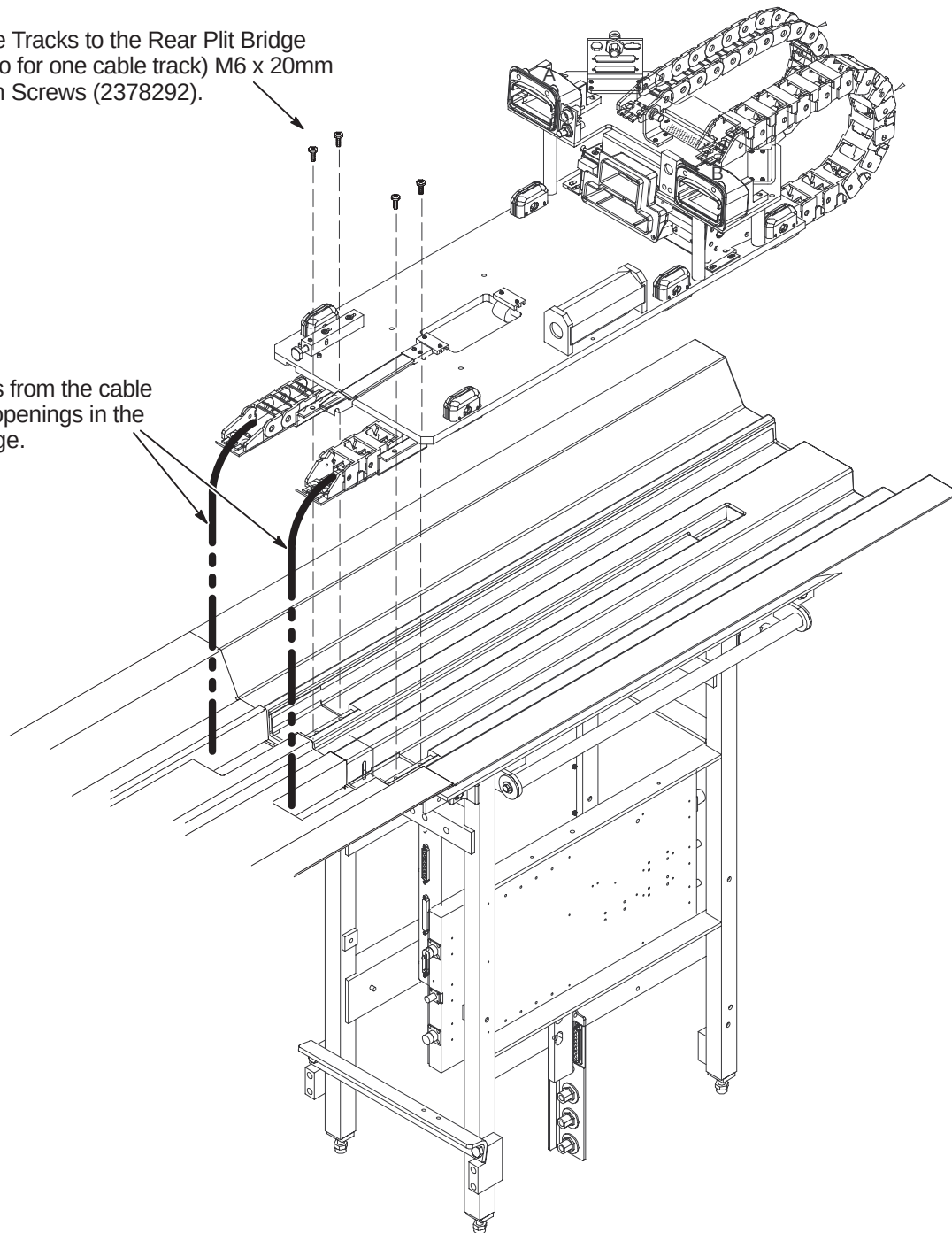
6-3-6-1 Remove LPCA Cover

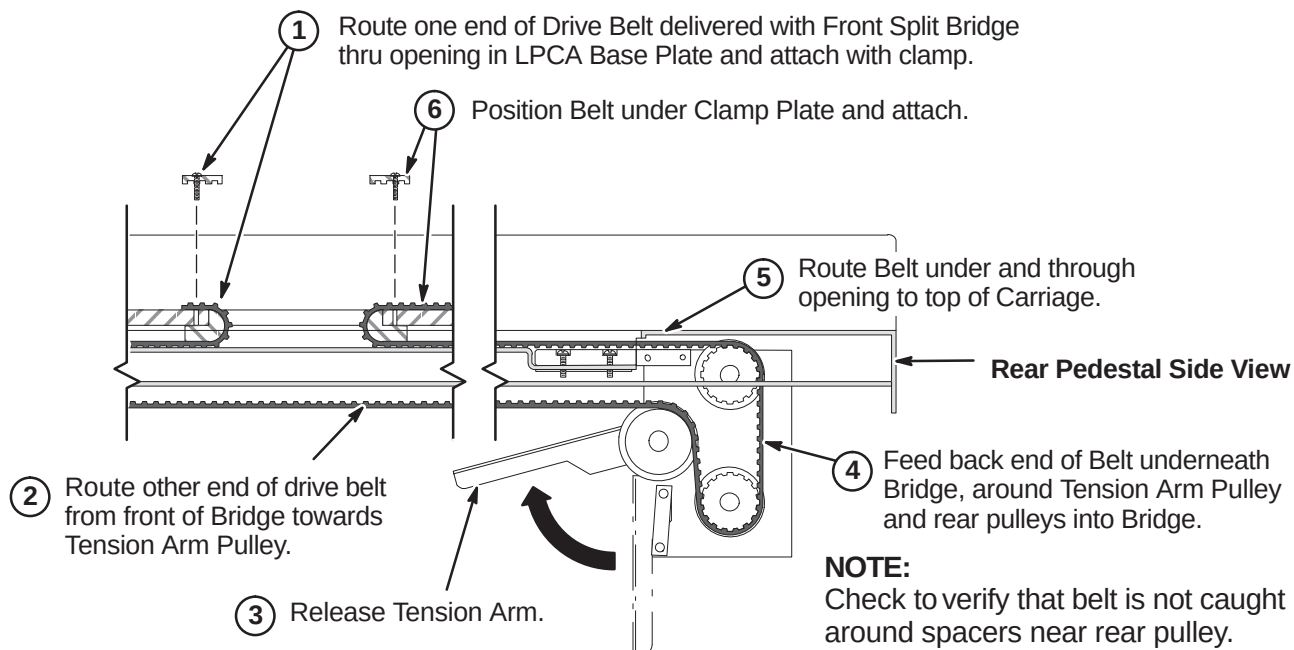
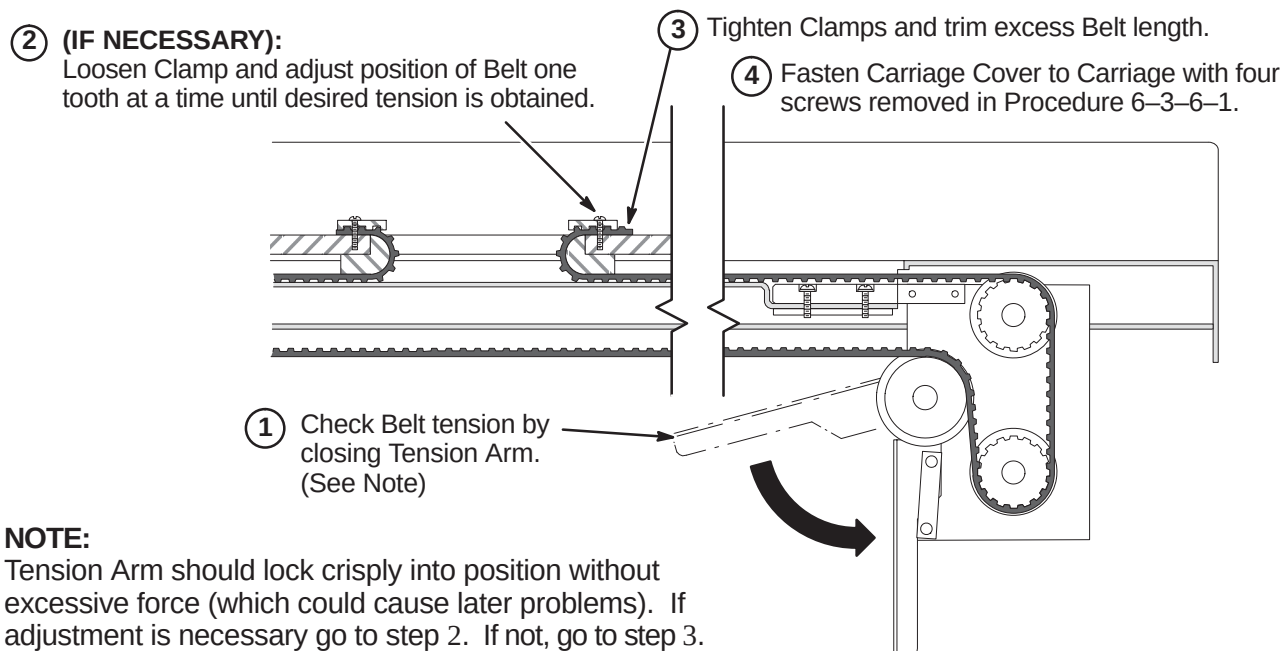


6-3-6-2 Attach Cable Track(s) to Bridge

- ② Attach the Cable Tracks to the Rear Plit Bridge using four (or two for one cable track) M6 x 20mm Pan Head Nylon Screws (2378292).

- ① Feed the cables from the cable tracks thru the openings in the Front Split Bridge.



6-3-6-3 Route and Attach Drive Belt**6-3-6-4 Final Adjustment of Drive Belt**

6-3-6-5 Installation of Sensor Flag

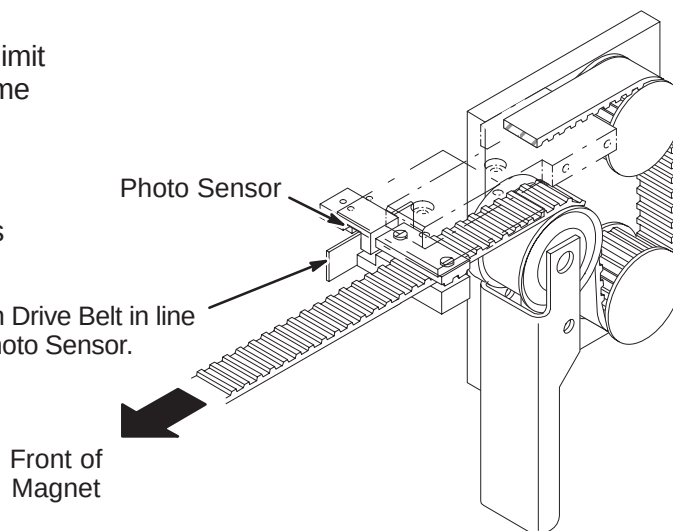
NOTE:

Function of the Flag is to interrupt the reflected limit switch sensor beam when the Carriage is in home position all the way forward in the bore.

NOTE:

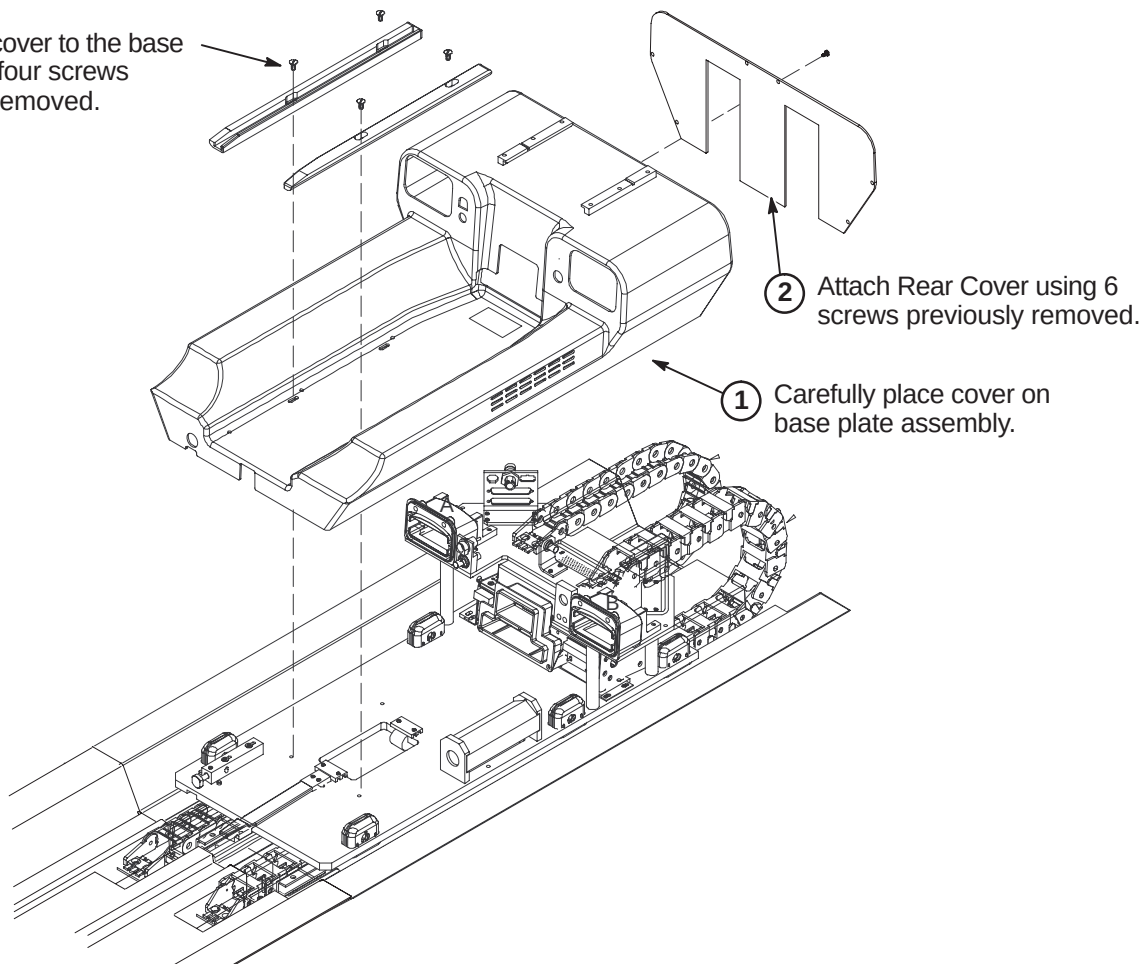
The following Step is needed if flag was removed during routing of drive belt.

- ① Install Flag Assembly on Drive Belt in line with Bridge mounted Photo Sensor.



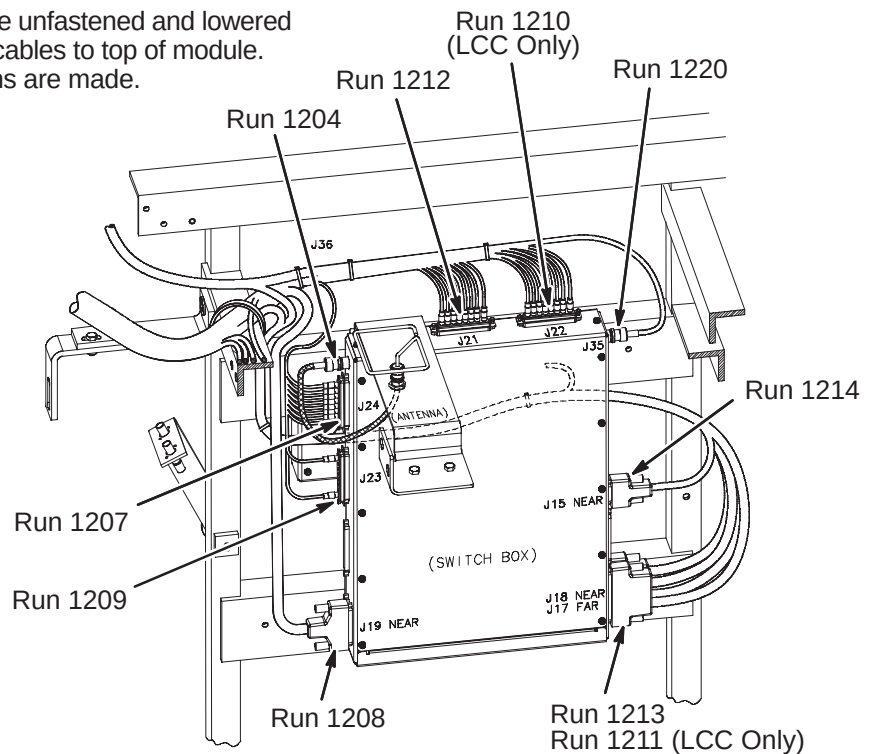
6-3-6-6 Installation of LPCA Cover

- ③ Attach the cover to the base plate using four screws previously removed.

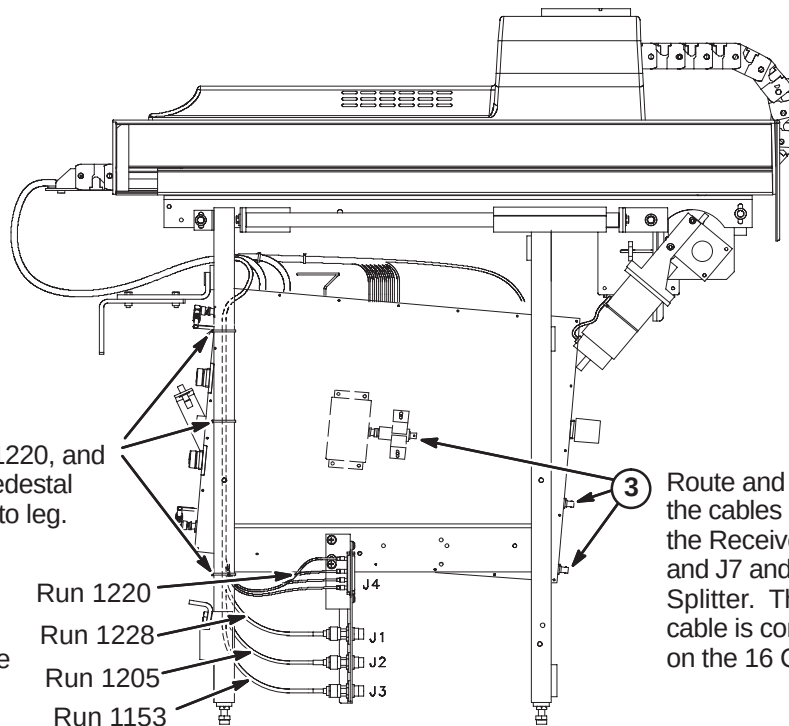


6-3-6-7 Connect Cables from Cable Track to 16 Channel Switch Module

- ① 16 Channel Switch may need to be unfastened and lowered from mounting position to connect cables to top of module. Re-mount module after connections are made.
- ② Connect cable Runs as indicated.

**6-3-6-8 Connect Cables from Cable Track to Bulkhead Interface**

- ② Route Runs 1153, 1205, 1220, and 1228 along leg of Rear Pedestal and ty-wrap as indicated to leg.
- ① Connect Runs to Bulkhead Interface as indicated.

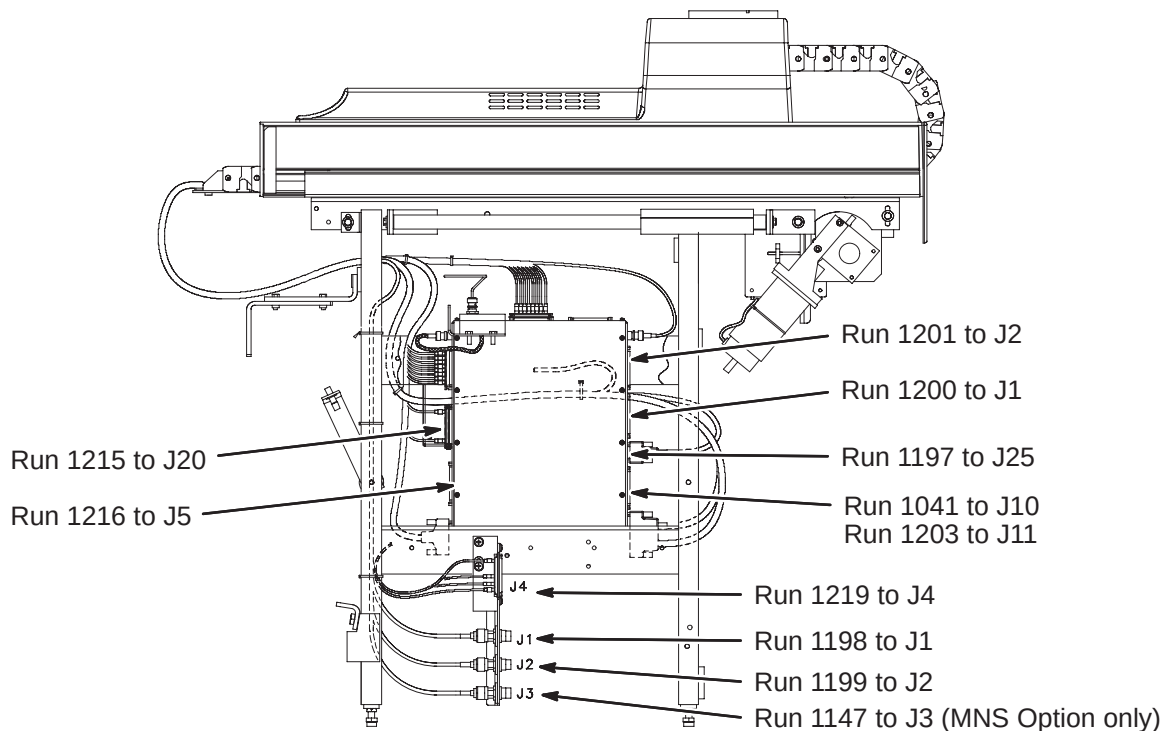


- ③ Route and connect three of the cables of Run 1220 to the Receive Filter on J11 and J7 and J8 on Hybrid Splitter. The remaining cable is connected to J35 on the 16 Channel Switch.

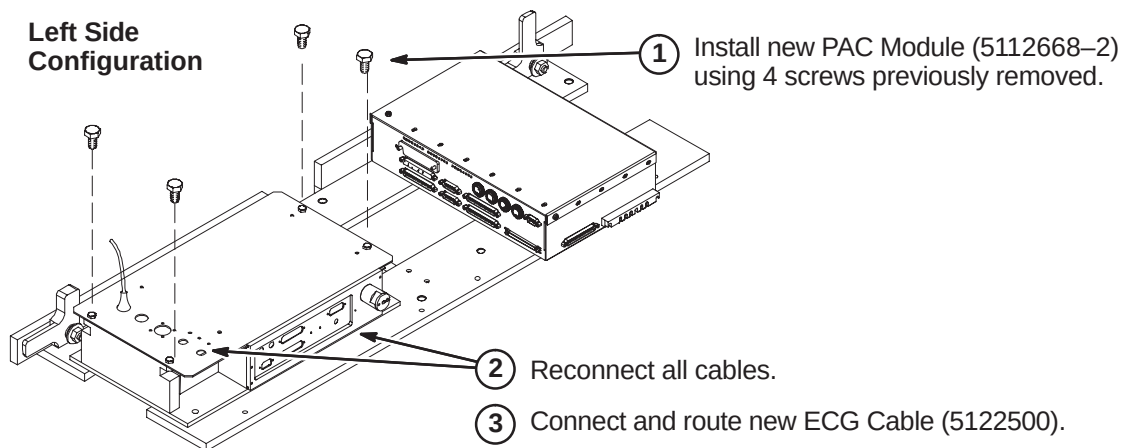
6-3-7 CONNECT REMAINING CABLES TO REAR PEDESTAL

- ① Route and connect Runs to 16 Channel Switch and Bulkhead Interface as indicated.

Note: Refer to Section 5-4 for instructions on connecting Runs 1200, 1201, and 1202 to Penetration Panel.

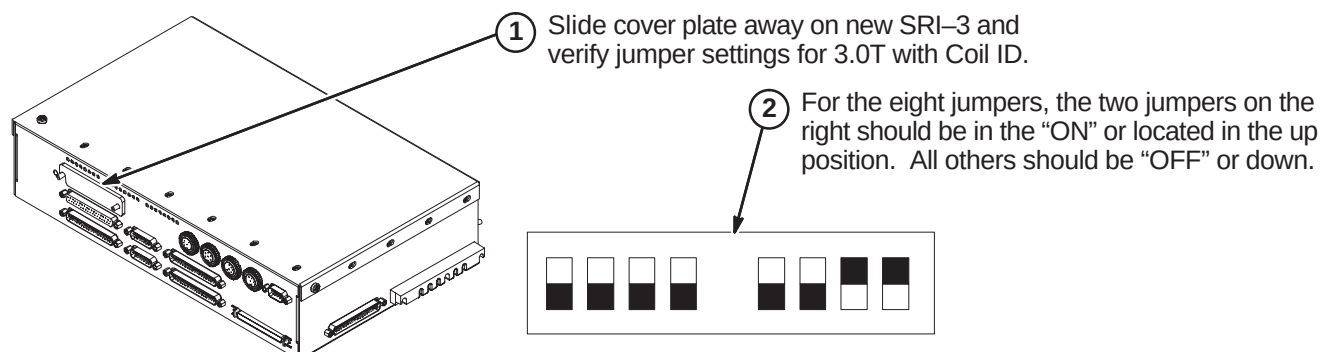
**6-3-8 INSTALL NEW PAC MODULE AND ECG CABLE**

Reverse the deinstallation steps for installation of the new PAC Module. Also install the new ECG Cable.

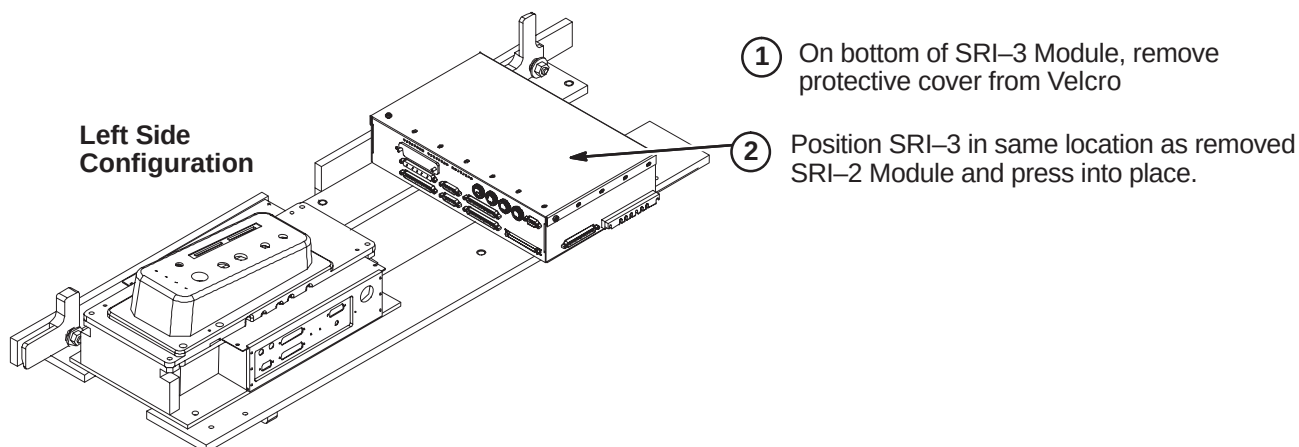


6-3-9 INSTALL SRI-3 MODULE

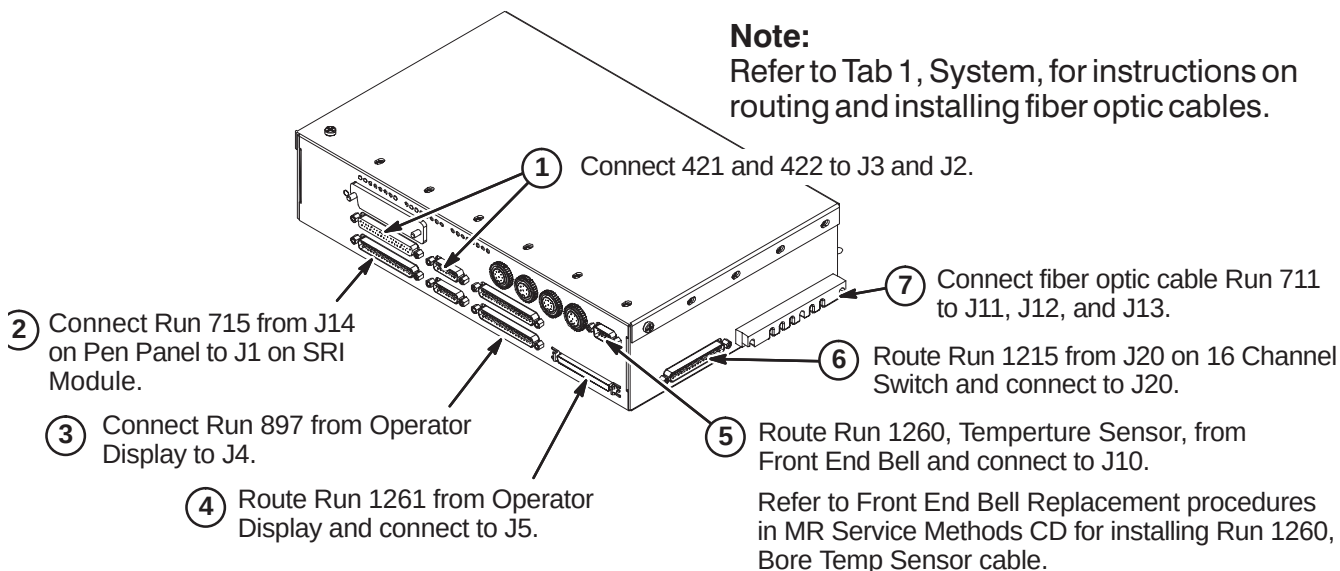
6-3-9-1 Verify SRI-3 Jumper Settings



6-3-9-2 Position SRI-3 Module



6-3-9-3 Connect Cables to SRI-3 Module



6-4 TIMES MICROWAVE RF CABLES INSTALLATION

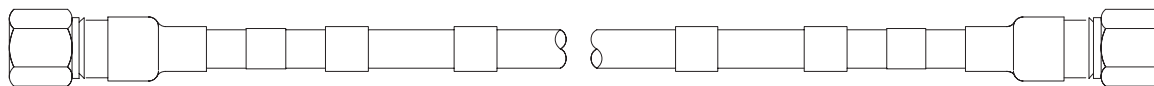
6-4-1 CABLE INFORMATION

Two cables are to be created and other associated parts to be installed are provided for this procedure. Also provided is a Times Microwave stripping tool for the LMR 600 (2352193), and a Times Microwave cutting tool (5111565).



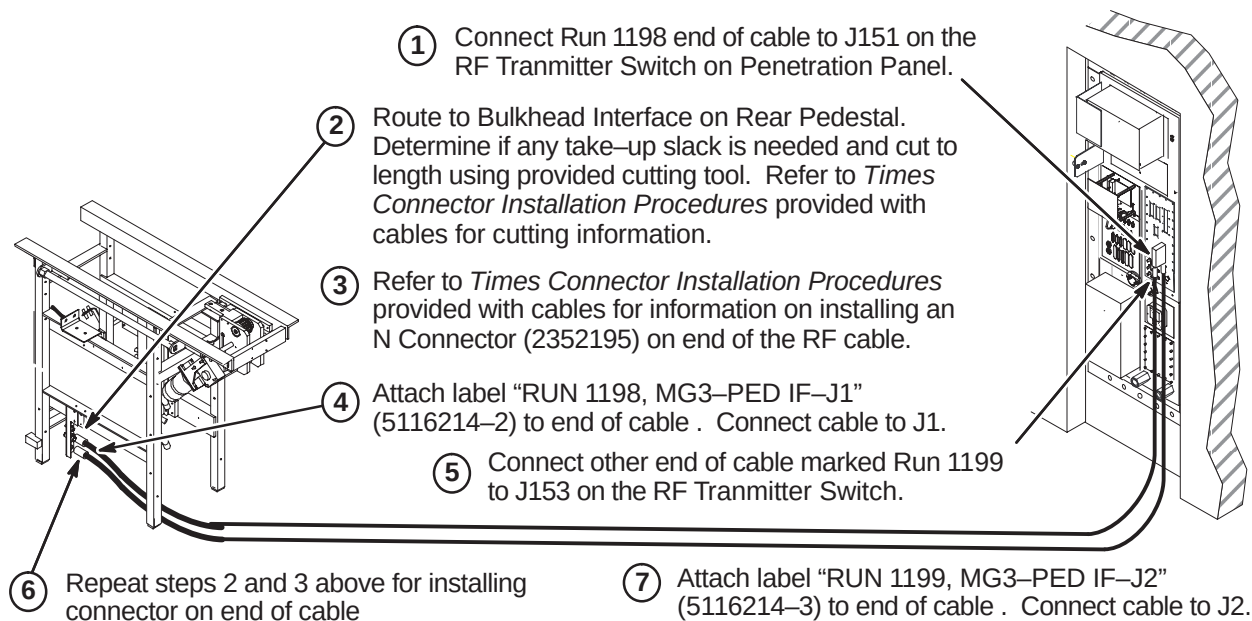
Do not subject LMR-600FR cable to a bend radius less than 3 inches (75mm). The cables will be permanently damaged if it is kinked by a bend radius less than the specified minimum.

Cable 5115915, will be cut to length to create two cables, Runs 1198 and 1199, and terminated for routing in the scan room. The remaining connectors, fasteners, and labels will be installed as directed in the following steps.



Run 1198 and Run 1199

6-4-2 ROUTE, TERMINATE, AND CONNECT RF CABLES



SECTION 7 – OPERATOR WORKSPACE

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7-1 INTRODUCTION

A new Monitor, SCIM, and Keyboard will replace existing components on the Operator Workspace Table. Also, the new Temperature and Humidity kit will be installed along with the new Z400 computer.

A new Z400 computer is also installed for this upgrade. Depending on site requirements, there are two options for this computer upgrade. They are:

- A replacement of the existing GOC with a new GOC containing the Z400 computer. Refer to Section 7-2.
- A replacement of only the existing computer with the new Z400 computer. Refer to Section 7-3.

After completing one of the above options, continue the Operator Workspace upgrade with Section 7-4.

7-2 INSTALL NEW GOC CONTAINING Z400 COMPUTER

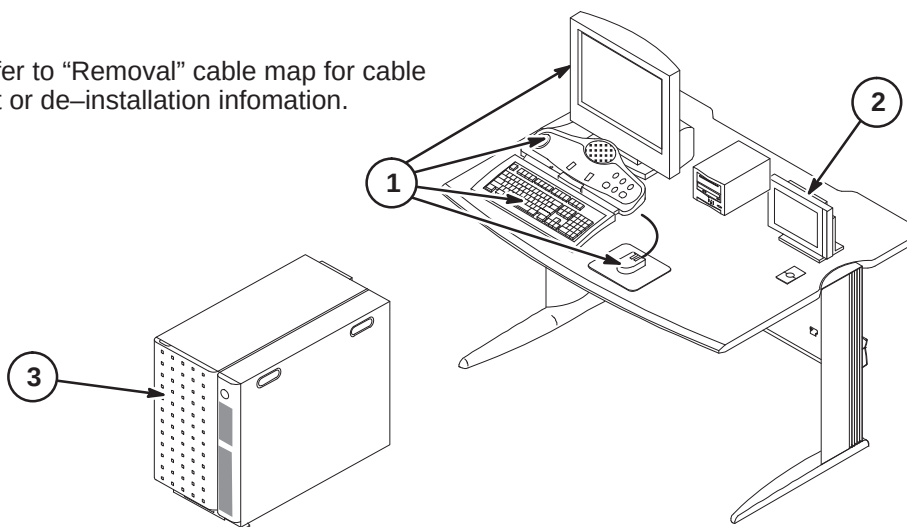
7-2-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

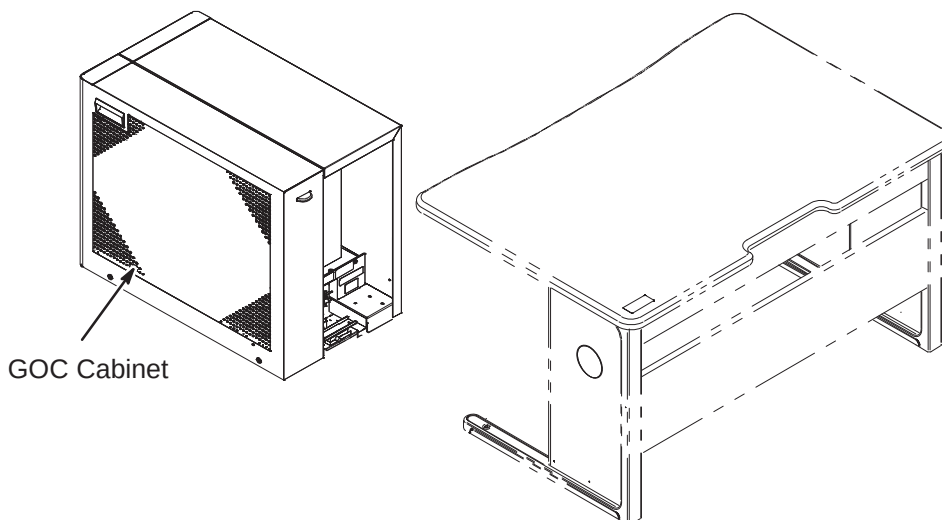
- 1 Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.
Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.
- 2 De-install LCD Display, Runs 1068 and 1069, and return for recycling.
- 3 Existing GOC will be replaced by a new GOC containing the Z400 computer:

Note: Refer to "Removal" cable map for cable retainment or de-installation information.



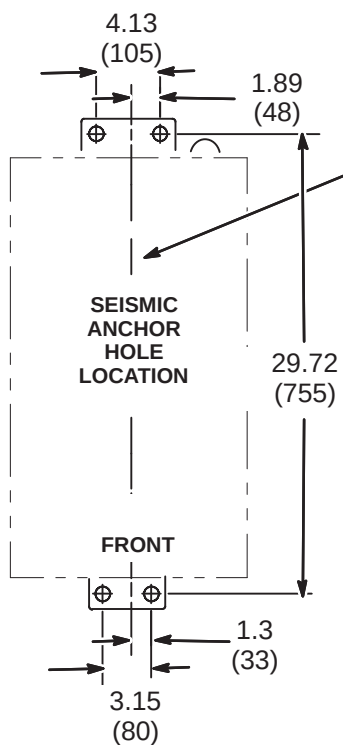
7-2-2 POSITION COMPUTER CABINET

Consult with Site customer for exact location of Global Operator Console (GOC). The location could be underneath the left or right side of the Operator Workspace table, or on the outside left or right side of the table.



7-2-3 SEISMIC ANCHORING (IF REQUIRED)

GOC Seismic Mount kit is shipped with GOC. Find kit inside of plastic bag.

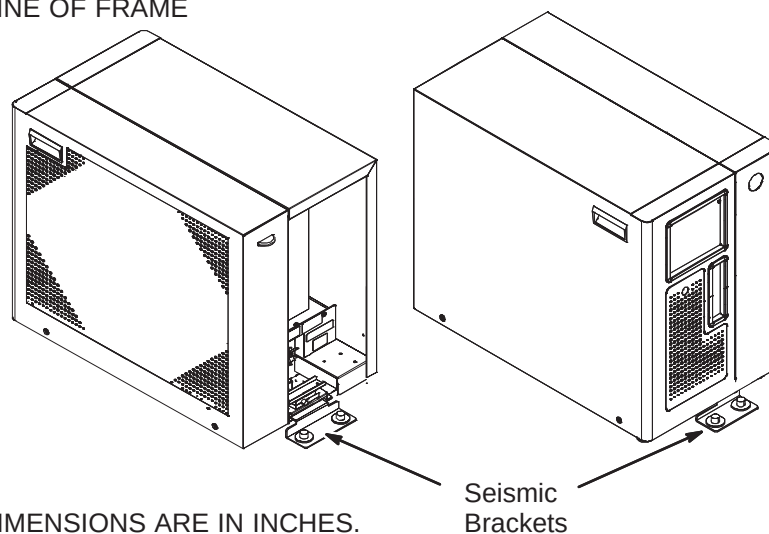


If seismic anchoring is required, refer to site architectural drawings for fastener type details or applicable Pre-installation manual. Illustration below indicates hole location for installing brackets

CENTERLINE OF FRAME

NOTE:

- ALL DIMENSIONS ARE IN INCHES. ALL BRACKETED () DIMENSIONS ARE IN MILLIMETERS.



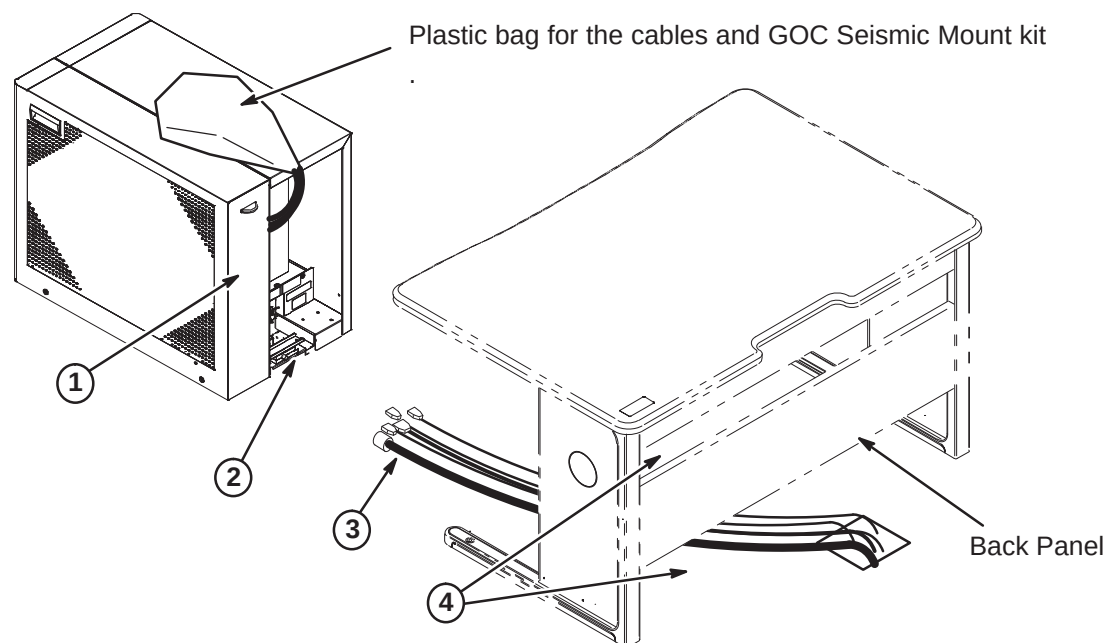
7-2-4 COMPUTER CABINET CABLE ROUTING AND CONNECTIONS

The GOC (Computer) Cabinet is shipped with cables connected to the computer that will be routed and connected to equipment that will be installed on the top of the OW Table. The cables are:

- Two power cords
- One video output cable
- Keyboard output cable
- SCIM output cable
- SCSI output cable
- Modem output cable

These cables are already connected to the computer and routed thru the opening located at the top of the rear of the GOC cabinet as shown in the illustration below. The remainder of the cables to be connected to designated modules within the GOC Cabinet will have to be routed thru the lower opening.

- 1 Runs 1064, 1070, E0015, E3045, E3046, E3047 and E3048 are pre-connected to computer and are routed at top of opening.
(These cables are inside of plastic bag)
- 2 Runs 1381, 1085, 1255, 1371, power cord for the Patient Alert system, and any other option cables will be routed thru lower opening.
- 3 Route all cables before connecting any cables. Refer to cable map for connection information.
- 4 If GOC Cabinet is to be located under OW Table, the cable coming out of the top opening of the cabinet will be routed above the Back Panel. The cables coming out of the lower opening will be routed below the Back Panel.



7-2-5 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

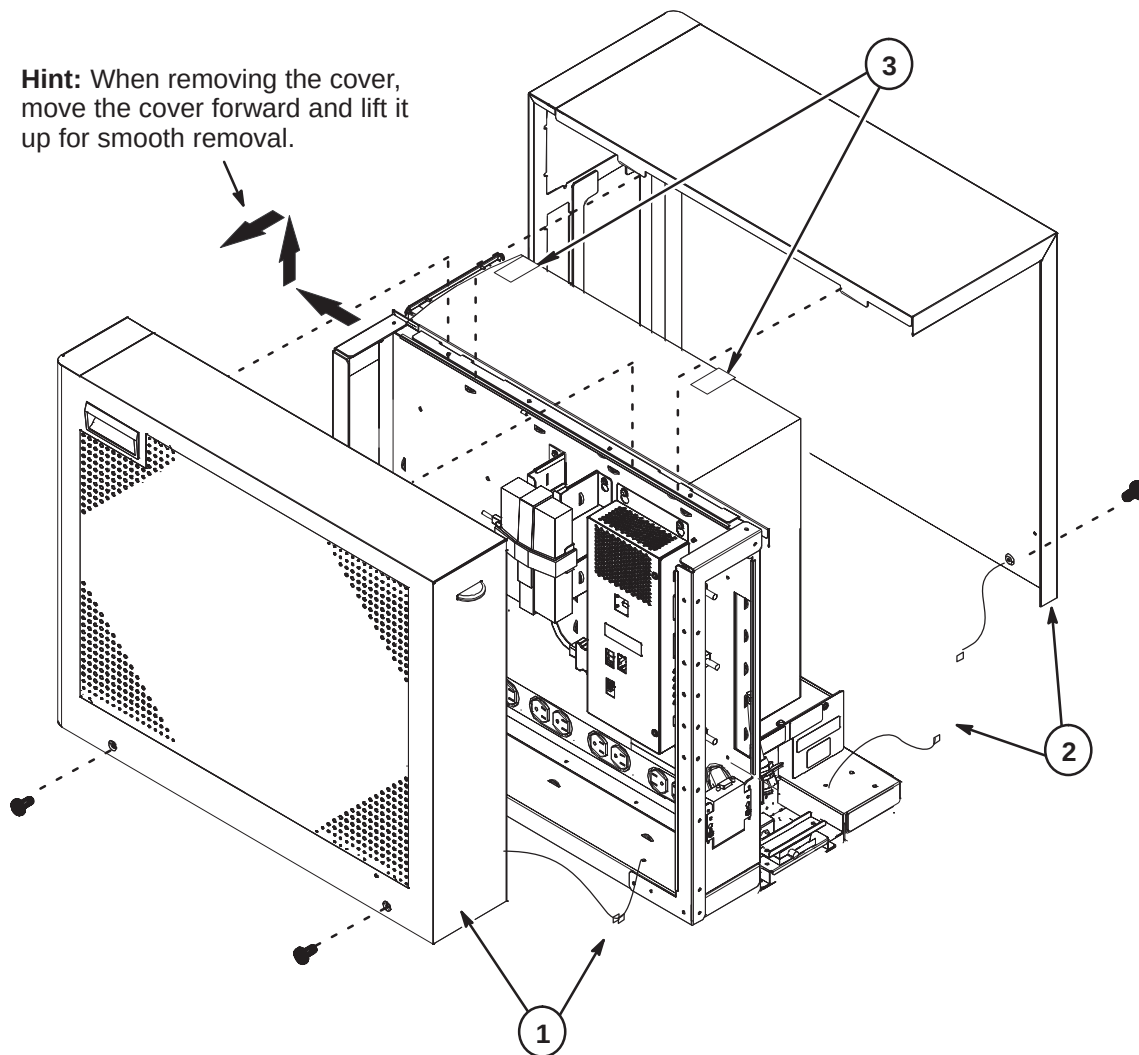
- 1 For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
- 2 For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

Note

A ground wire is connected to each side cover. Use care when removing side covers.

- 3 Peel the two shipping tapes off from PC.

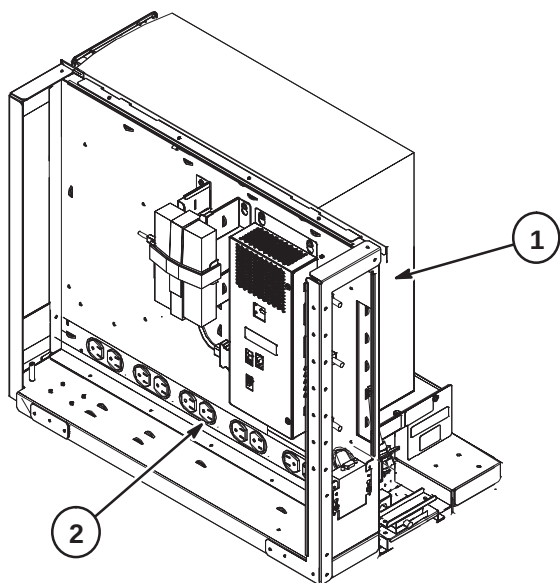
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



7-2-6 INSTALL MODEM CABLE TO GOC CABINET (If Applicable)

If Modem is part of system installation or upgrade, it will be placed on Operator Table or GOC.

- 1 Modem cable to the Linux PC is provided. Connect Run E3045 to Modem.
- 2 Modem power cord must be connected to J7 on GOC PDU Module. Route this cable to the rear top of the GOC Cabinet.



7-2-7 GOC CABLE CONNECTIONS

Refer to cable map for information on cables to be connected thru lower area of rear opening.

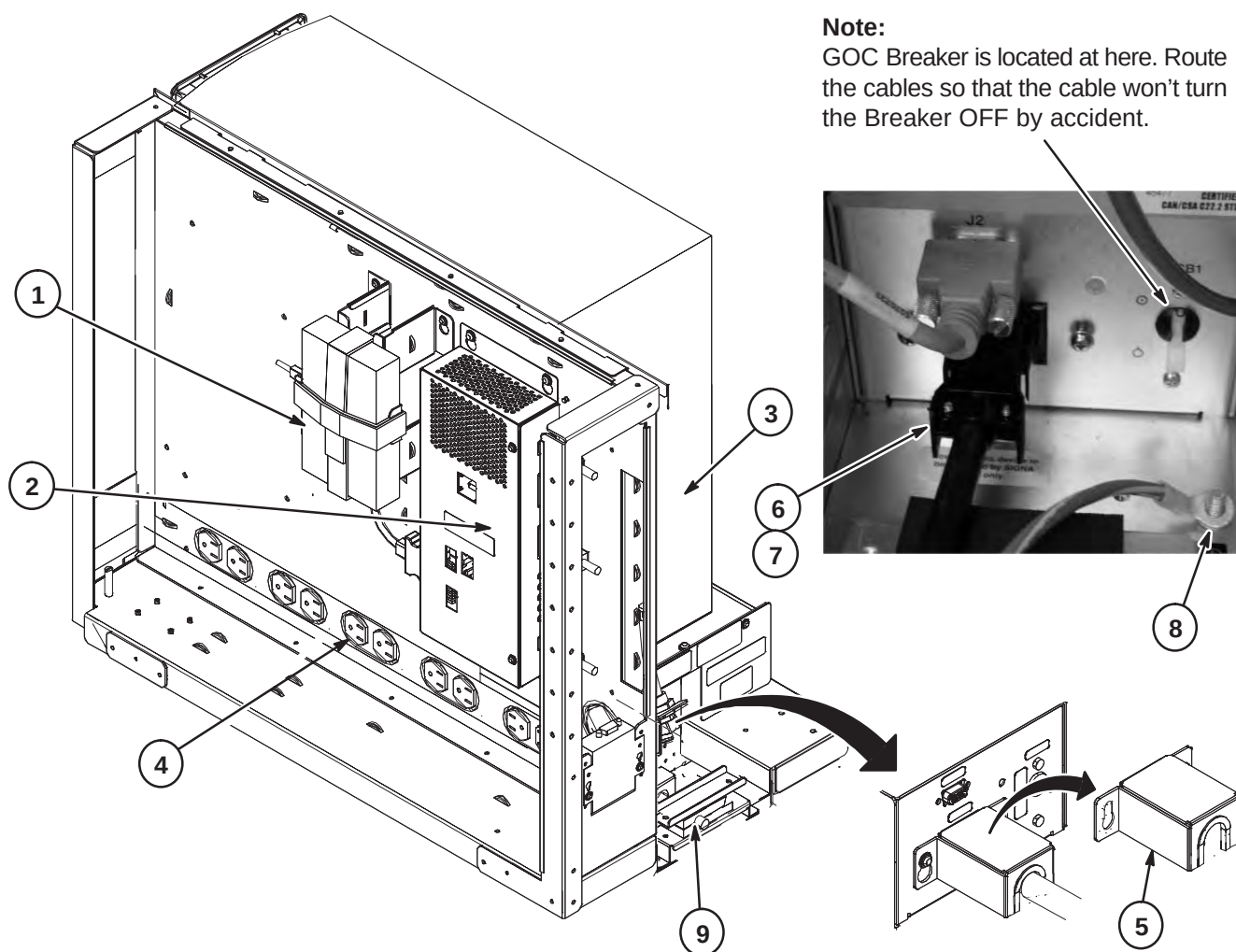
- 1 Route and connect Ethernet cables to Lan Hub.

Note Use ports 3, 7 or 8 for connecting site ethernet cables such as AW. Don not use any other ports for site ethernet cables.

- 2 Connect Run 1085 to J7 and Run 1255 to J18 on GOC AA.
- 3 Refer to Section 10-6-9 for all cables connected to Linux PC computer.
- 4 Refer to Section 7-2-8 for power cable connections to GOC PDU. Make sure Patient Alert power is connected to J3. If Modem is installed, power cable is connected to J7.
- 5 Before connecting GOC power cable Run 1381 to J1, loosen screws and remove cover.
- 6 Route and connect GOC power cable Run 1381 to J1 on PDU.
- 7 Confirm that GOC Power Cable is firmly connected to J1 by pulling the cable once. Re-install cover removed in Step 5 after connecting cable and tighten screws.
- 8 Connect Run 1382 to GND. Make sure nut securing GND cable is firmly tightened to GND stud.
- 9 Tighten clamp to secure cables in place.

Note:

GOC Breaker is located at here. Route the cables so that the cable won't turn the Breaker OFF by accident.



7-2-8 POWER CABLE CONNECTIONS

The illustration below show the power cable connection locations to the power strip in the GOC cabinet. Make sure power cables are attached as shown.

Also refer to cable map for information on power cable conections to OW1-PDM.

- 1
- If Modem is installed, route power cable and connect to J7.
- 2
- Tie-wrap point for Modem power cable.
- 3
- Route power cables connected to J3 thru J7 through access hole.
- 4
- route and connect Patient Alert power cable to J3 and tie-wrap at this point.
- 5
- Tie-wrap power cables J4 to J7 at this point.

The diagram shows a power strip with 12 outlets labeled J1 through J12. J1 and J2 are on the right side, while J3 through J12 are on the left. Arrows indicate cable routing: J1 and J2 connect to the back of the cabinet; J3, J4, J5, J6, and J7 route through an access hole at the bottom; J8, J9, J10, J11, and J12 connect to the back of the cabinet. A separate photograph shows the internal routing of these cables, with a red dashed circle highlighting the tie-wrap point for J3-J7.

This photograph shows the physical implementation of the cable connections. Callout 1 points to the top of the cabinet where cables enter. Callout 2 points to the tie-wrap point for the J3-J7 cables. Callout 3 points to the J1 and J2 connections. Callout 4 points to the J3 connection. Callout 5 points to the tie-wrap point for J4-J7.

POSITION	NAME
J1	GOC Input
J2	GOCAA
J3	Patient Alert
J4	Host Computer
J5	Host LCD
J6	SCSI Tower
J7	Modem
J8	
J9	LAN Hub
J10	
J11	
J12	One Wire Hub

7-3 INSTALL NEW Z400 COMPUTER IN EXISTING GOC

7-3-1 DE-INSTALL OPERATOR WORKSPACE COMPONENTS

CAUTION

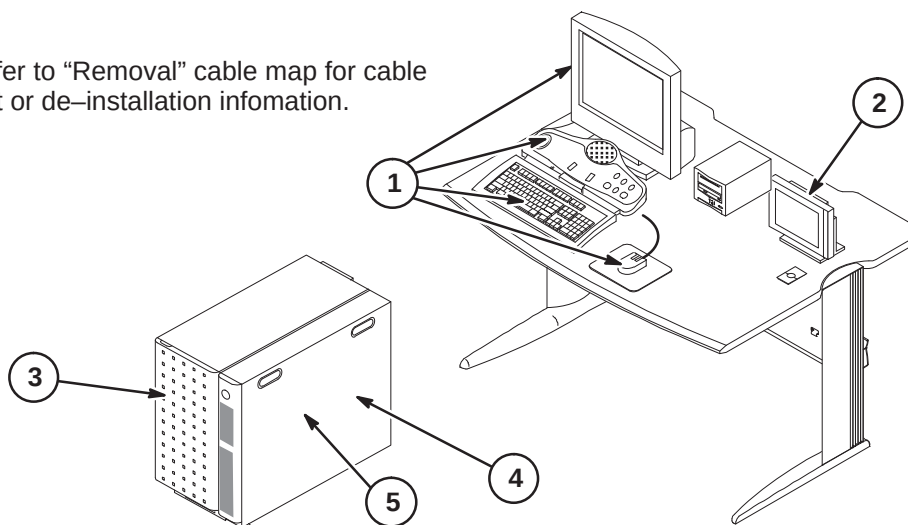
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

- 1 Existing Monitor including Runs 1063 and 1065, SCIM, Keyboard, and Mouse will be de-installed and returned for recycling.

Note: If Run 1062 is connected to the SCIM, remove and return for recycling. Install new Run 2053 which is provided with this upgrade (A few early versions of the cable upgrade kit provided a new Run 1256, without the short cables for connecting the keyboard and mouse. If received at this site, relabel as Run 2053). If existing SCIM cable is Run 1256, retain and reconnect to new SCIM.

- 2 De-install LCD Display, Runs 1068 and 1069, and return for recycling.
- 3 Existing PC computer in the GOC will be replaced by a new computer:
 - If the existing GOC is old style (screws securing covers from rear), secure the new PC using straps.
 - If the existing GOC is new (screws securing covers from side), the new PC must be secured to the GOC using the conversion kit, part number 5389531.
- 4 The GOC AA module has been delivered as part of the upgrade kit.
 - If the existing GOC already has a GOC AA module installed, the new GOC AA is not required.
 - If the existing GOC has a WIM installed, it must be replaced by the new GOC AA module.
- 5 An 8-Port 10/100/1000 Mbps Gigabit Ethernet Switch has been delivered as part of the upgrade kit.
 - If the existing GOC already has this Switch installed, the new Switch is not required.
 - If the existing LAN Hub Switch is an 8-Port 10/100, then it must be replaced by the new 10/100/1000 Switch.

Note: Refer to "Removal" cable map for cable retainment or de-installation information.

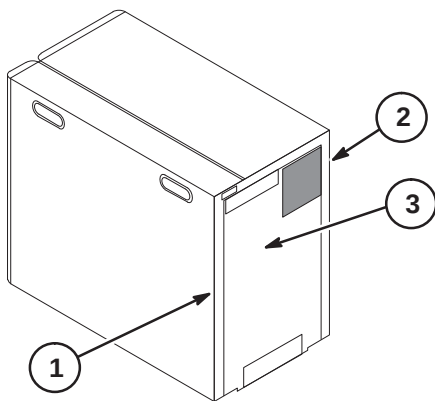


7-3-2 GOC CABINET COVER REMOVAL

The Computer Cabinet covers need to be removed to access the cable connection points.

7-3-2-A OLDER STYLE GOC COVERS

- 1 For right side cover, remove three screws at rear of cover, slide toward rear of cabinet and then lift up.
- 2 For left side cover, remove four screws at rear of cover, slide toward rear of cabinet and then lift up.
- 3 Loosen four screws holding the back cover. Lift and remove cover.

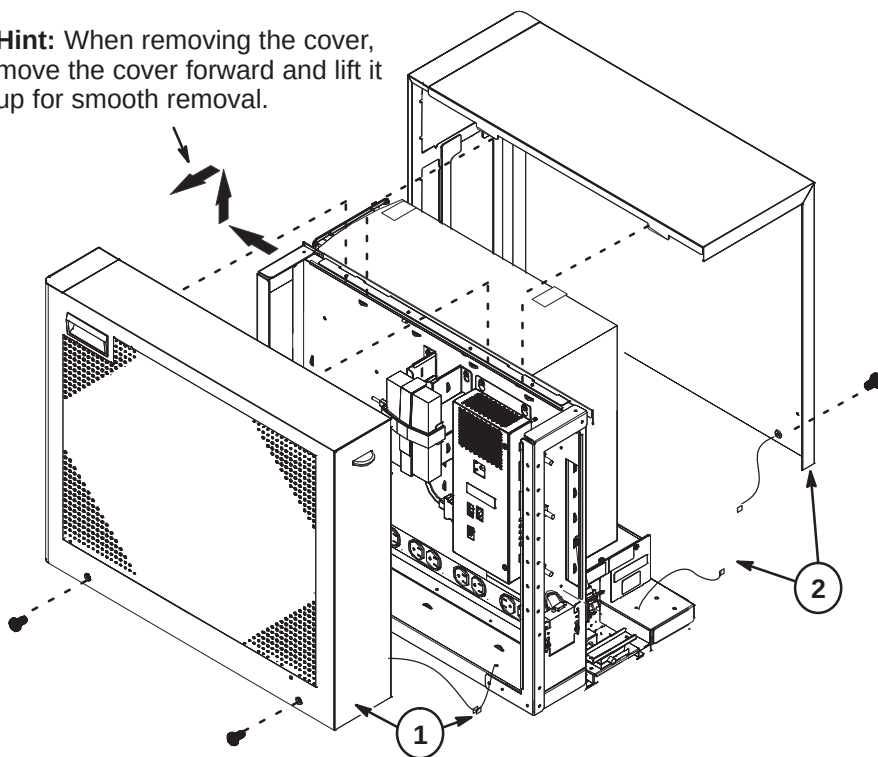


7-3-2-B NEWER STYLE GOC COVERS

Note: A ground wire is connected to each side cover. Use care when removing side covers.

- 1 For right side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.
- 2 For left side cover, remove two screws and lift up. Disconnect Ground cable connector and remove the cover.

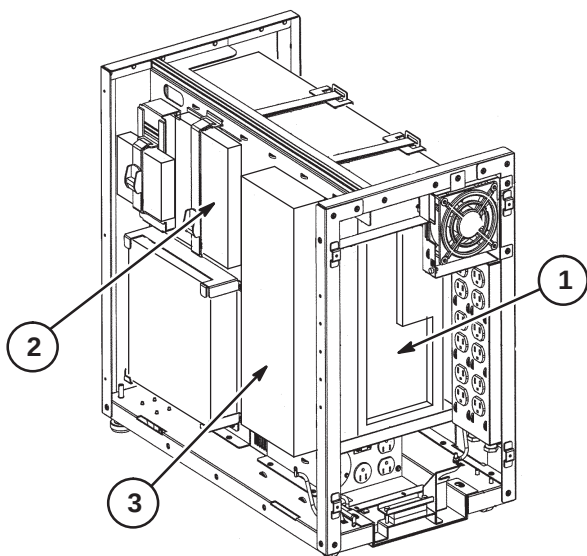
Hint: When removing the cover, move the cover forward and lift it up for smooth removal.



7-3-3 GOC COMPONENT REPLACEMENT

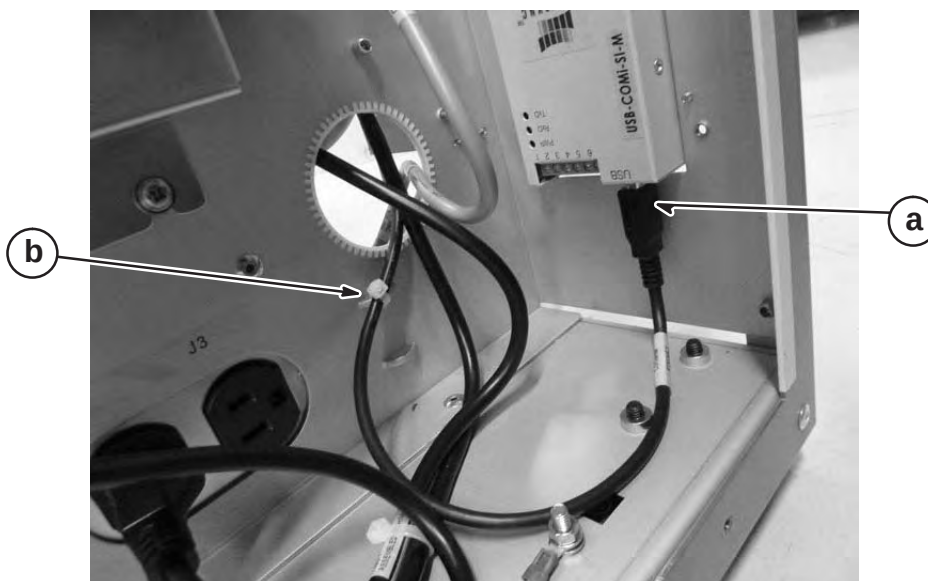
Refer to applicable cable removal map for de-installation of existing cables that will be replaced by new cables. The cable maps also show which cables will be disconnected from existing components and reconnected to new components.

- 1 Disconnect all cables from existing computer. Make sure they are properly labeled.
- 2 If required, replace existing LAN Hub Switch with new 10/100/1000 Mbps Gigabit Ethernet Switch (5139480-2).
- 3 If required, replace existing WIM with new GOC AA module (2395043).



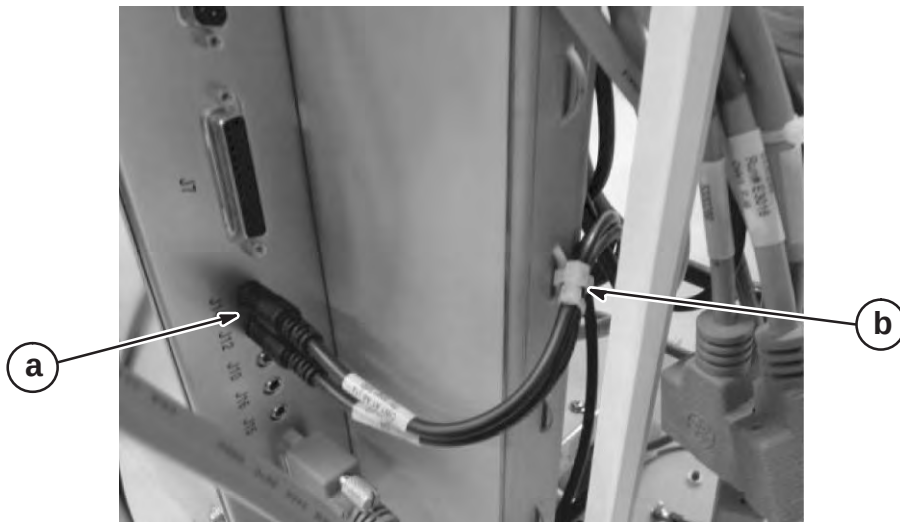
7-3-4 INSTALLATION OF NEW GOC COMPONENTS

- 1 Replace the Host to USB–Serial Converter USB Cable.
 - a. Replace the Host to USB–Serial Converter USB Cable with the new cable (5330295-2) supplied in the Conversion Kit.
 - b. Cut and restore tie-wrap(s) as required during cable replacement.

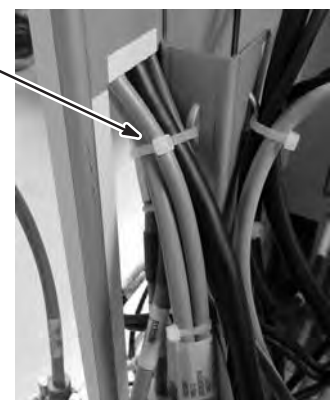
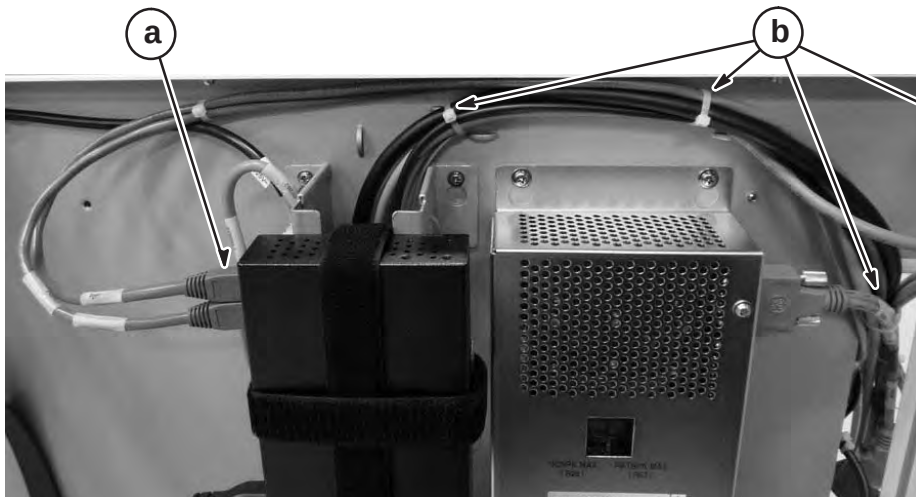


7-3-4 INSTALLATION OF NEW GOC COMPONENTS (Continued)**2 Replace the Audio Cable Assembly.**

- a. Replace the Audio Cable Assembly, connected to J12 and J14 on GOCAA with the new cable (5330301) supplied in the Conversion Kit.
- b. Cut and restore tie-wrap(s) as required during cable replacement.

**3 Replace the Host-Hub Ethernet Cable.**

- a. Replace the Ethernet Cable between the Host and Port 1 of Ethernet Switch Hub with the new Ethernet Cable (5330309) provided in the Conversion Kit.
- b. Cut and restore tie-wrap(s) as required during cable replacement.



View from GOC rear side.

7-3-5 GOC COMPUTER REPLACEMENT

Dependent on model of Host PC Computer being replaced, there are two different procedures for upgrading to a Z400 computer.

- For sites with an existing older style GOC, refer to Section 5-5-1 for upgrade instructions.
- For sites with an existing newer style GOC, refer to Section 5-5-2 for upgrade instructions.

IMPORTANT!

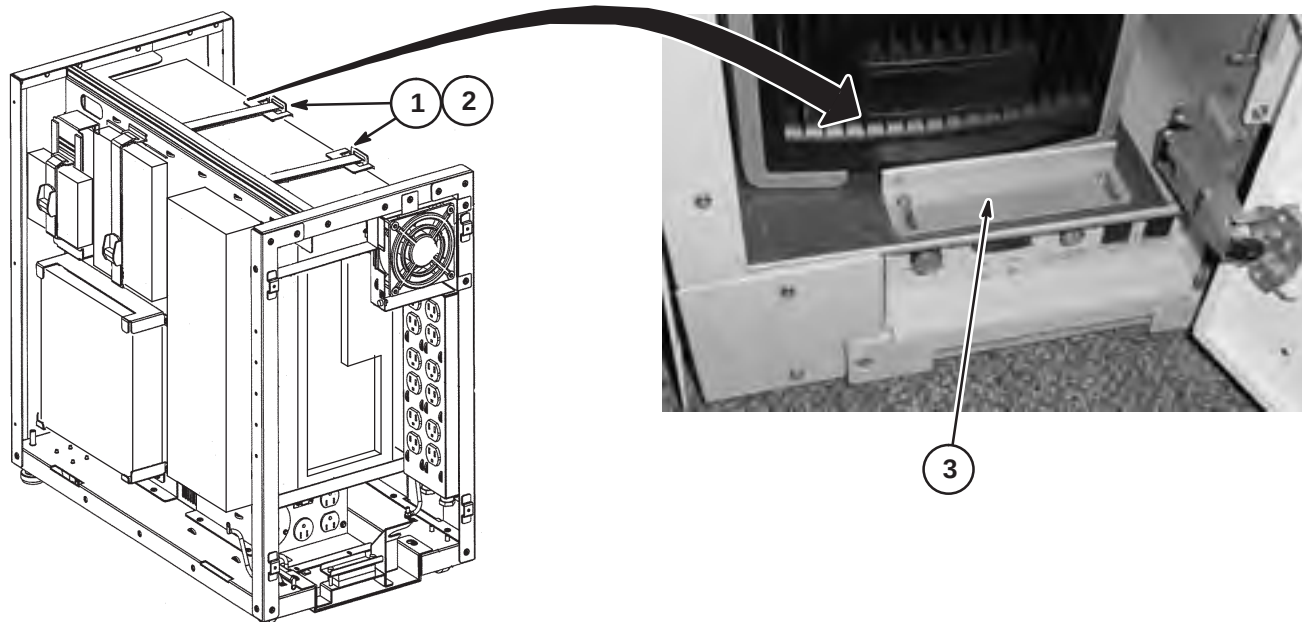
XW8000, 8200, 8400, or 9300 PC internal hardware, such as any hard drive, is not compatible with the Z400 computer. Do not attempt to transfer any internal hardware between any of these and the Z400 computer.

CAUTION

The computer weighs about 50 pounds (23Kg). This requires two people to handle the computer.

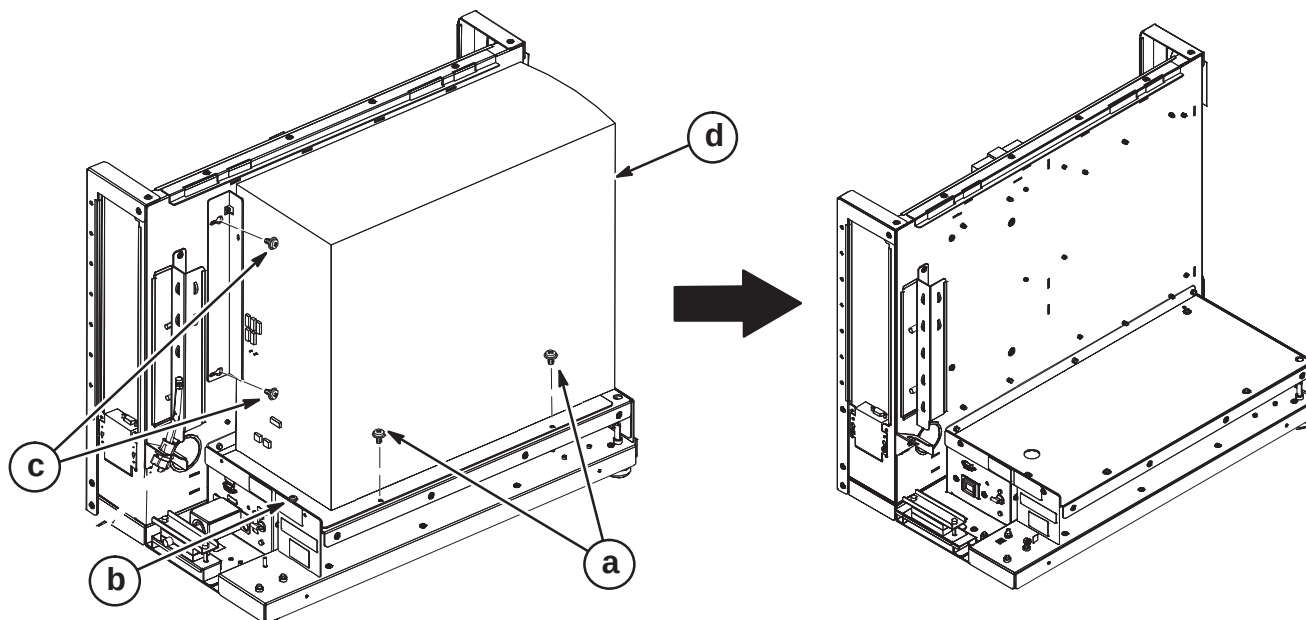
7-3-5-A COMPUTER REPLACEMENT IN OLD STYLE GOC

- 1 Unfasten straps and carefully remove computer.
- 2 Because the new Z400 computer is shorter than the original PC being replaced, install the new computer to the rear-most position and secure in place with straps.
- 3 If required, install PC Stopper Bracket (5127954) at front of computer using existing screws.

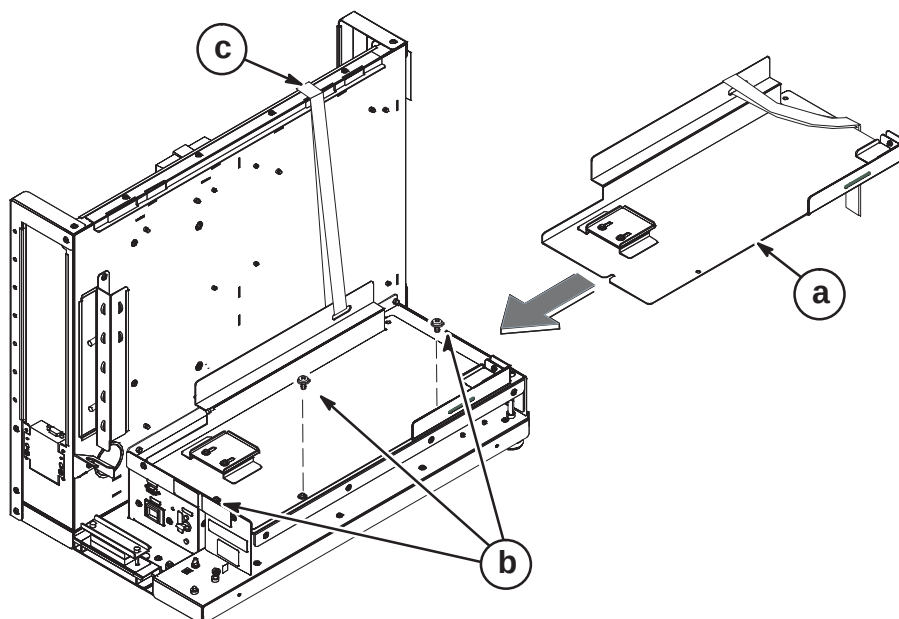


7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC

- 1 Remove the XW8400 Computer that is secured with back and bottom brackets as follows:
 - a. Remove the two screws securing the side of the bottom plate of the PC to the GOC frame.
 - b. Loosen the one screw securing the rear of the bottom plate of the PC to the GOC frame.
 - c. Remove the two screws from the back bracket.
 - d. Carefully remove the computer with back and bottom brackets from the GOC

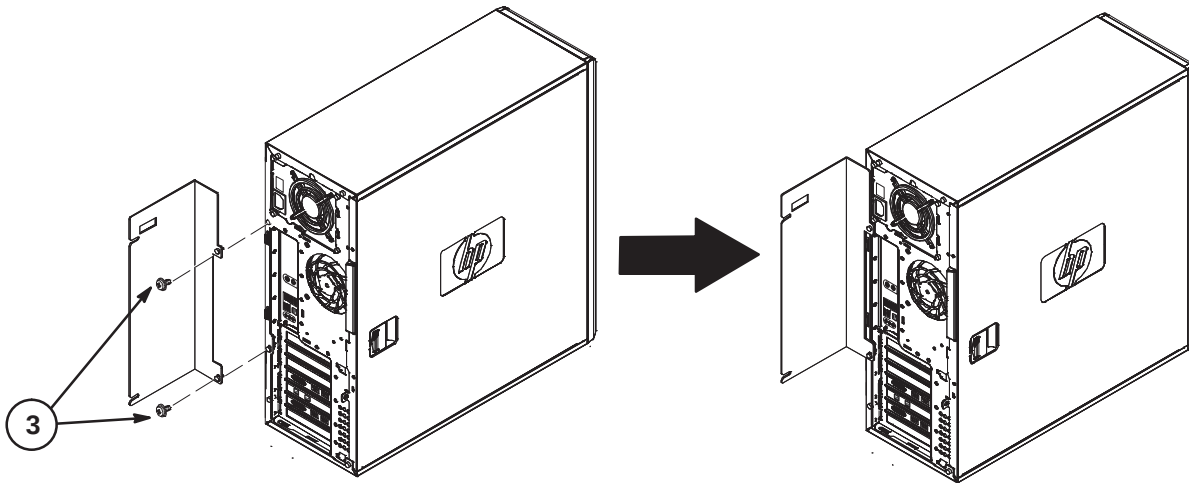


- 2 Install the Host Mounting Plate for the Z400 computer as follows:
 - a. Insert the Host Mounting Plate so that the GOC chassis screw holes are visible through the plate.
 - b. Tighten the one screw at the rear and the two screws on the left as shown below.
 - c. Temporarily hang the strap on the opposite side.



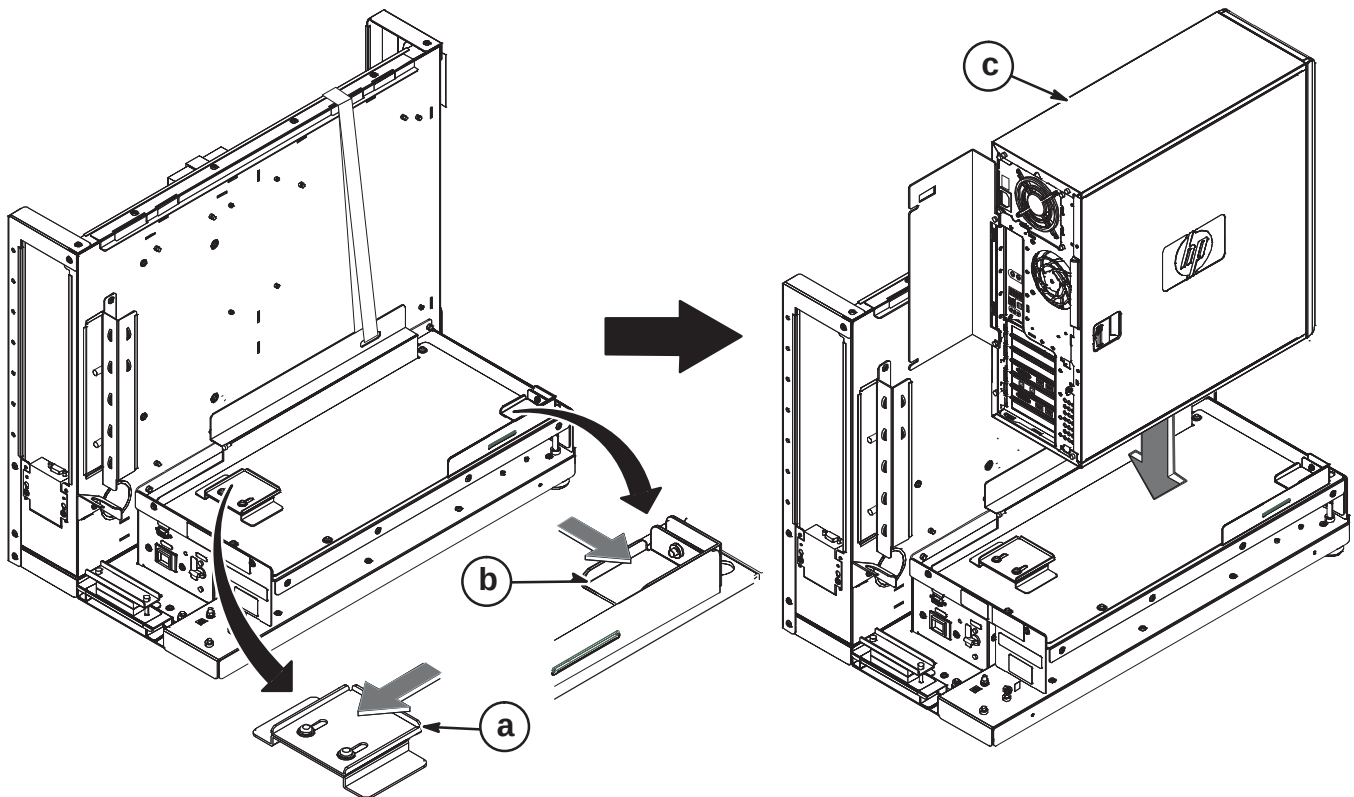
7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC (Continued)

- 3 Install the back bracket to the computer with the two screws.



- 4 Position the PC as follows:

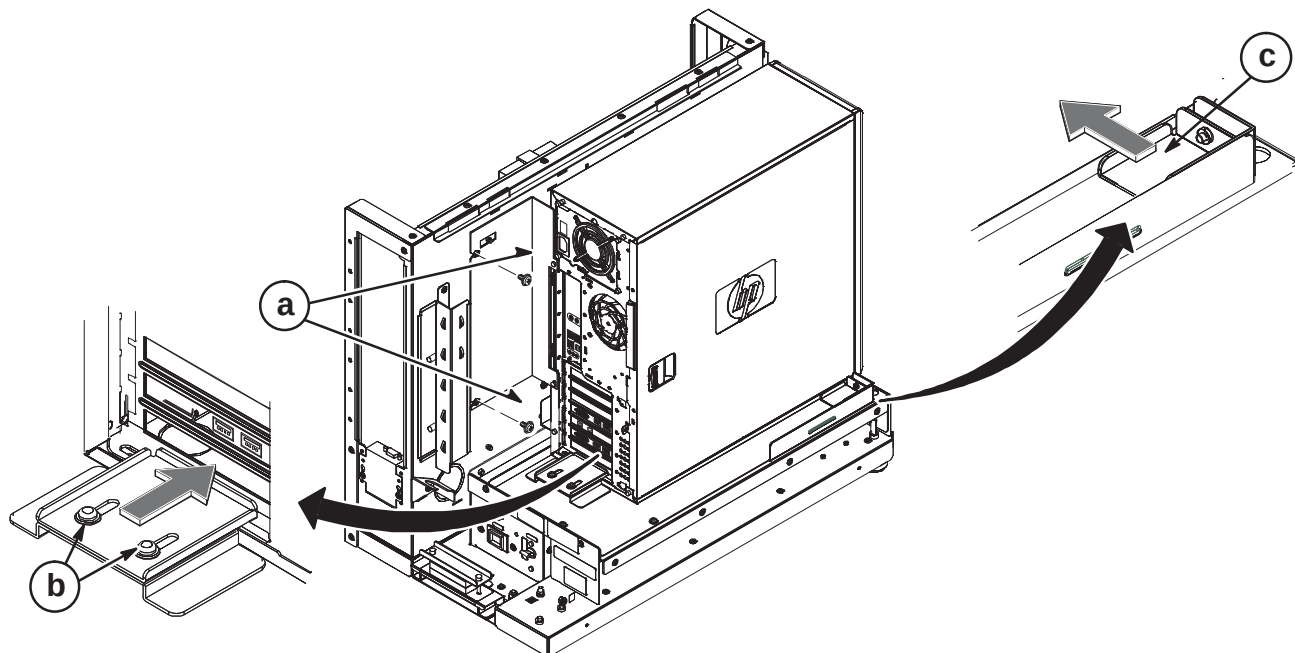
- Loosen the two screws on the back slide plate, and slide the plate backward.
- Loosen the two screws on the side slide plate, and slide the plate in the direction of the arrow as shown.
- Carefully place the Z400 Computer on the mounting plate.



7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC (Continued)

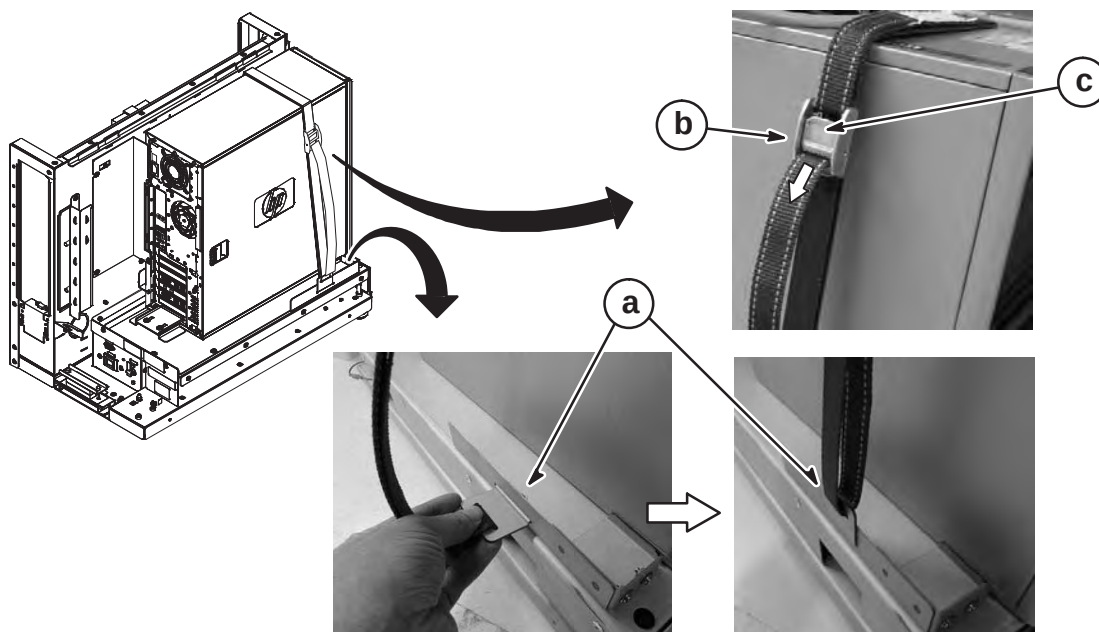
5 Secure the computer as follows:

- a. Secure the back bracket to GOC frame with two screws.
- b. Slide the back slide plate forward until it makes contact with the computer, and tighten the two screws.
- c. Slide the side slide plate until it makes contact with the computer, and tighten the two screws.



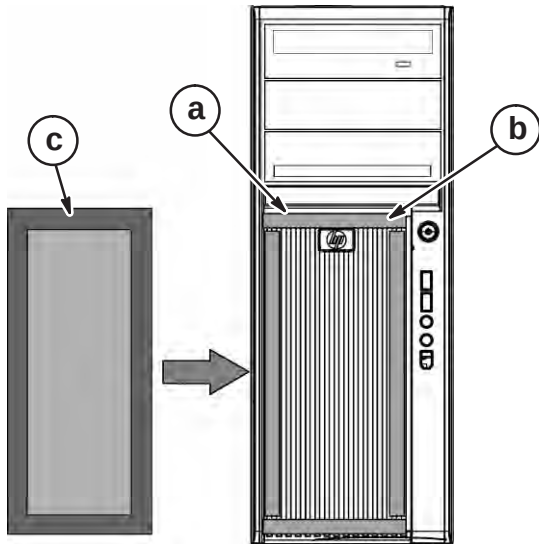
6 Tighten the strap around the computer as follows:

- a. Hook the bracket onto the slot of bottom plate.
- b. Pull and tighten the strap to secure the computer.
- c. **Note:** To loosen the strap, press the release lever on the strap.



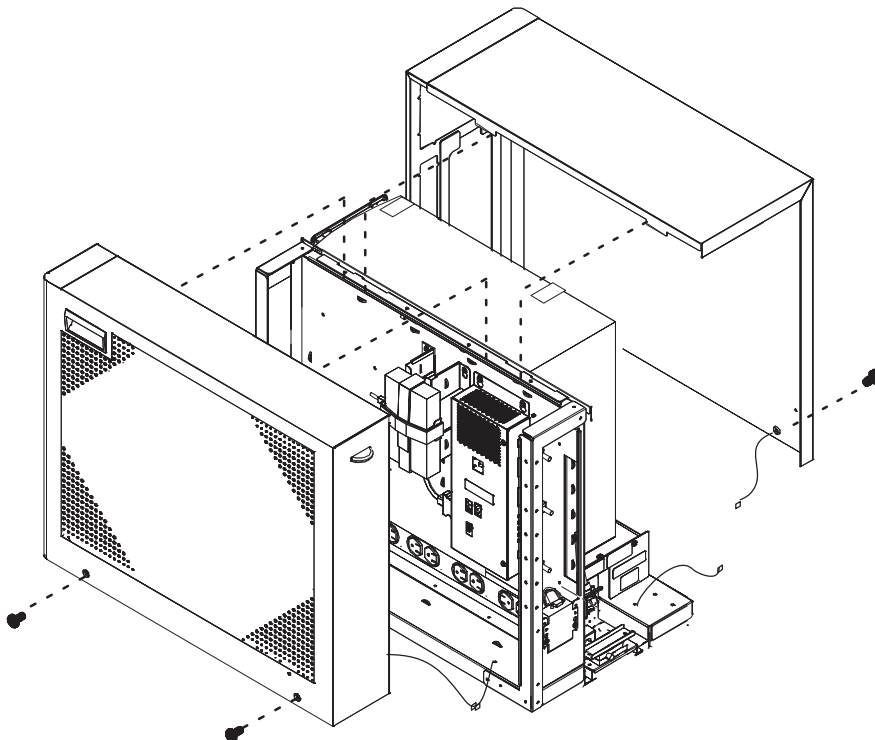
7-3-5-B COMPUTER REPLACEMENT IN NEW STYLE GOC (Continued)

- 7 Attach the Dust Filter to the front panel after following the cleaning instructions below.
- Clean the velcro area of front panel with alcohol and cloth. Peel the velcro from backside of filter.
 - Attach the velcro around front panel.
 - Attach the filter to front panel.



- 8 Replace the GOC covers, using the new left cover for the Z400 computer that shipped with the GOC Conversion Kit.

IMPORTANT! Make sure the Z400 computer is installed at the rear-most position before installing covers. This position will aid in connecting existing cables to rear of Z400 computer.



7-4 TEMPERATURE AND HUMIDITY SENSOR KIT INSTALLATION

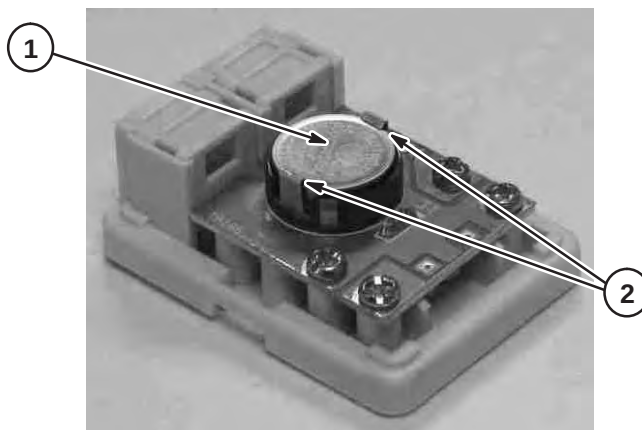
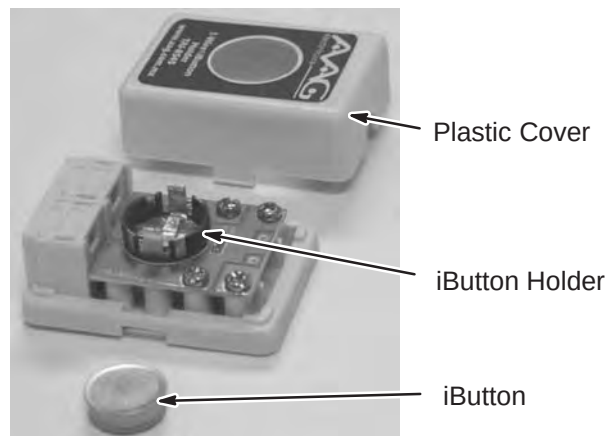
The components of the Temperature and Humidity Sensor Kit (5147136-2) will be installed on the back of the GOC cabinet and on the back of the System Cabinet with one cable routed from the GOC and connected to it.

7-4-1 INSTALL iBUTTON INTO HOLDER

Install the iButton into the holder in the orientation shown in the illustration.

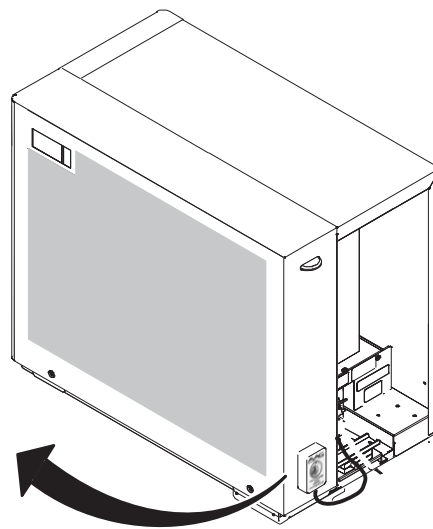
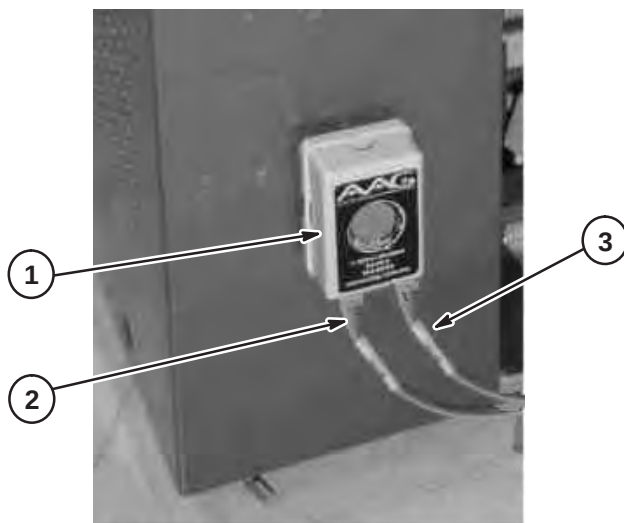
Note: The pictures below show the plastic cover removed although this is not necessary to install the iButton.

- 1 Install the iButton in the holder so that the solid part of the case is exposed.
- 2 Make sure the “fingers” latch over the top of the iButton and secure it in the holder.
- 3 If removed, reinstall the Plastic Cover.



7-4-2 INSTALL TEMPERATURE & HUMIDITY SENSOR ON GOC

- 1 Attach iButton Sensor assembly (Holder, 5141856, and iButton 5141857-5), to back of GOC Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
- 2 Route and connect Run 1322 (5143794-2) from System cabinet to Sensor at back of GOC.
- 3 Connect cable Run 1321 (5143794-2) to Sensor and route other end thru bottom opening of GOC and connect to USB to 1-Wire Adapter (5307784) installed in GOC USB port. Refer to step 10 in Section 5-5-2.



7-4-3 INSTALL TEMPERATURE & HUMIDITY SENSOR ON SYSTEM CABINET

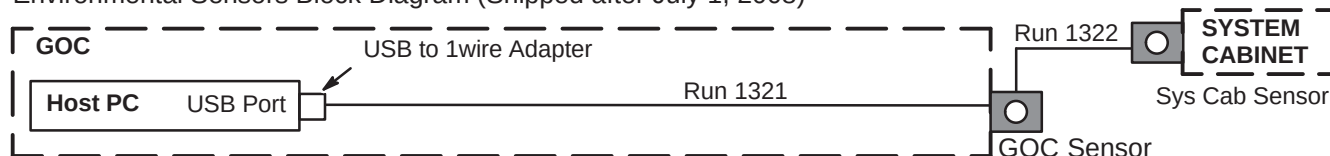
Note: iButton Holders are unique to each location. Make sure to install 5141857-4 to System Cabinet and 5141857-5 to GOC.

- 1 Attach iButton Sensor assembly (Holder, 5141856, and Humidity iButton (5141857-4), to back of System/RFS Cabinet at this location using Dual Locking Velcro Tape (2341779 & 2341779-2).
- 2 Route and connect cable Run 1322 (5143794) from GOC rear cover.

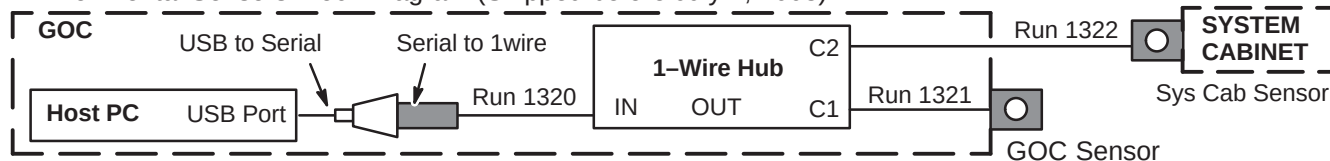


7-4-4 TEMPERATURE & HUMIDITY SENSOR WIRING SCHEMATICS

Environmental Sensors Block Diagram (Shipped after July 1, 2008)



Environmental Sensors Block Diagram (Shipped before July 1, 2008)

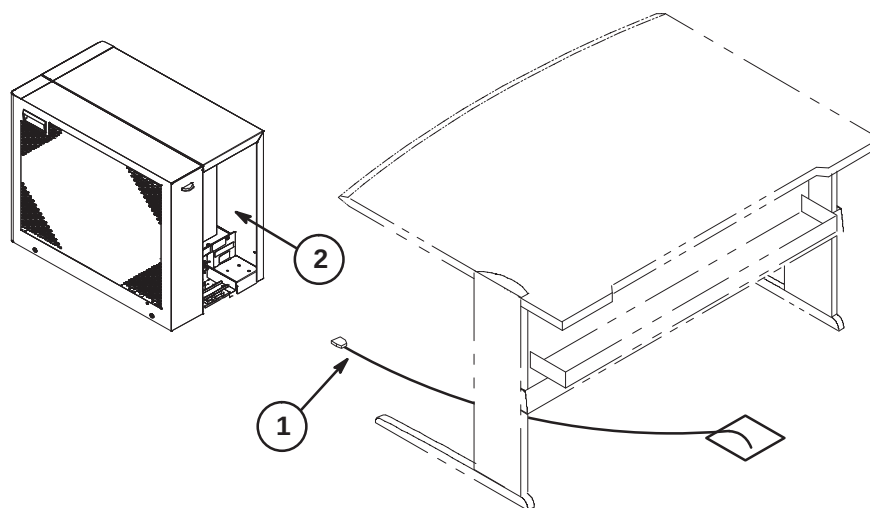


7-5 ROUTE RUN 1323

- 1 Route Run 1323 from RFS or System Cabinet to location of GOC. This is a new CAT6 cable that must be installed. Existing CAT5 cables cannot be reused.

NOTE: For optimal efficiency, route Run 1322, Section 7-4-3, at the same time as Rune 1323. It is also routed between RFS/System Cabinet and GOC.

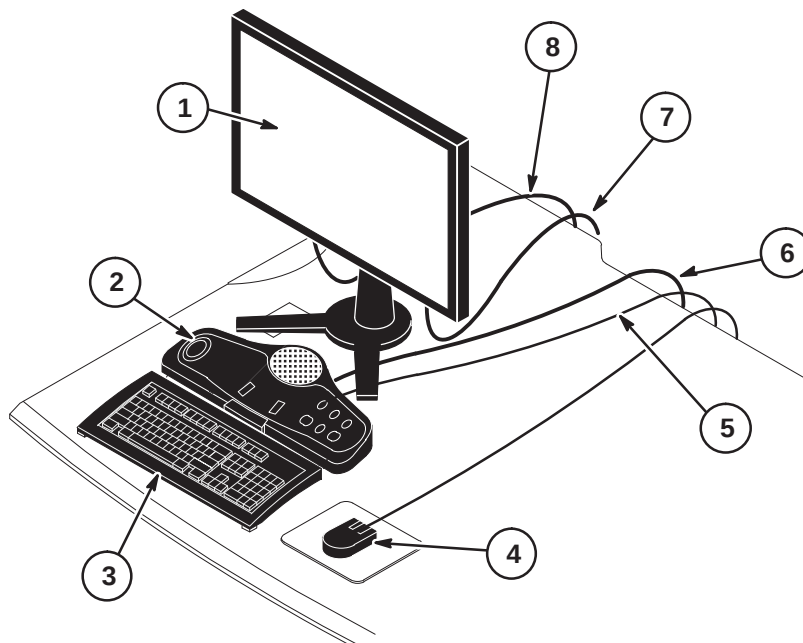
- 2 Cable will be routed thru the lower rear opening.



7-6 INSTALL NEW OW TABLE TOP COMPONENTS

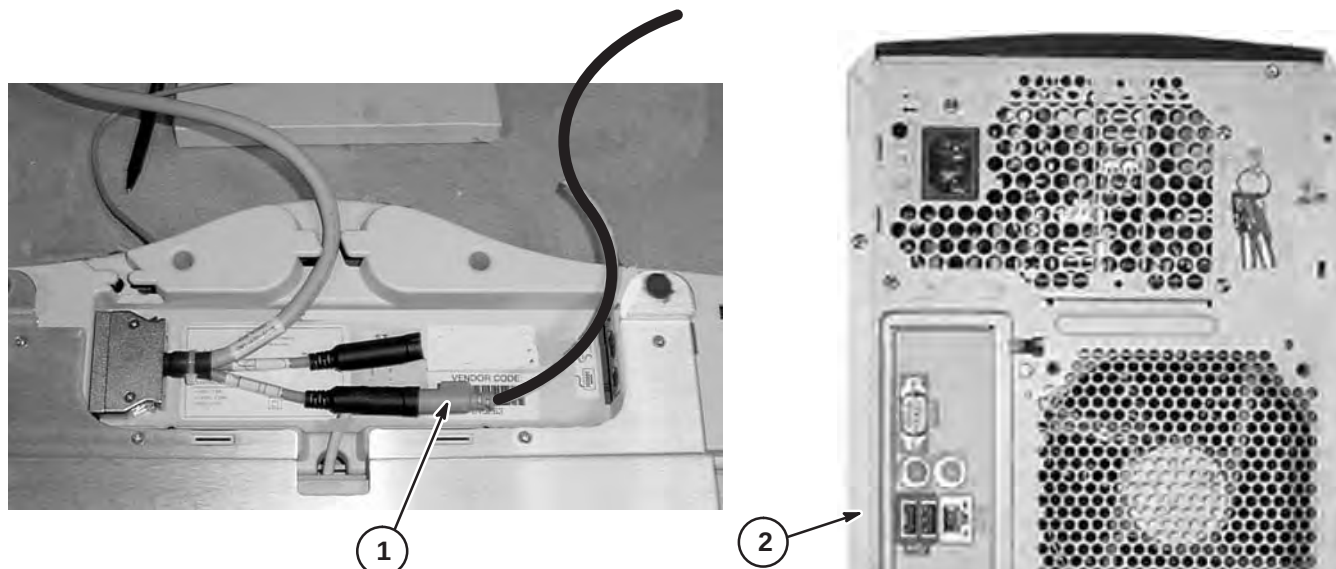
- 1 Install new 23 Inch Monitor on table.
- 2 Place new SCIM (5148388-x) on table.
- 3 Place new Keyboard (5143798-x) on table.
- 4 Install new Mouse (5143798-6) on table and route USB cable to computer in GOC. Attach provided extension USB cable if necessary.
- 5 Connect extension cable Run 2038 to USB Keyboard cable and route to computer in GOC.
- 6 Connect Run 2053 (or 1256) to SCIM and route to rear of GOC.
- 7 Connect provided monitor power cable to monitor. Connect other end of cable to Run 1065, which is connected to OW power strip.
- 8 Connect DVI cable, Run 2037, to monitor and route to rear of computer in GOC.

Note: Do not use or connect provided VGA cable.

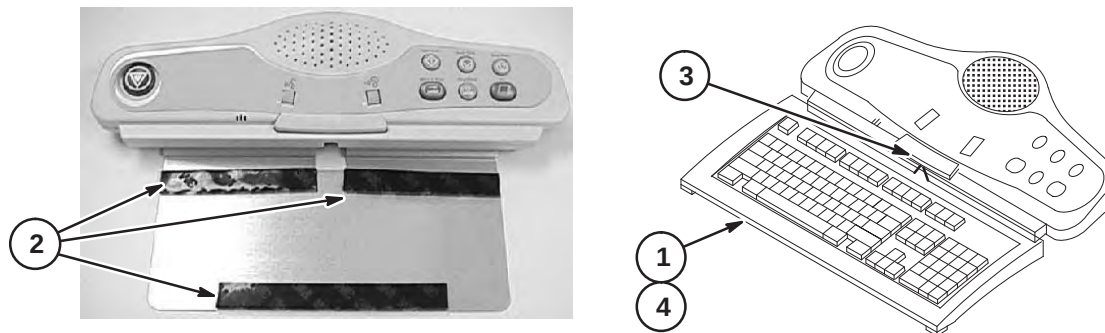


7-7 CONNECT BAR CODE READER (If part of site configuration)

- 1 If Run1256 is present – Disconnect the Bar Code Reader cable from Run 1256 Y-connector under SCIM.
- 2 Connect cable Run 1379 to rear of computer and Bar Code Reader.

**7-8 INSTALL SCIM/KEYBOARD**

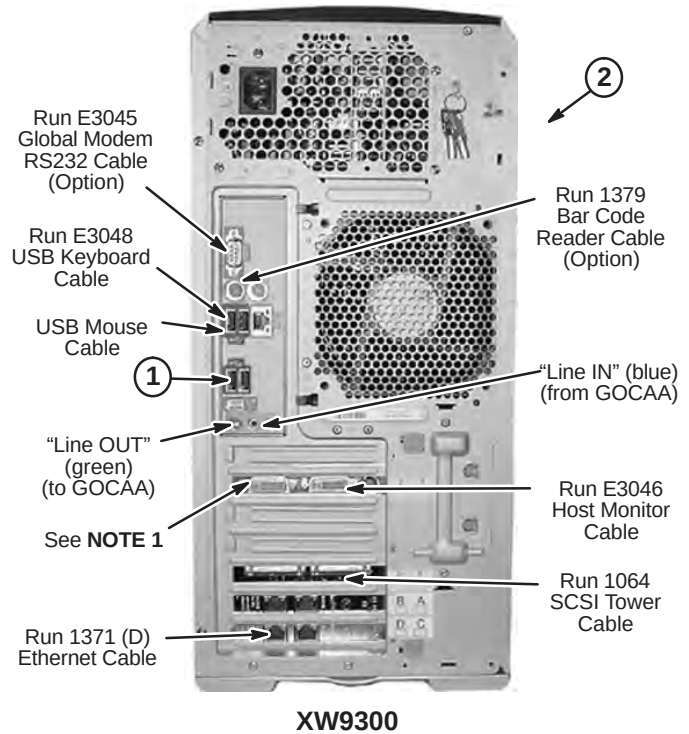
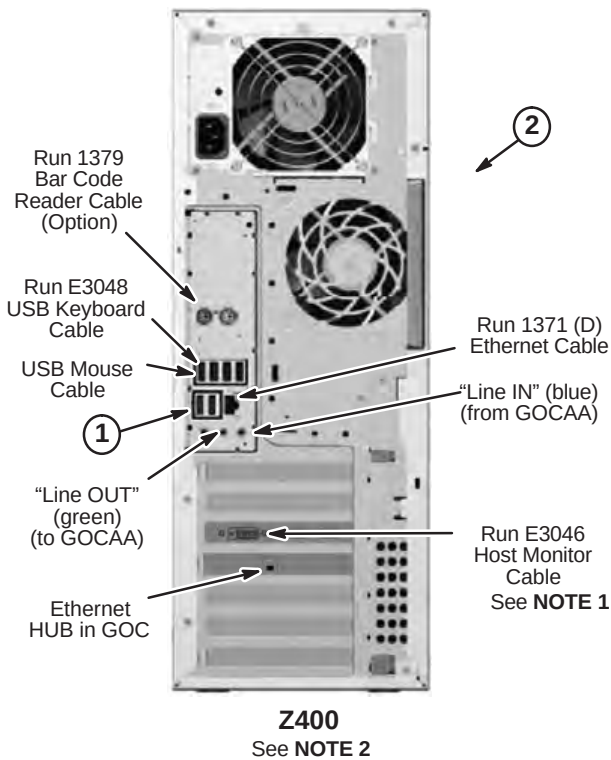
- 1 Place keyboard in front of SCIM, pull the keyboard forward, flip it, and place on top of the SCIM up side down.
- 2 Remove the plastic strips from the velcro located on SCIM plate, exposing the adhesive surface.
- 3 Guide the cable back into the center notched opening until all the cable is under the SCIM base.
- 4 Place the keyboard tight against the SCIM, Center it as best as possible, and drop into place. The Velcro pads will now be sticking to the bottom of the keyboard.

**Note**

The adhesive on the velcro takes 24 hours to completely adhere to the bottom of the keyboard. Allow the adhesive to dry completely before removing keyboard from SCIM base.

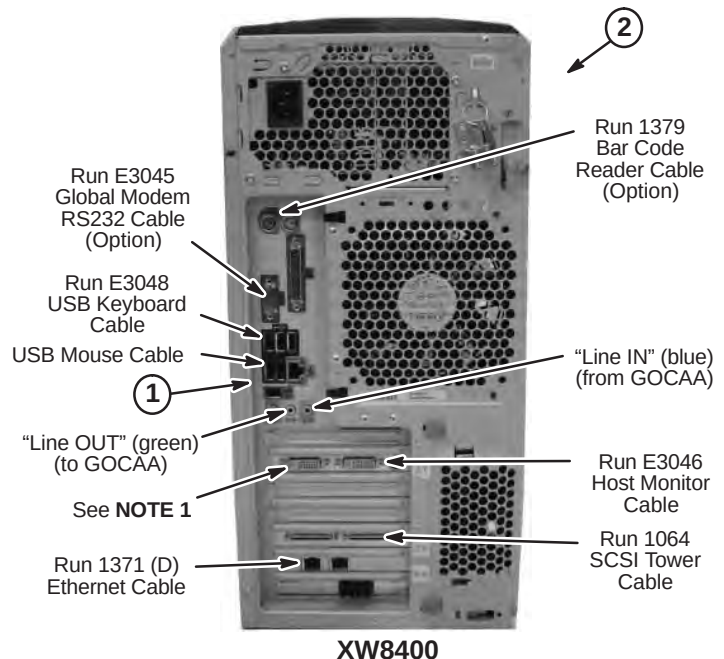
7-9 CONNECT CABLES TO z400 COMPUTER

- 1 Refer to Section 7-4 for installing the USB to 1-Wire Adapter (5307784) to USB port.
- 2 Connect remaining cables as marked at rear of computer.



NOTE 1:
Do not connect any additional monitors prior to loading operating system software. Doing so will cause boot-up to stop.

NOTE 2:
SCSI Tower and DASM are not supported for Z400PC.



7-10 FINAL POSITIONING OF GOC CABINET

Complete positioning of GOC Cabinet.

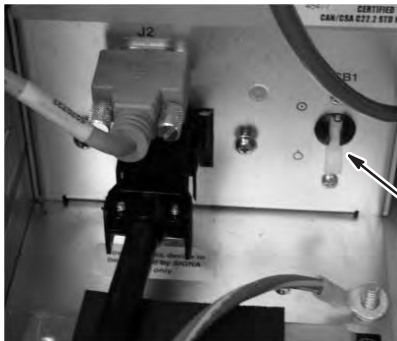
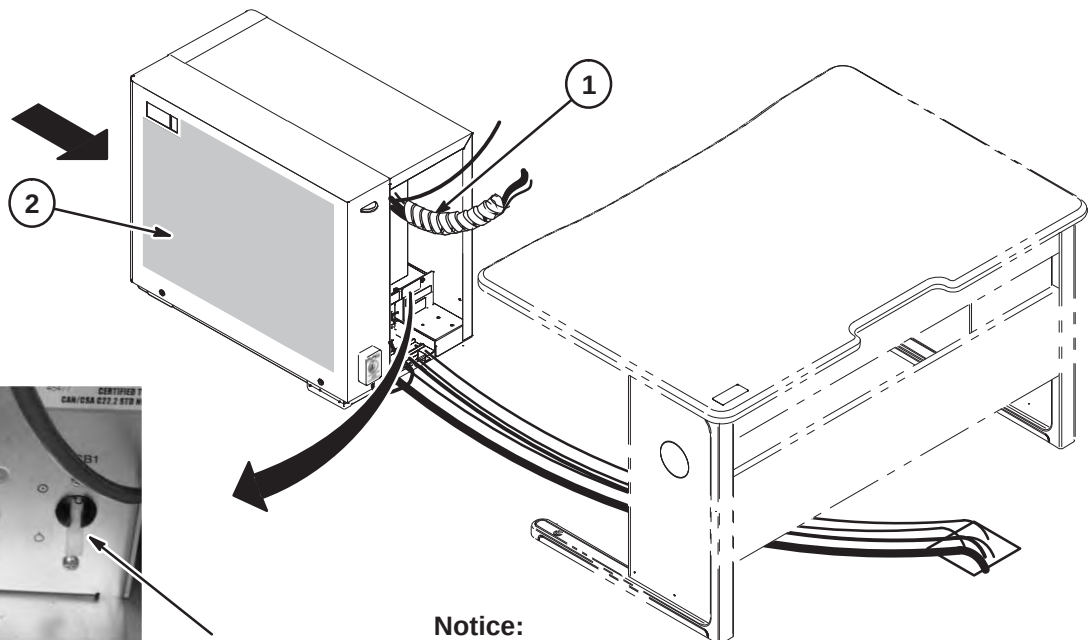
- 1 Use the spiral tube to bind the following cables. (Length of spiral tube is about 1 m).

Notice

Do Not include Modem power cable in Spiral Tube.

- Two power cords
- One video cable
- Keyboard cable
- SCIM cable
- Modem cable (if applicable)

- 2 Place cabinet in final position as determined by customer.



Notice:

GOC circuit breaker location. To avoid turning the Breaker OFF by accident, do not place the cables inside of GOC Cabinet near the circuit breaker.

7-11 POWER-ON INFORMATION

- 1 Remove LOTO from the PDU.

IMPORTANT

When turning on the circuit breaker in the PDU, both the PC and Host circuit breakers must be energized.

Note

When powering up the GOC, there is a Main breaker located in the front of the GOC. There are also two switches for the GOC AA and Modem power.

- 2 Restore power to the GOC, boot software, and ensure proper system operation.

NOTE: If necessary, reconnect the optional UPS accessory and ensure power is restored.

- 3 Perform the applicable Loading Host System Software procedure to load software and applications onto the operating system.

NOTE: The Load from Cold must be performed to properly transfer customer options

- 4 Update the elicense with the new Host ID.
- 5 Repackage the replaced computer for return
- 6 Perform a scan to ensure the system and applications are working properly.

SECTION 8 – SYSTEM CABINET

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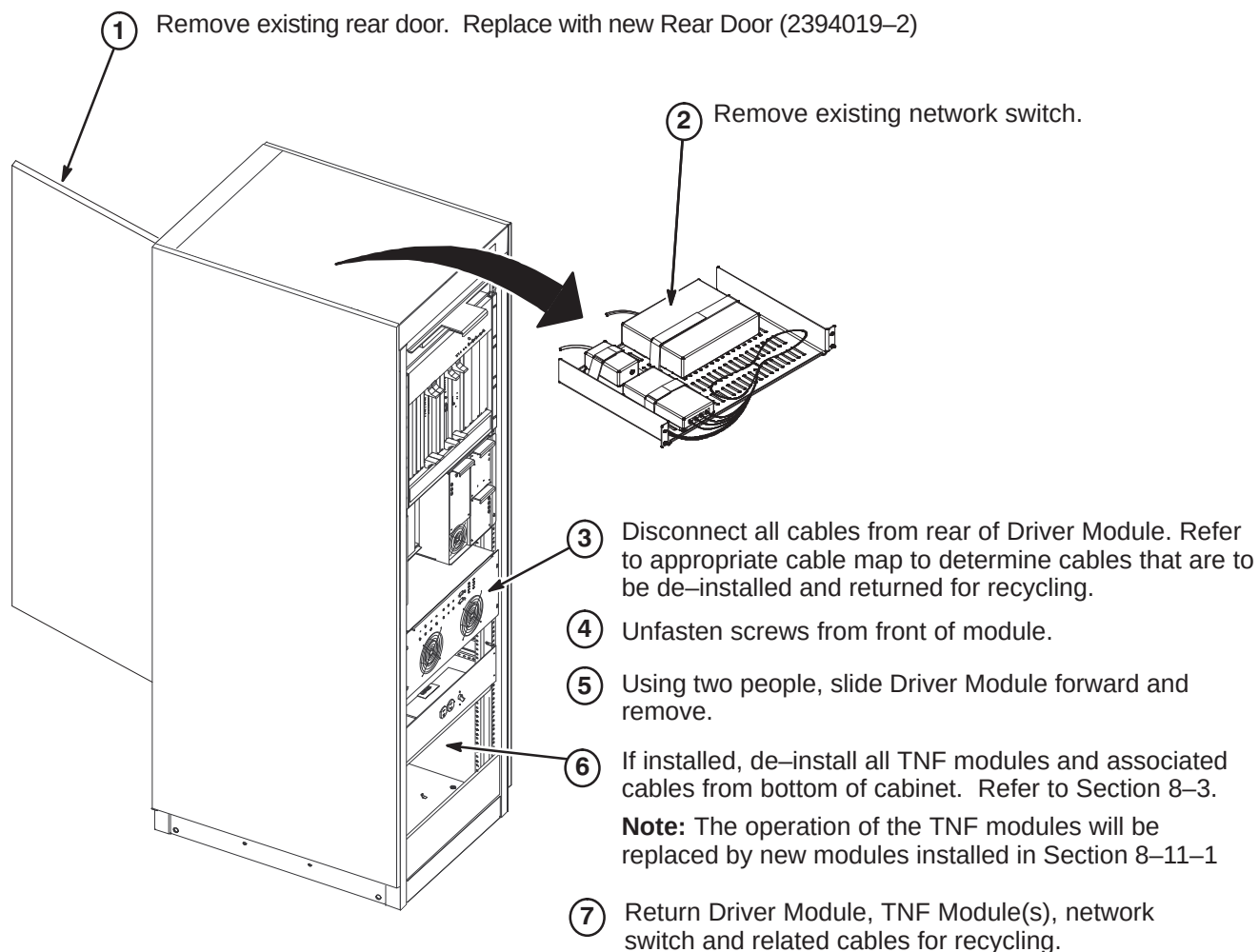
<u>SECTION</u>	<u>PAGE</u>
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8-10 INSTALL NEW REAR DOOR AND RATING PLATE	8-12

CAUTION

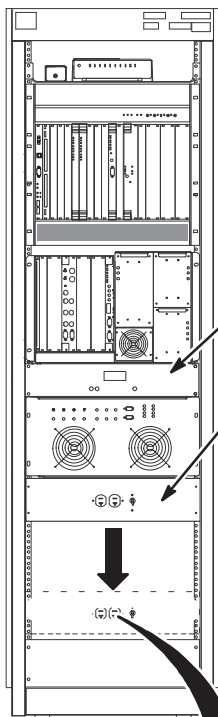
POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

8-1 INTRODUCTION

The System cabinet will be modified for the upgrade to HDx Release 14.x. Refer to cable maps in Section 4 for disconnection and removal of existing cables and installation of new cables. Refer to the following steps for the upgrade procedures that pertain to your site.

8-2 PREPARE SYSTEM CABINET FOR UPGRADE

8-3 RELOCATE SERVICE POWER PANEL

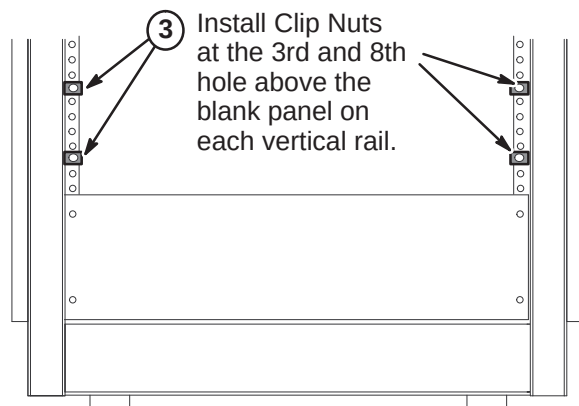


① If Cryogen Meter is part of cabinet, de-install and return for recycling.

② Service Power Panel has to be relocated down to new location.

⑤ Cut tie-wraps holding Service Power Panel wires to vertical rail, re-position cables and tie-wrap as shown.

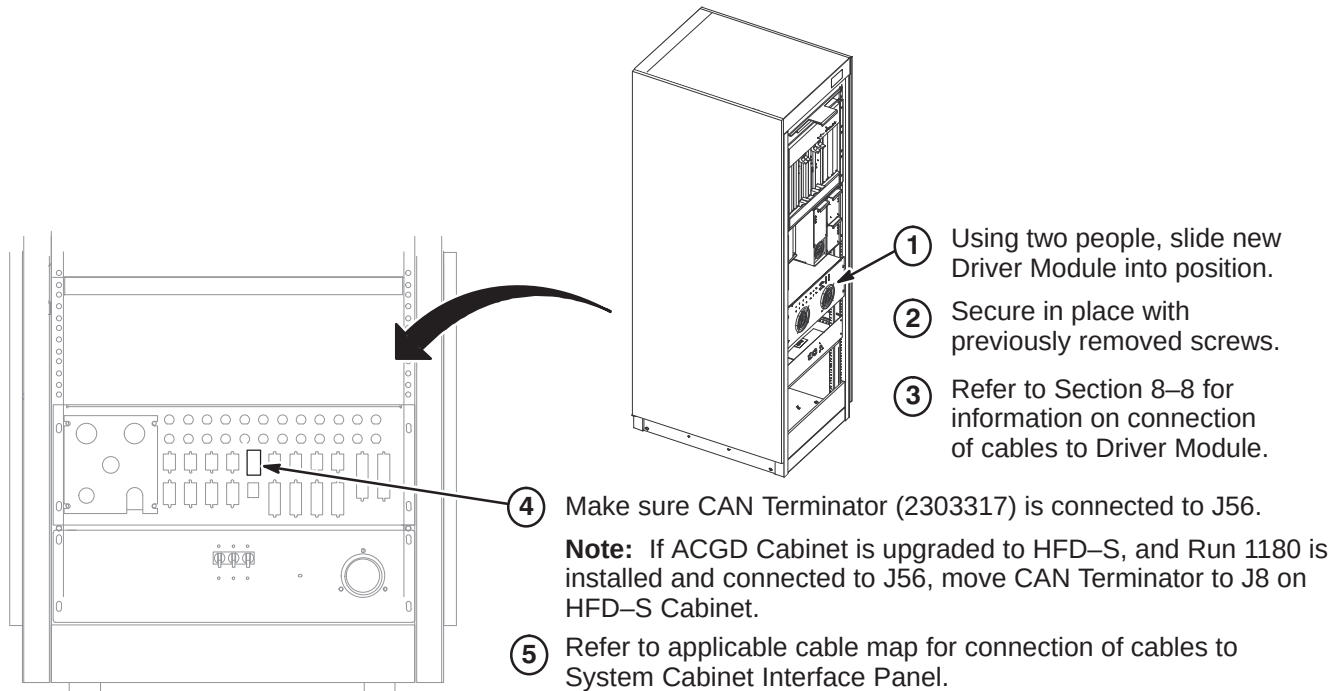
④ Fasten the Panel in place using the same mounting hardware. There will be a gap between Service Power Panel and the blank panel below.



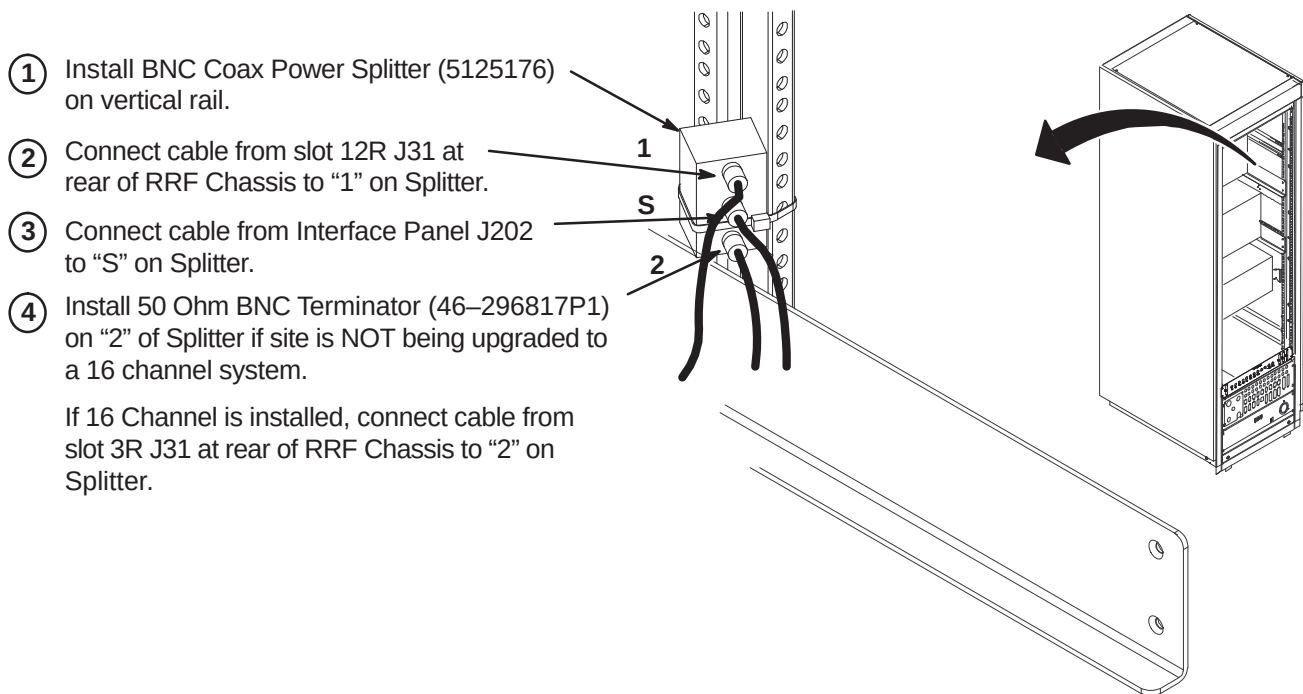
③ Install Clip Nuts at the 3rd and 8th hole above the blank panel on each vertical rail.

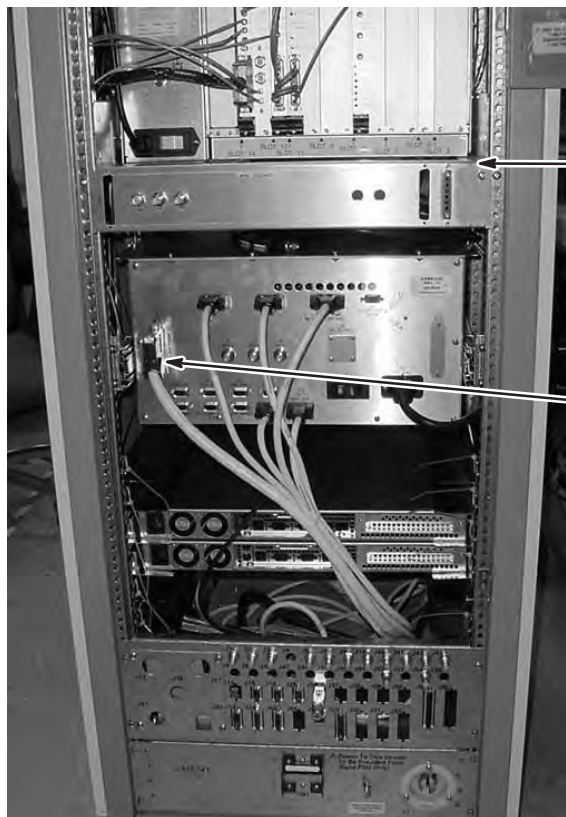


8-4 INSTALL NEW DRIVER MODULE



8-5 INSTALL COAX POWER SPLITTER



8-6 INSTALL NEW INTERFACE PANEL

- ① At rear of cabinet, install 3.0T upgrade Interface Panel Assembly (5151361-2). Align top of Panel to with bottom of RRF Chassis.
- ② Fasten with four (4) M5 x 16mm Pan head screw (2279742-2).
- ③ Reconnect door ground wire.
- ④ Disconnect and remove cable routed from Driver Module J5 to Interface Panel J61. New cable Run 1330 will connect directly to J5.
- ⑤ Use table below for connecting cables in the System Cabinet. All the cables connected to the new I/O Panel are pre-installed on the I/O Panel.

CABLE PART #	ONE END	OTHER END
5117295	Driver Module J4	MGD Slot 11R J21
2394870-3	RF Slot 14R J5	I/O Panel J200
2394870-4	RF Slot 5R J5	I/O Panel J201
5115416	Splitter S RF Slot 14R J13 I/O Panel J100 I/O Panel J101	I/O Panel J202

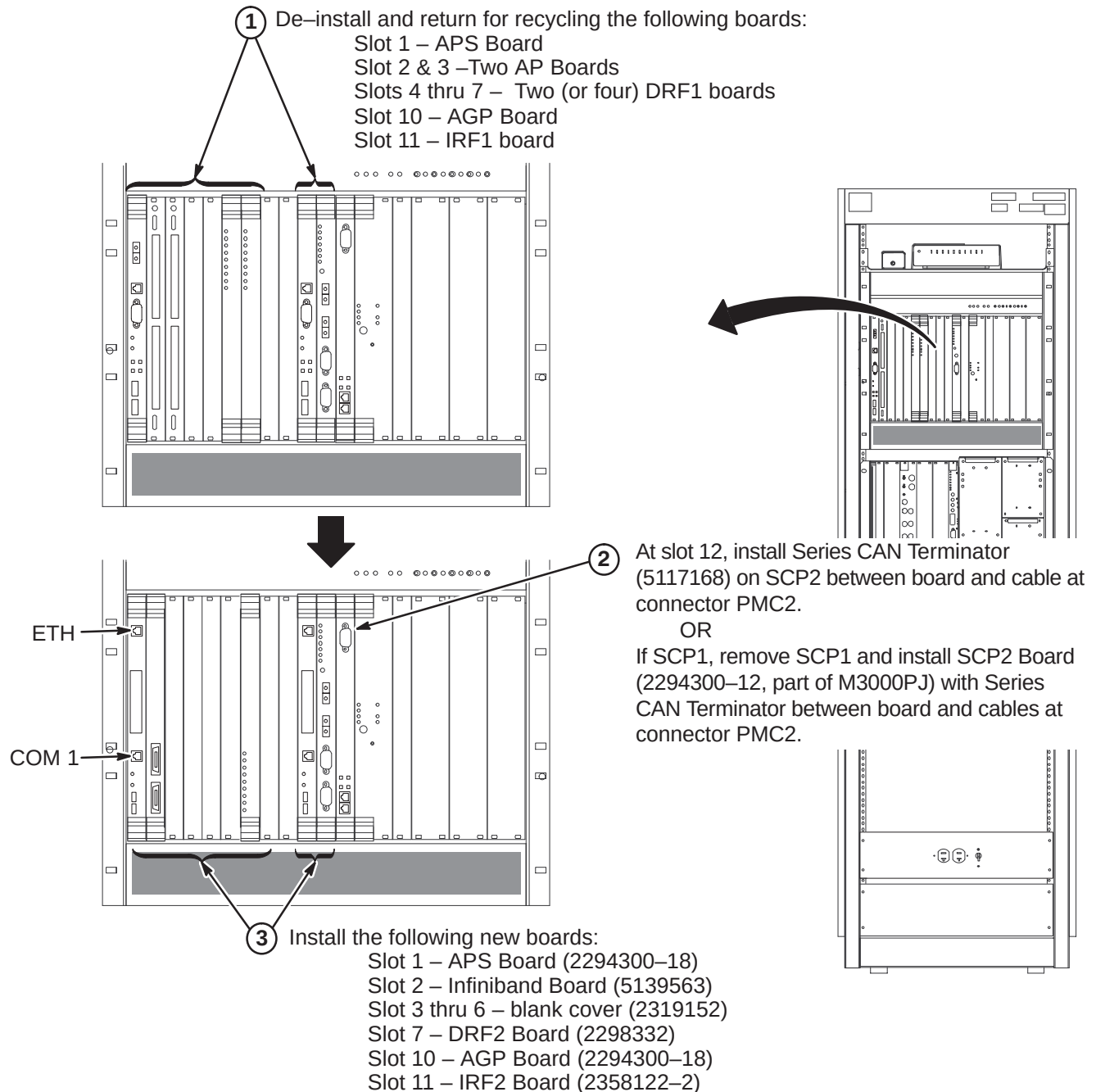
8-7 UPGRADE BOARDS IN MGD CHASSIS

DRF2 and IRF2 boards will replace DRF1 and IRF1 in the MGD Chassis (Pinot) when upgrading to HDx (Release 14.x) or HDxt (Release 15.x). If a pre-EXCITE HD site has four channels, it will have 2 DRF1 boards. If a pre-EXCITE HDx site has 8 channels, it will have 4 DRF1 boards. All existing sites will have one IRF1 board.



Note: Modules are static sensitive components and need to be handled as such (wrist strap).

Note: to remove modules – loosen the screws in the ejector pins, then push the top and bottom tabs to eject the boards from the chassis.



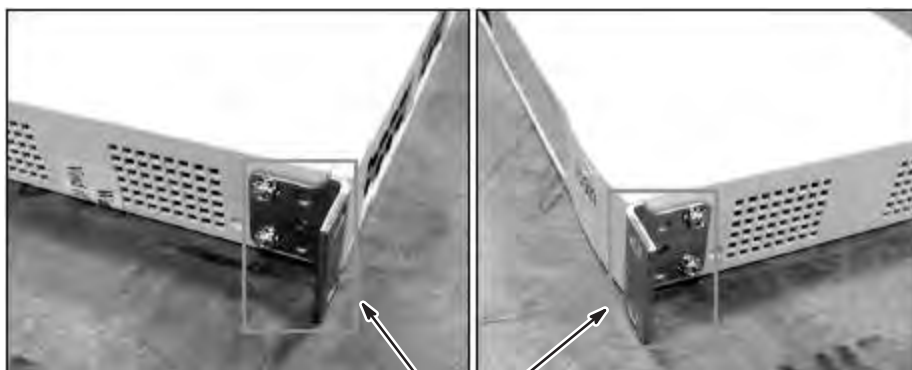
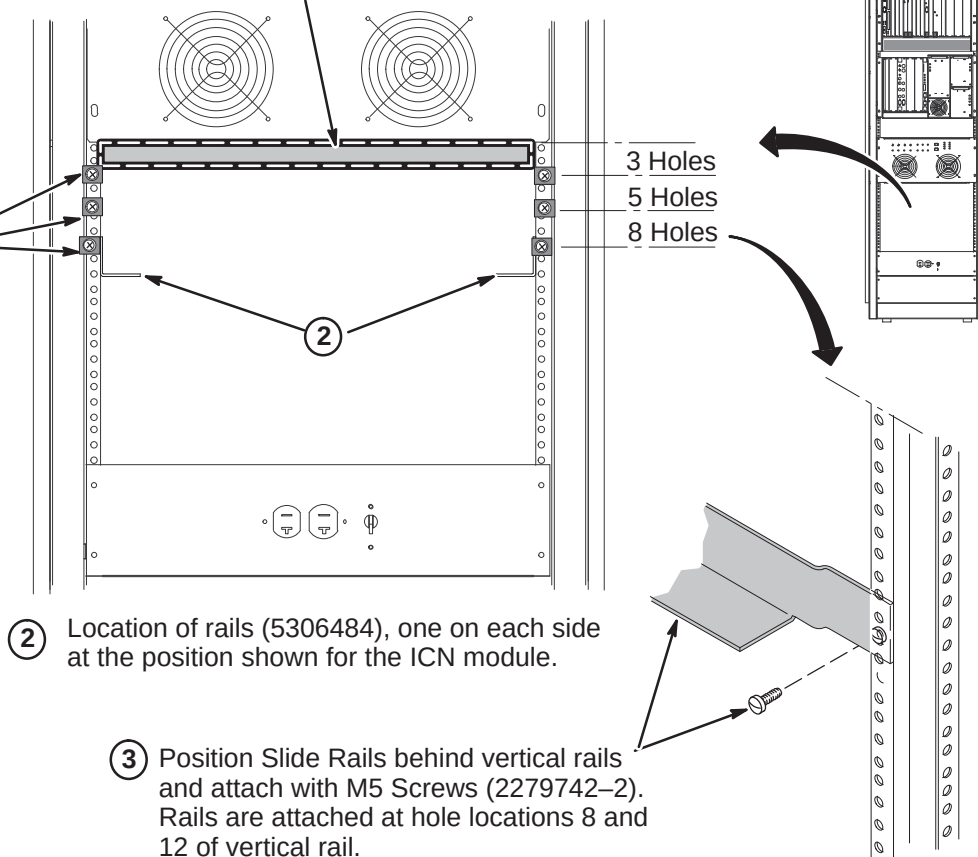
8-8 INSTALL VRE COMPONENTS

8-8-1 INSTALL VRE MODULE RAILS

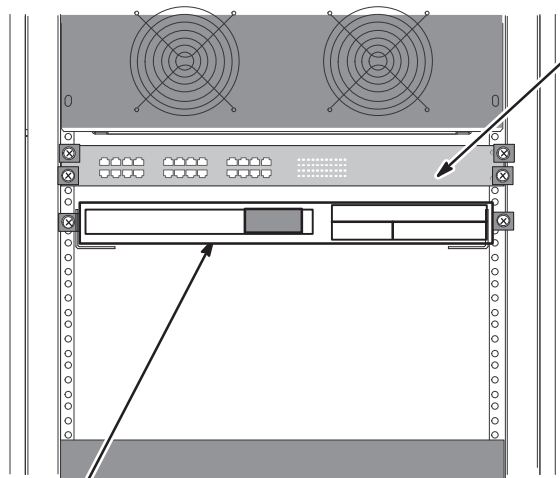
- ① Make sure that the top clipnuts are located below two mounting holes on the vertical rail. This area between Driver Module and Ethernet Switch will be used for routing cables from the Ethernet Switch to applicable modules.

- ④ Install 3 Clip Nuts (2180698-2) on each side at locations indicated.

- ⑤ Install 2 (5306487) L-brackets at the back of the cabinet.



- ⑥ Install brackets on front corners of Ethernet switch as shown. Hardware kit comes with unit.

8-8-2 INSTALL VRE MODULES

① Install Ethernet Switch (5162073) at this location using Clip Nuts (2180698-2) and M5 Screws (2279742-2). Make sure power switch is turned on.

② Install 2 (5306485) brackets on the ICN. The brackets attach by snapping them into place. On the left side, the bracket is installed downward and snapped into place.

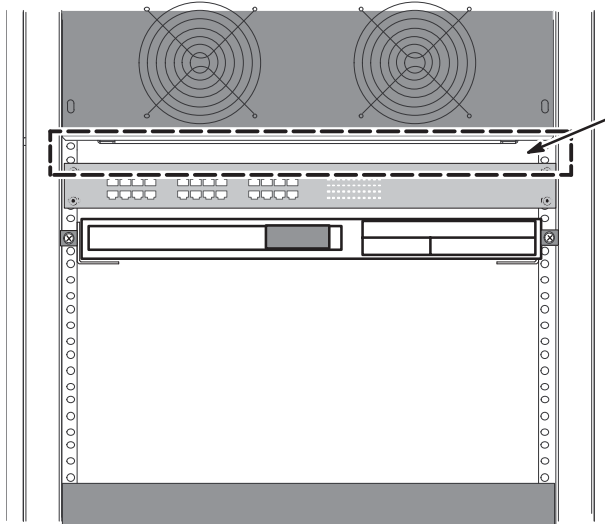


On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.



8-8-3 INSTALL VRE CABLES

Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables.



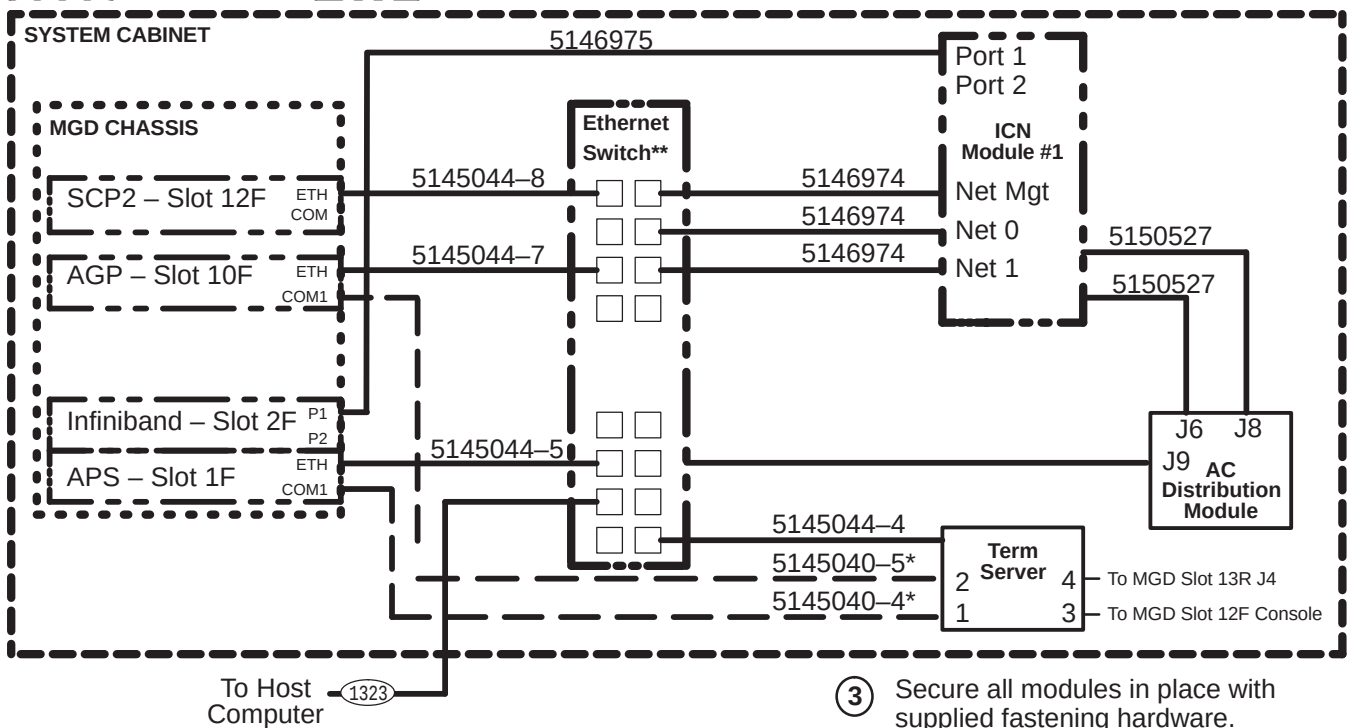
- ① All cables connecting between Ethernet Module and ICN Modules must be routed thru area above Ethernet Switch. Ethernet cables routed to Ethernet Switch can be connected to any ethernet port.

NOTE: Any existing cables connected to the MGD Chassis, or any other modules, that are not identified below should remain connected. They are not affected by this upgrade.

- ② Route and connect new VRE Cables as shown below. Cables 5145040-4 and 5145040-5 are replacing existing cables. Make sure 5146975 is connected to P1 (Port 1) and not P2 on the back of the ICN.

- ③ Use power cable supplied with Ethernet Switch and connect to J9.

LEGEND:



- ③ Secure all modules in place with supplied fastening hardware.

* = New cable replacing existing cable

** = Ethernet cables routed to Ethernet Switch can be connected to any port.

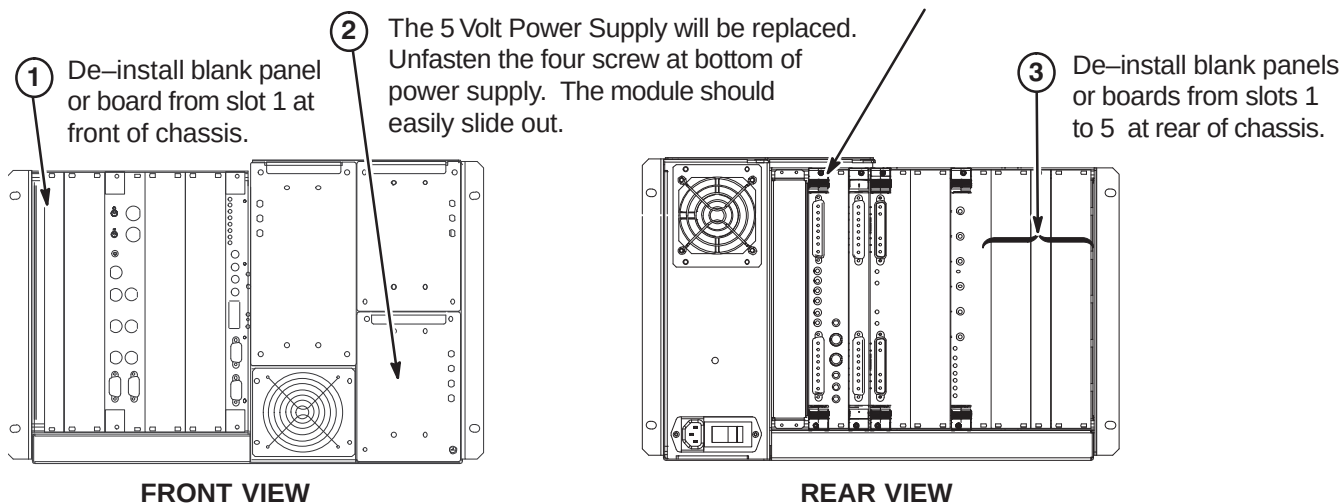
8-9 PROCEDURES FOR UPGRADING TO 16 CHANNEL (M3000PE)

8-9-1 REMOVE EXISTING COMPONENTS FROM RRF CHASSIS



Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

Note: It is recommended that the existing MUX board be at the same part number and revision level as the new MUX board being installed in slots 4 & 5.

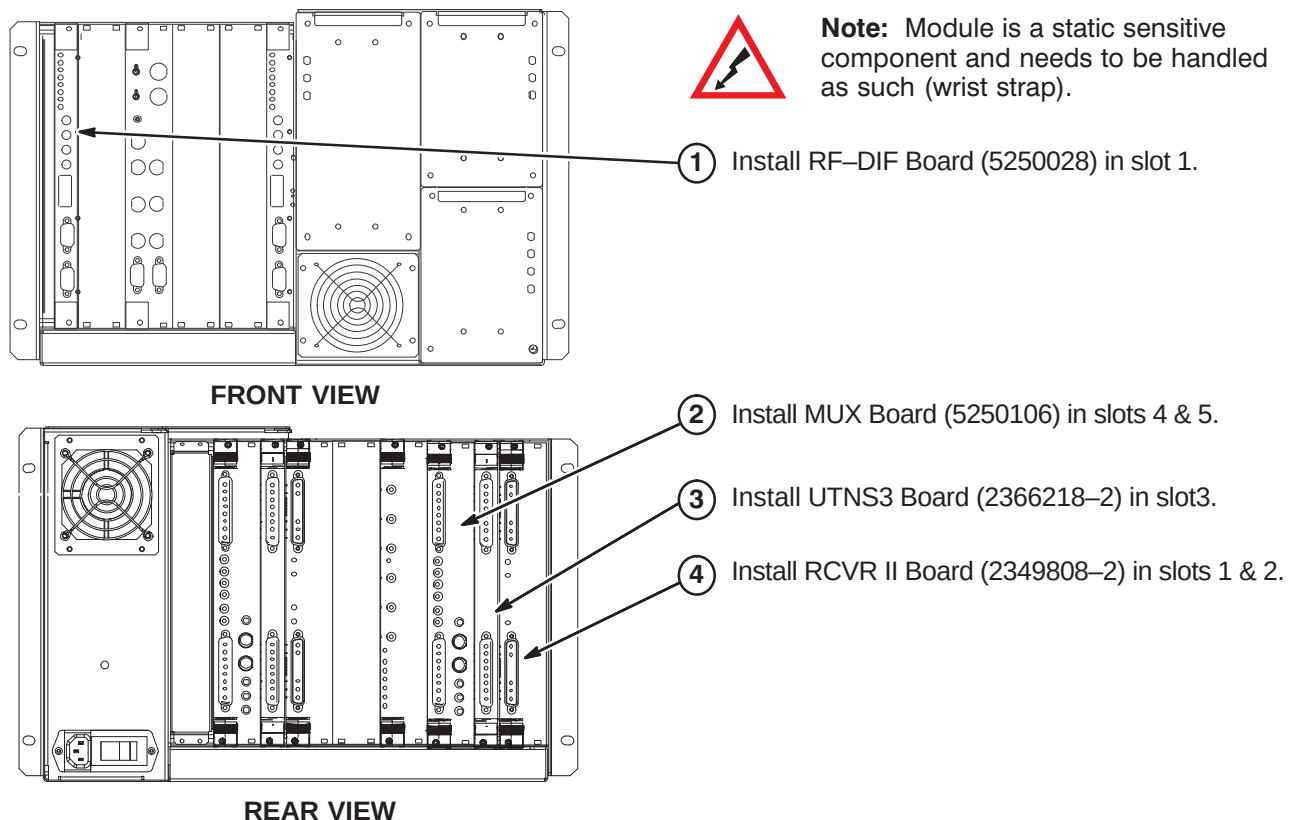


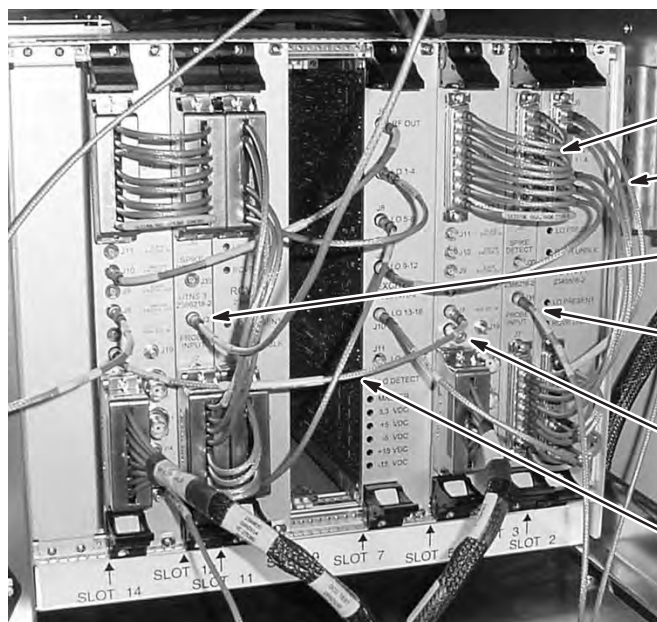
Note: All de-installed boards are to be returned for recycling.

8-9-2 INSTALL NEW MODULES IN RRF CHASSIS

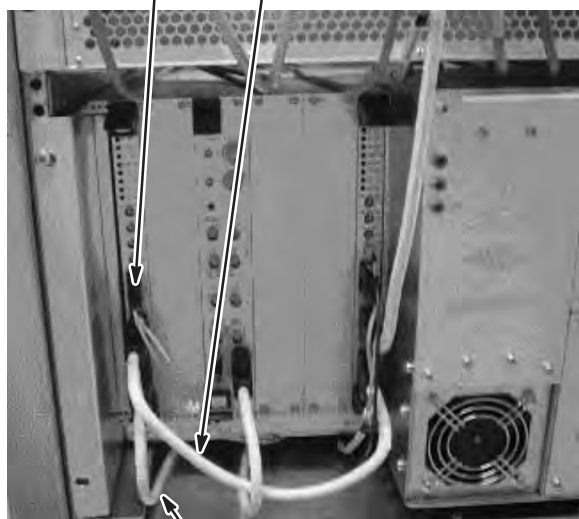


Note: Module is a static sensitive component and needs to be handled as such (wrist strap).

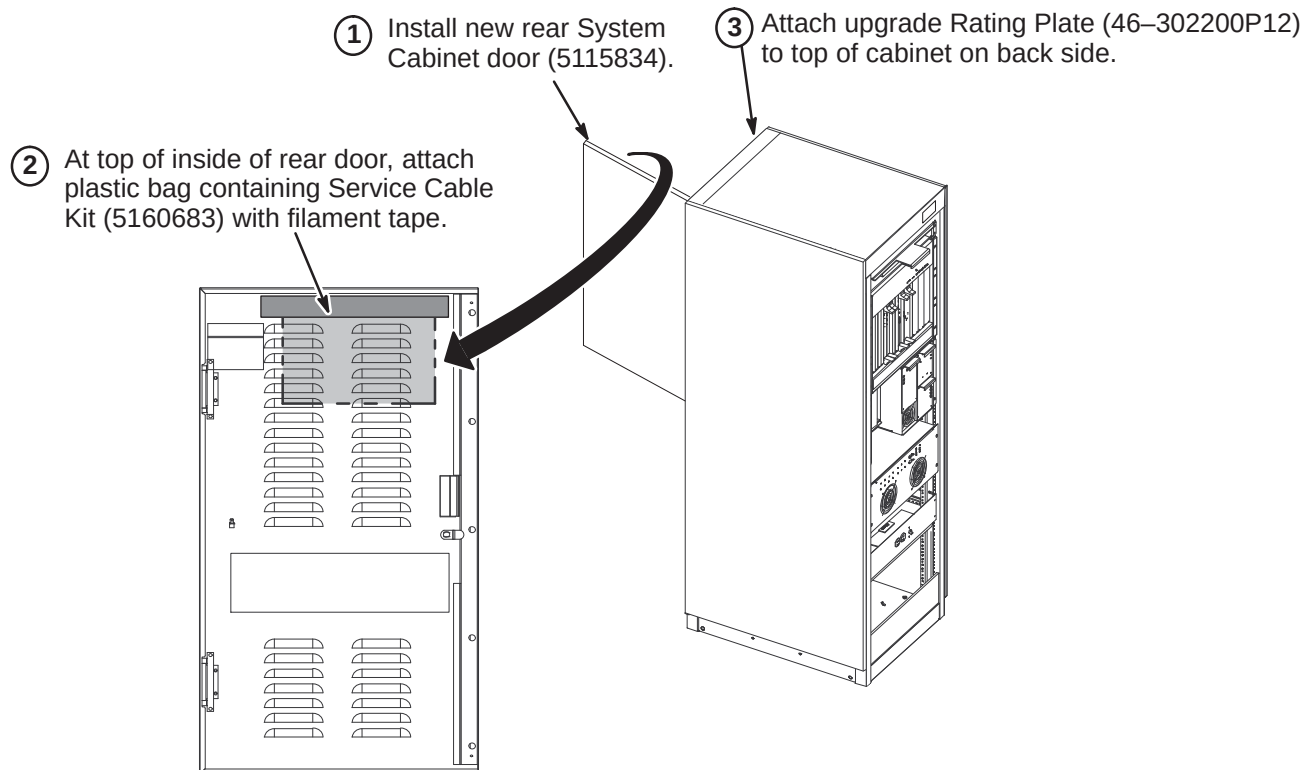


8-9-3 INSTALL CABLES AT REAR OF RRF CHASSIS

- ① Connect cable 5123706 as marked.
- ② Connect cable 5113193-2 as marked.
- ③ Make sure cable 5115417 slot 12R J31 is connected as marked. it should be routed and connect to Coax Power Splitter "1".
- ④ Connect cable 5115417-2 as marked. Route and connect to Coax Power Splitter "2". Remove 50 Ohm terminator if installed at "2".
- ⑤ Move 50 Ohm Terminator from Mux1 J6 (slots 13 & 14) to Mux2 J6 (slots 4 & 5).
- ⑥ Connect cable 5115630 as marked.

8-9-4 INSTALL CABLES AND POWER SUPPLY AT FRONT OF RRF CHASSIS

- ③ Connect cable 5121122-2 as marked to RF Slot 1F J21. Route and connect other end to MGD Slot 11F J7.
- ② Connect cable 5113182-2 as marked.
- ④ Slide in new 5V Power Supply (2292267-2). Make sure that it is properly seated before tightening the four screws.
- ① Disconnect cable (2323878-3) from Slot 10F J23 and connect to Slot 1F J23.

8-10 INSTALL NEW REAR DOOR AND RATING PLATE

SECTION 9 – RF ACCESSORY CABINET

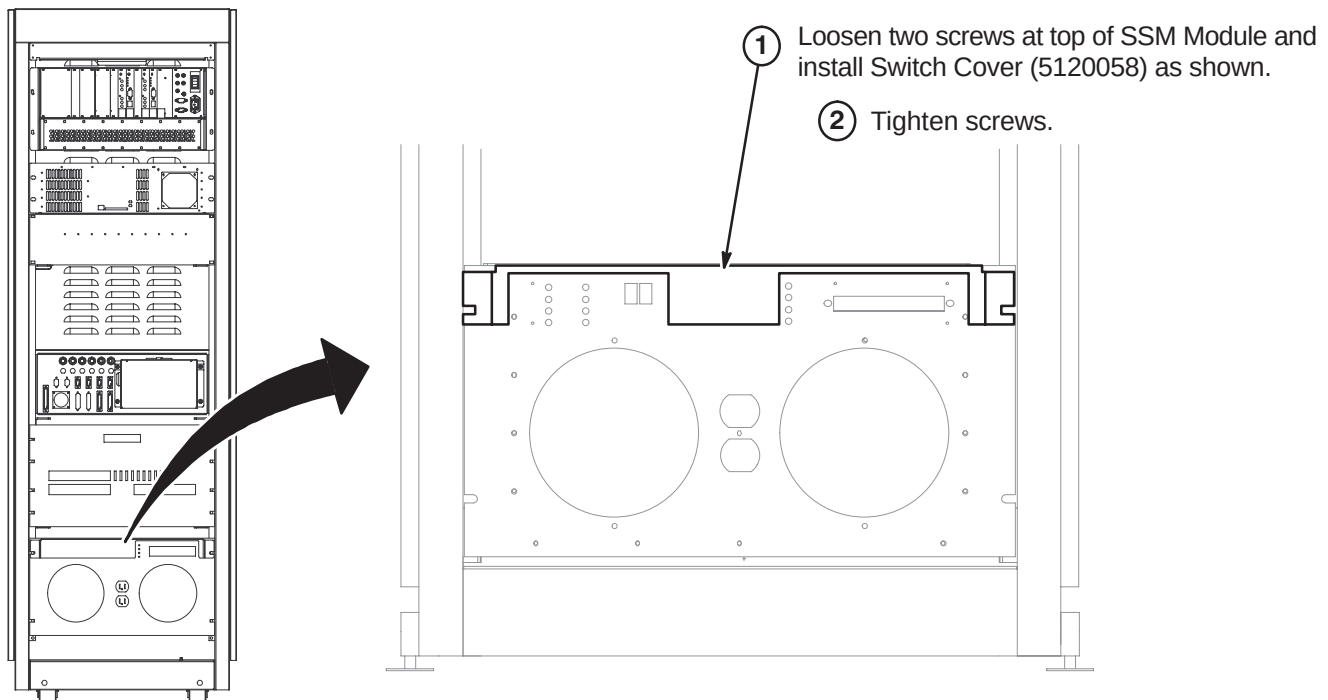
CAUTION

POWER TO SYSTEM MUST BE OFF, LOCKED AND TAGGED, BEFORE STARTING ANY UPGRADE PROCEDURES. REFER TO SECTION 3 FOR POWER OFF PROCEDURES.

9-1 INTRODUCTION

The existing RF Accessory Cabinet will not be affected by the EXCITE upgrade to 12.x. Perform the following procedure only if cover is not already installed.

9-2 INSTALL SWITCH COVER



SECTION 10 – COMPLETION

10-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 10-2, 10-3, and 10-4.

10-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 10-3. Recycling Operation will route them to appropriate location.

10-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-7. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

10-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Replace Magnet Enclosure covers and Rear Pedestal covers
3. Dock Patient Transport in position at front of bore.
4. Record and enter applicable data into applicable site configuration files and records.
5. Complete Product Locator information. See Section 1-8.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

6. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
7. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
8. Verify that coordination of application support for instruction of site users on Software Release 14.x has been made before returning system to the users. Turn site over to applications who will instruct users.

10-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the ACGD Gradient power modules, switch all cabinet circuit breakers to ON. The ACGD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.